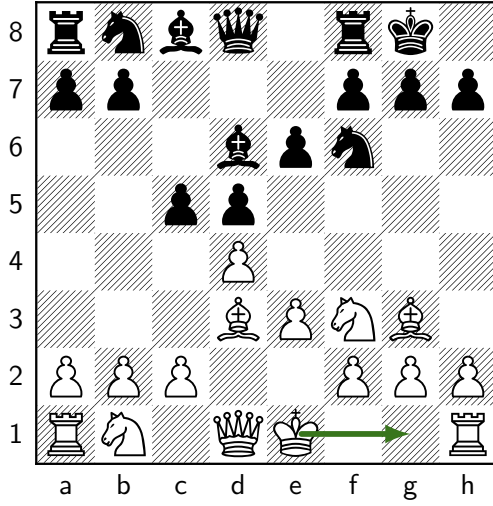


Melanie-MASTERS (human masters) — London System

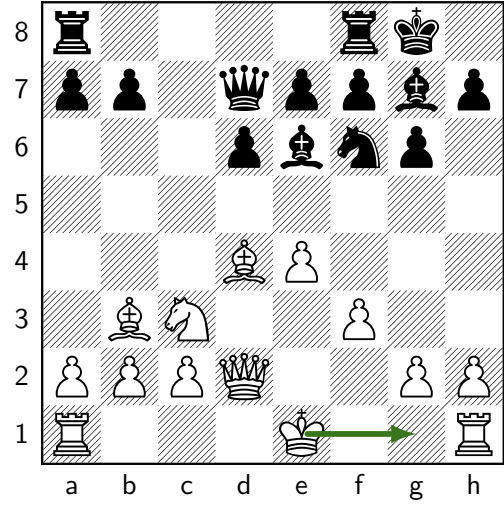
Melanie (Deep_Vision)

Abstract

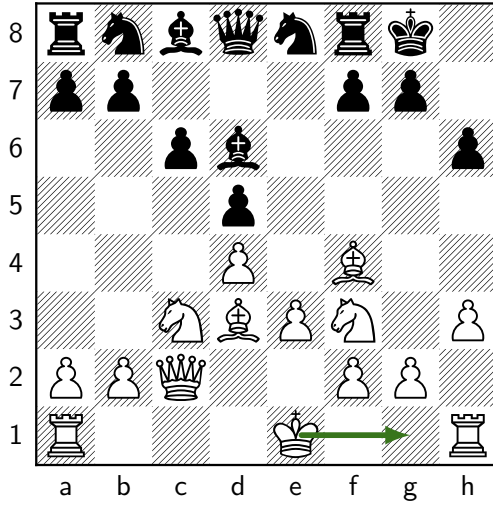
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie's move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen's role) are called out, as no material/eval engine explains a move that way.



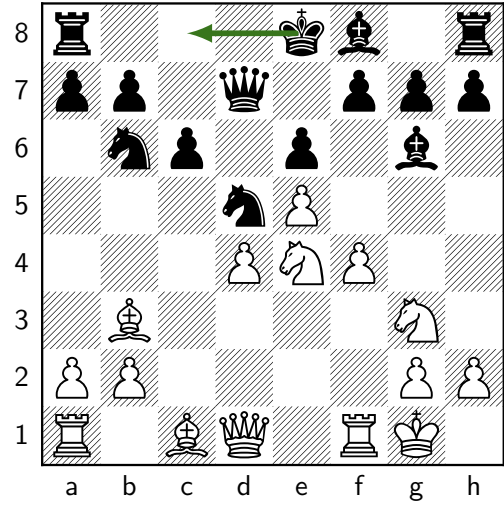
(a)



(b)

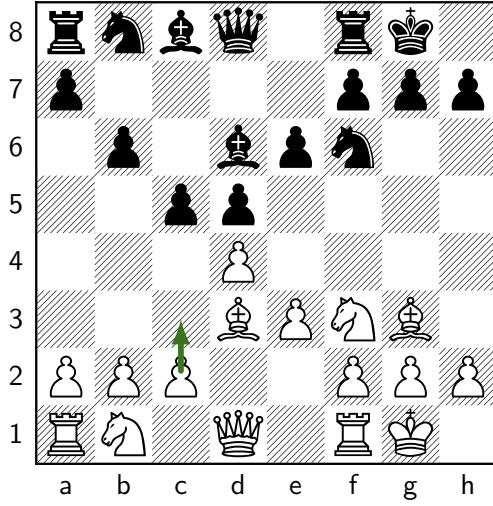


(c)

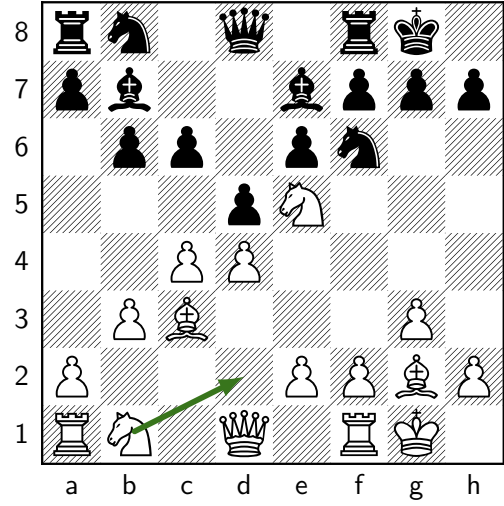


(d)

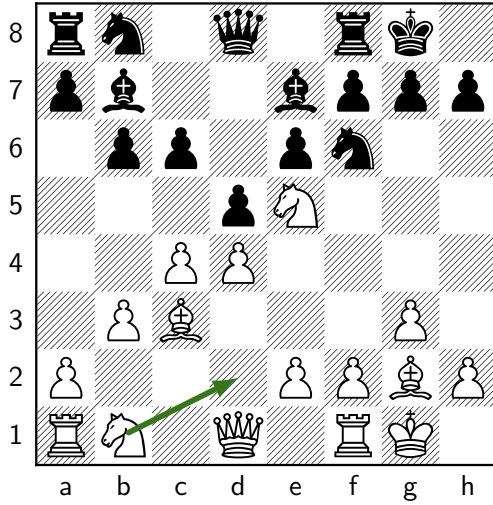
Figure 1: **Move 7, White: 0-0.** (a) Our position; Melanie plays 0-0 (e1g1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0028#2646` (ply 22) [structural distance $d=68$, QAP obj=54]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Nf3* as their *Ra1*, our *Nb1* as their *Ph2*; transferred move e1g1 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0031#3304` (ply 20) [structural distance $d=66$, QAP obj=46]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *pd5* as their *Rh1*, our *Nb1* as their *Pa2*; transferred move e1g1 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0025#3637` (ply 23) [structural distance $d=82$, QAP obj=35]: the master played 0-0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Qd1* as their *Ra8*, our *Ph2* as their *Bf8*; transferred move e1g1 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to 0-0 (summed QAP objective 161, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



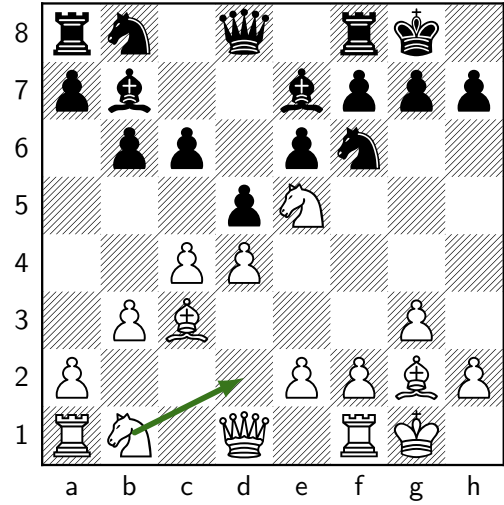
(a)



(b)

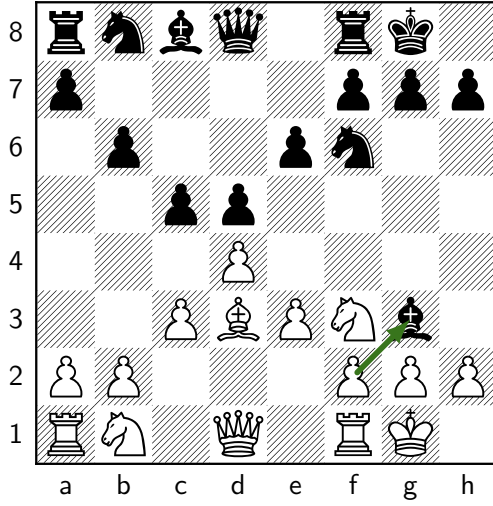


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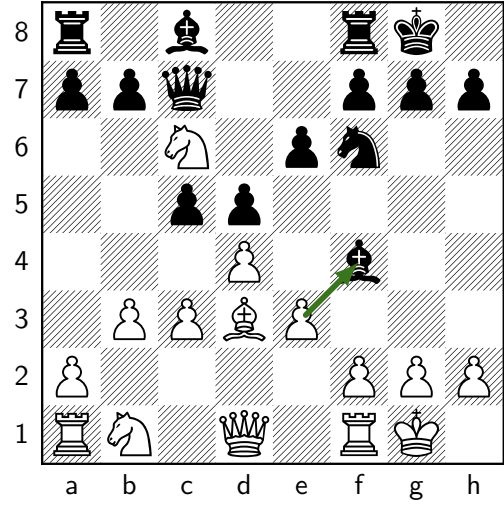


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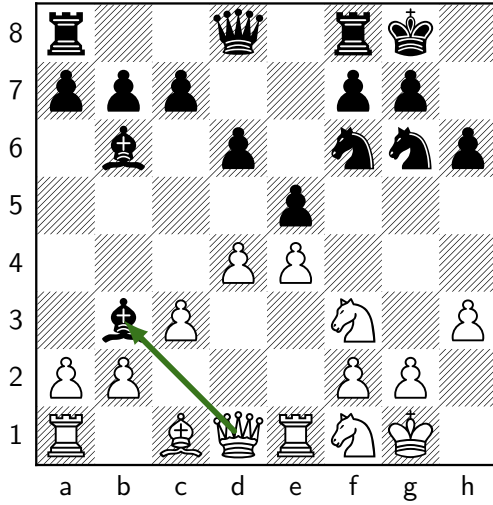
Figure 2: **Move 8, White: c3.** (a) Our position; Melanie plays c3 (c2c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0003#824` (ply 20) [structural distance $d=81$, QAP obj=63]: the master played Nd2; under μ our *Pc2* plays the role of that game’s *Nb1* (a **cross-type** correspondence); cross-type roles: our *Qd1* as their *Pf2*, our *Nb1* as their *Ph2*; transferred move c2c3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0003#1180` (ply 20) [structural distance $d=81$, QAP obj=63]: the master played Nd2; under μ our *Pc2* plays the role of that game’s *Nb1* (a **cross-type** correspondence); cross-type roles: our *Qd1* as their *Pf2*, our *Nb1* as their *Ph2*; transferred move c2c3 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0013#2762` (ply 20) [structural distance $d=81$, QAP obj=63]: the master played Nd2; under μ our *Pc2* plays the role of that game’s *Nb1* (a **cross-type** correspondence); cross-type roles: our *Qd1* as their *Pf2*, our *Nb1* as their *Ph2*; transferred move c2c3 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to c3 (summed QAP objective 189, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



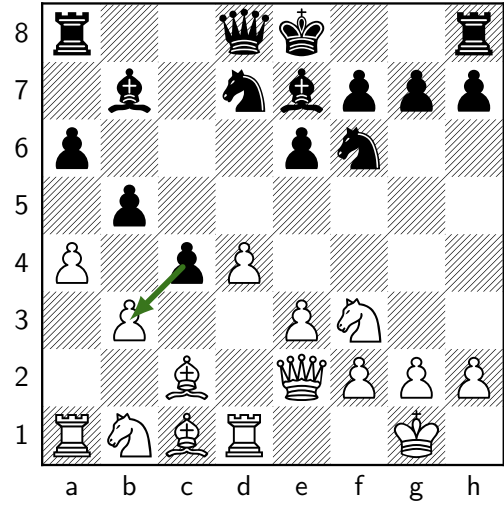
(a)



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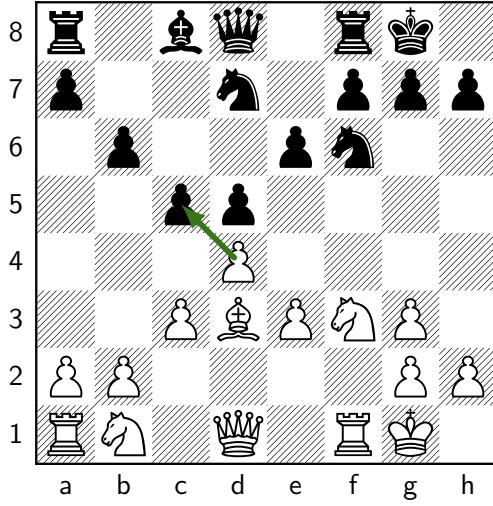


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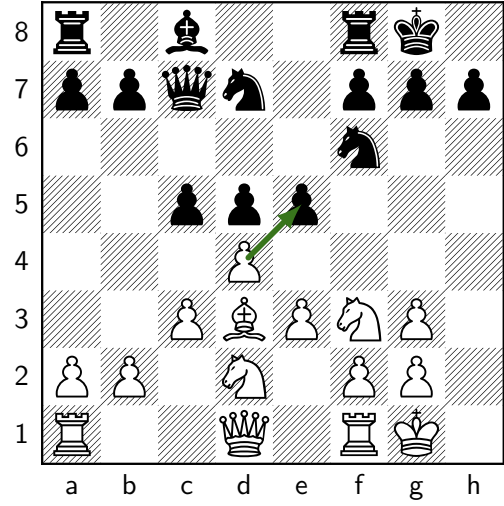


(d)

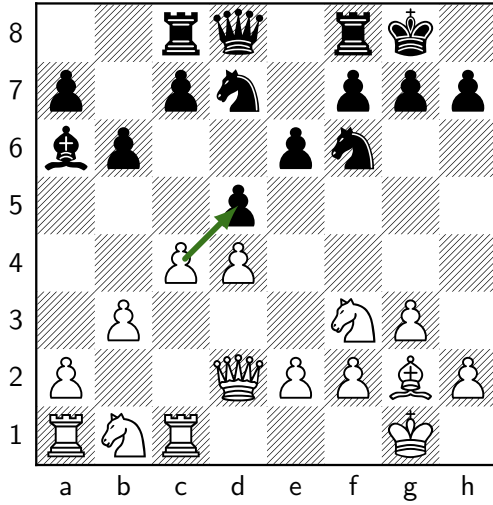
Figure 3: **Move 9, White: f2g3.** (a) Our position; Melanie plays f2g3. Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0031#3330` (ply 20) [structural distance $d=89$, QAP obj=70]: the master played `exf4`; under μ our *Pf2* plays the role of that game’s *Pe3*; cross-type roles: our *pd5* as their *Ra1*, our *pa7* as their *Nb1*; transferred move `f2g3` (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0025#2726` (ply 24) [structural distance $d=85$, QAP obj=43]: the master played `Qxb3`; under μ our *Pf2* plays the role of that game’s *Qd1* (a **cross-type** correspondence); cross-type roles: our *Qd1* as their *Ra1*, our *ra8* as their *Bc1*; transferred move `f2g3` (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0019#3664` (ply 23) [structural distance $d=90$, QAP obj=33]: the master played `cxb3`; under μ our *Pf2* plays the role of that game’s *Pc5*; cross-type roles: our *Pg2* as their *Ra8*, our *Rf1* as their *Qd8*; transferred move `f2g3` (**endorses this move**). Synthesis: 5 retrieved analogs transferred to `f2g3` (summed QAP objective 186, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



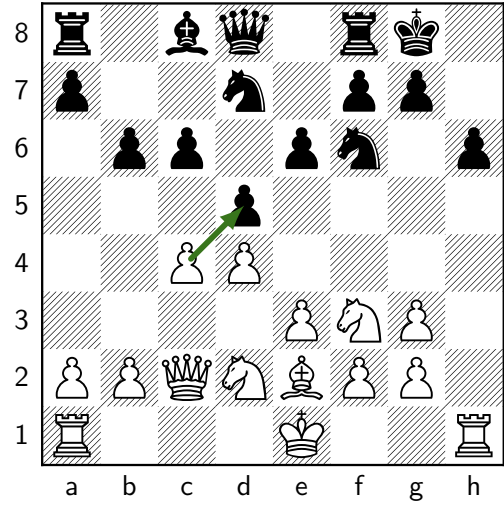
(a)



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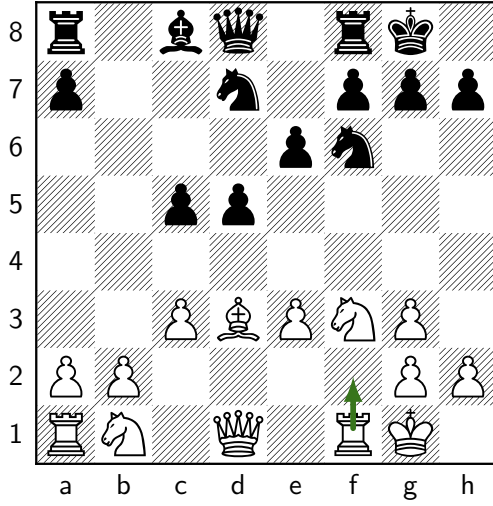


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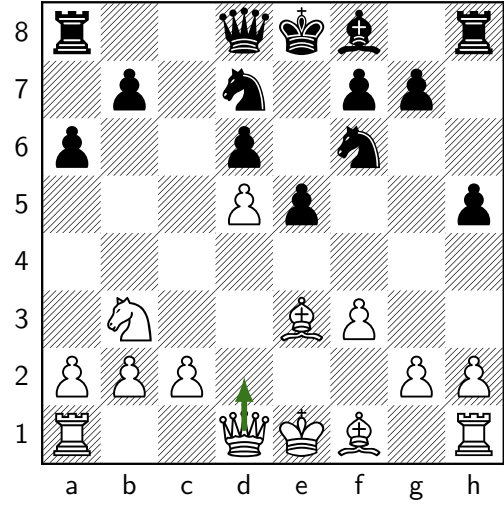


(d)

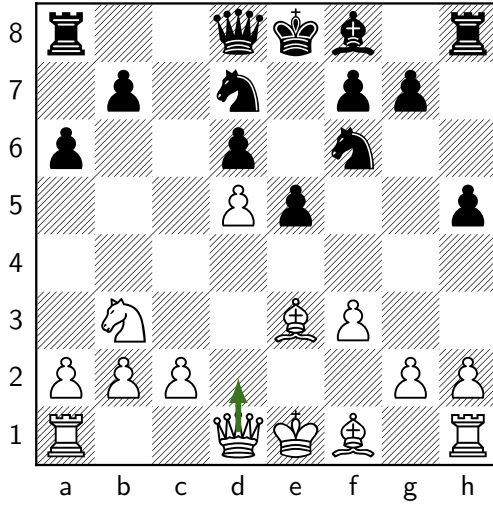
Figure 4: **Move 10, White: dxc5.** (a) Our position; Melanie plays dxc5 (d4c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0031#3095` (ply 20) [structural distance $d=84$, QAP obj=107]: the master played dxe5; under μ our $Pd4$ plays the role of that game’s $Pd4$; cross-type roles: our $Nf3$ as their $Qd1$, our $Nb1$ as their $Pa2$; transferred move d4c5 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0005#2875` (ply 20) [structural distance $d=87$, QAP obj=55]: the master played cxd5; under μ our $Pd4$ plays the role of that game’s $Pc4$; cross-type roles: our $Nb1$ as their $Ra1$, our $Pb2$ as their $Nb1$; transferred move d4c5 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0025#1517` (ply 20) [structural distance $d=81$, QAP obj=45]: the master played cxd5; under μ our $Pd4$ plays the role of that game’s $Pc4$; cross-type roles: our $Nb1$ as their $Pa2$, our $Pg2$ as their $Qc2$; transferred move d4c5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to dxc5 (summed QAP objective 207, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



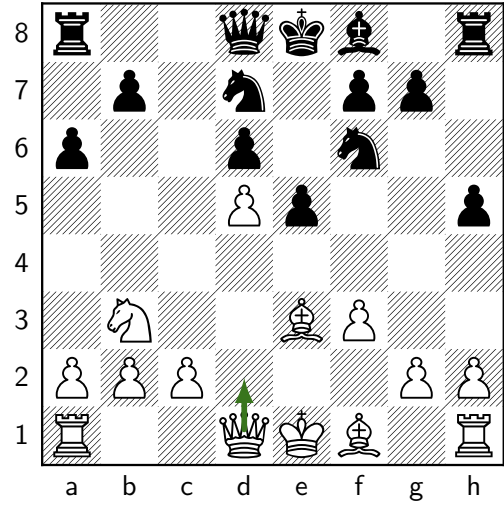
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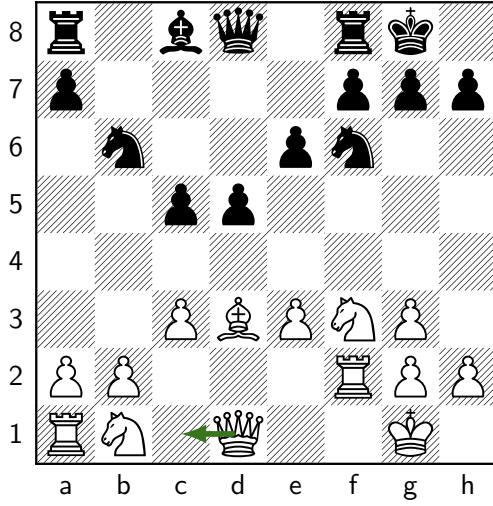


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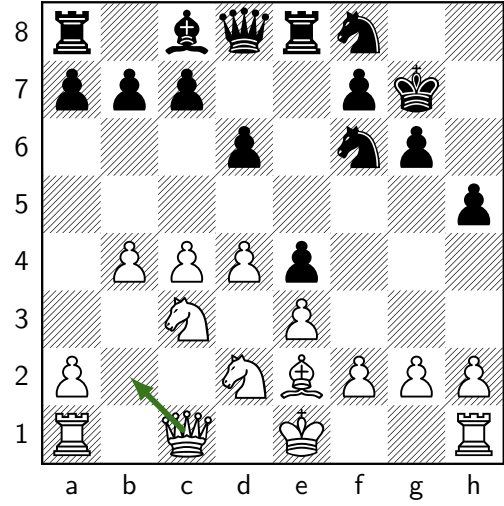


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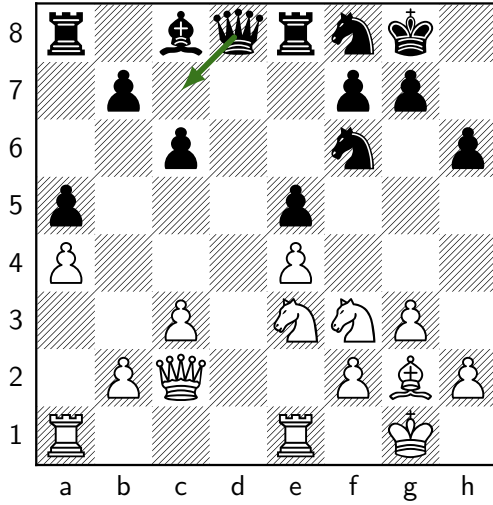
Figure 5: **Move 11, White: Rf2**. (a) Our position; Melanie plays Rf2 (f1f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0005#1831` (ply 20) [structural distance $d=85$, QAP obj=16]: the master played Qd2; under μ our Rf1 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our pe6 as their Bf1, our Qd1 as their Rh1; transferred move f1f2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0024#343` (ply 20) [structural distance $d=85$, QAP obj=16]: the master played Qd2; under μ our Rf1 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our pe6 as their Bf1, our Qd1 as their Rh1; transferred move f1f2 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0024#3827` (ply 20) [structural distance $d=85$, QAP obj=16]: the master played Qd2; under μ our Rf1 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our pe6 as their Bf1, our Qd1 as their Rh1; transferred move f1f2 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rf2 (summed QAP objective 48, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



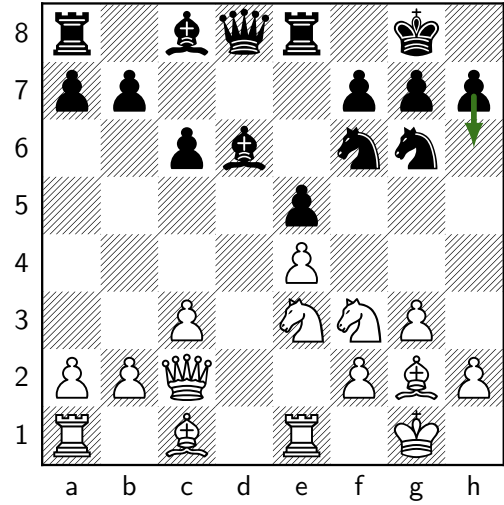
(a)



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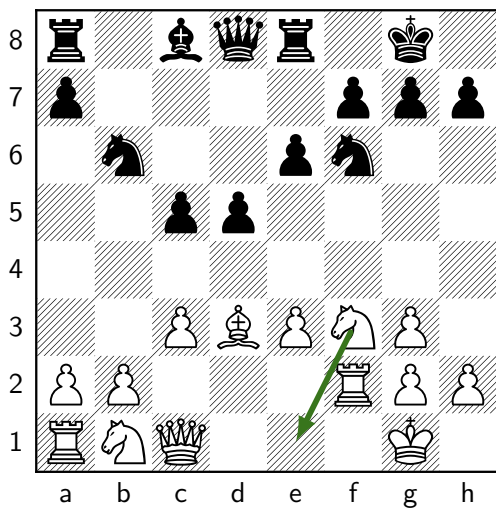


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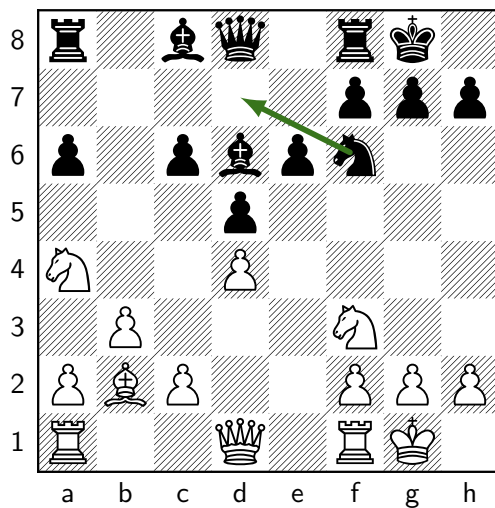


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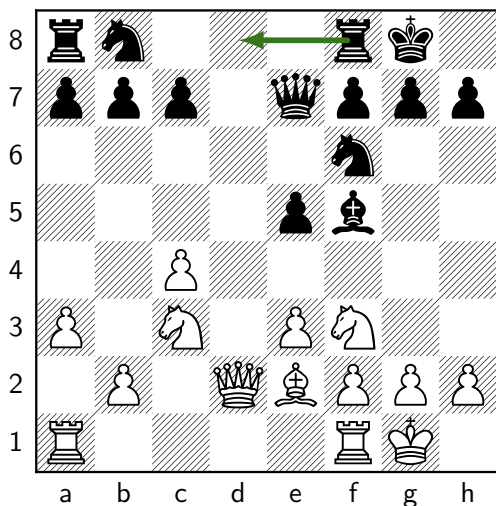
Figure 6: **Move 12, White: Qc1**. (a) Our position; Melanie plays Qc1 (d1c1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0013#1218` (ply 24) [structural distance $d=84$, QAP obj=23]: the master played Qb2; under μ our Qd1 plays the role of that game’s Qc1; cross-type roles: our Pc3 as their Ra1, our Nb1 as their Rh1; transferred move d1c1 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0019#1750` (ply 29) [structural distance $d=52$, QAP obj=14]: the master played Qc7; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our Nb1 as their Bc8, our nb6 as their Re8; transferred move d1c1 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0019#3136` (ply 25) [structural distance $d=81$, QAP obj=14]: the master played h6; under μ our Qd1 plays the role of that game’s Ph2 (a **cross-type** correspondence); cross-type roles: our pd5 as their Bc8, our Pg2 as their Qd8; transferred move d1c1 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Qc1 (summed QAP objective 52, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



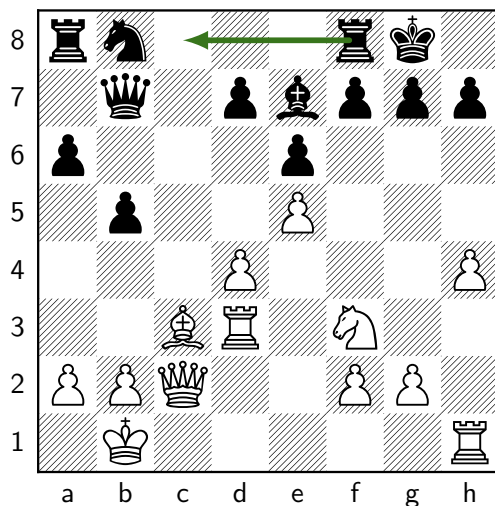
(a)



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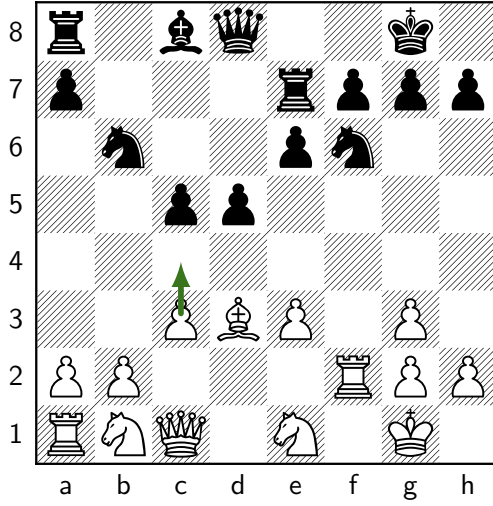


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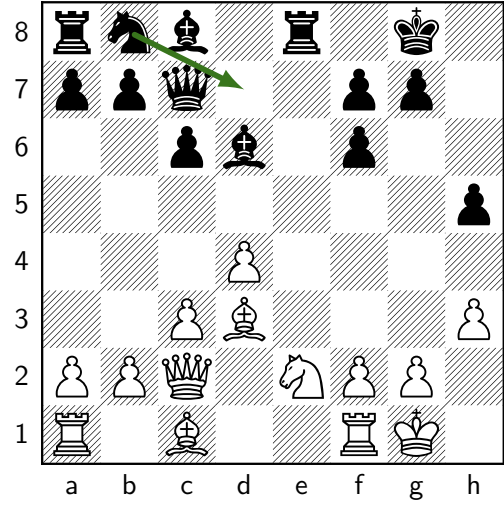


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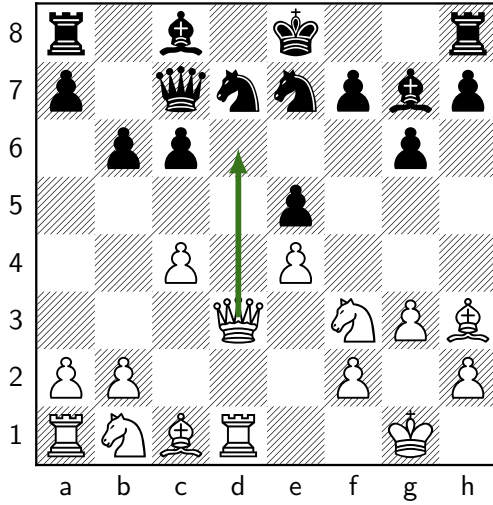
Figure 7: **Move 13, White: Ne1**. (a) Our position; Melanie plays Ne1 (f3e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0013#2365` (ply 21) [structural distance $d=101$, QAP obj=26]: the master played Nd7; under μ our Nf3 plays the role of that game’s Nf3; cross-type roles: our nb6 as their Bc8, our nf6 as their Qd8; transferred move f3e1 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0019#1963` (ply 21) [structural distance $d=83$, QAP obj=16]: the master played Rd8; under μ our Nf3 plays the role of that game’s Rf1 (a **cross-type** correspondence); cross-type roles: our nb6 as their Ra8, our Pg3 as their Nb8; transferred move f3e1 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0020#2831` (ply 33) [structural distance $d=97$, QAP obj=13]: the master played Rc8; under μ our Nf3 plays the role of that game’s Rf1 (a **cross-type** correspondence); cross-type roles: our pd5 as their Qb7, our Ra1 as their Pd7; transferred move f3e1 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Ne1 (summed QAP objective 55, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



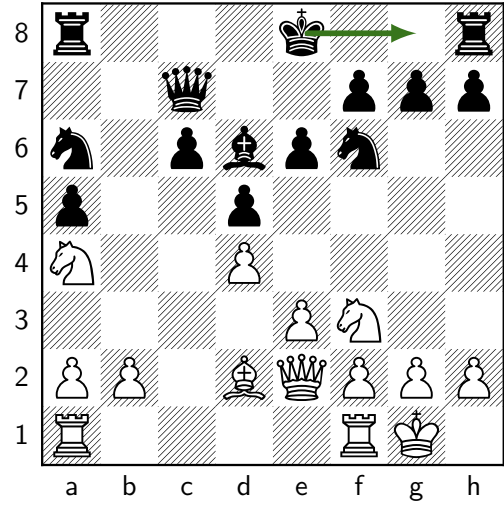
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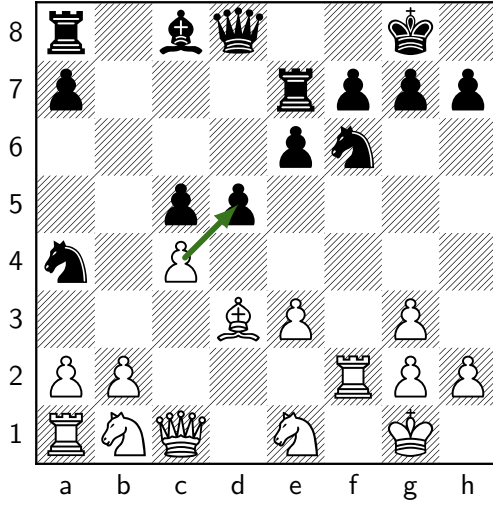


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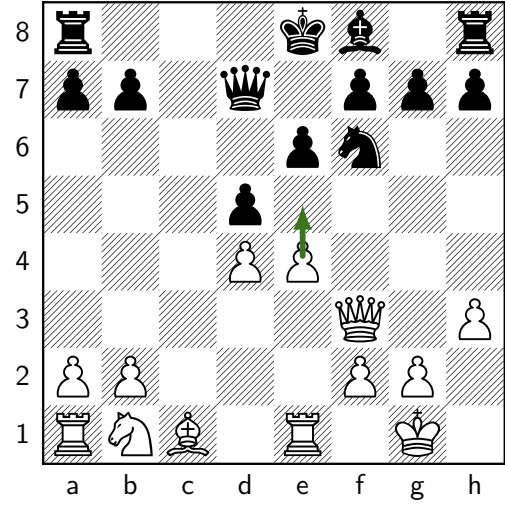


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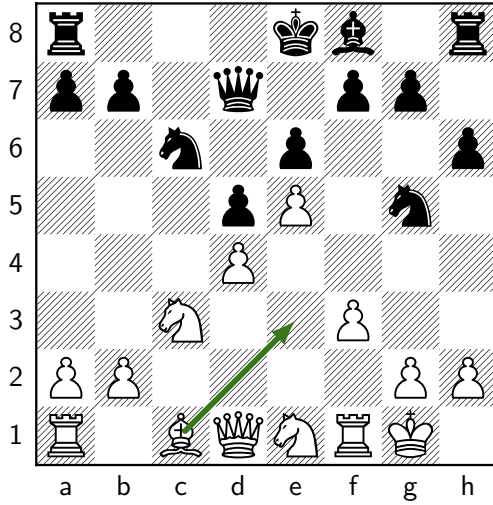
Figure 8: **Move 14, White: c4**. (a) Our position; Melanie plays c4 (c3c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0010#392` (ply 21) [structural distance $d=116$, QAP obj=25]: the master played Nd7; under μ our *Pc3* plays the role of that game’s *Nb1* (a **cross-type** correspondence); cross-type roles: our *Nb1* as their *Pa7*, our *Ne1* as their *Qc7*; transferred move c3c4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0010#2882` (ply 24) [structural distance $d=102$, QAP obj=40]: the master played Qd6; under μ our *Rf2* plays the role of that game’s *Qd3* (a **cross-type** correspondence); cross-type roles: our *Nb1* as their *Ra1*, our *Qc1* as their *Nb1*; transferred move f2f6. (d) Retrieved from `lichess_elite_2024-09_p0017#1017` (ply 25) [structural distance $d=111$, QAP obj=32]: the master played O-O; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Nb1* as their *Rh8*, our *pg7* as their *Qc7*; transferred move g1f1. Synthesis: 1 retrieved analog transferred to c4 (summed QAP objective 25, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



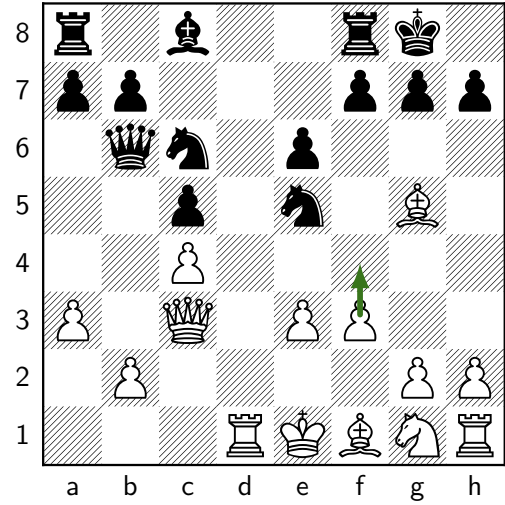
(a)



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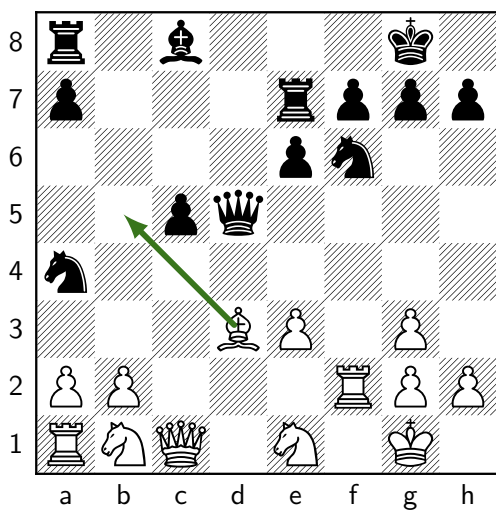


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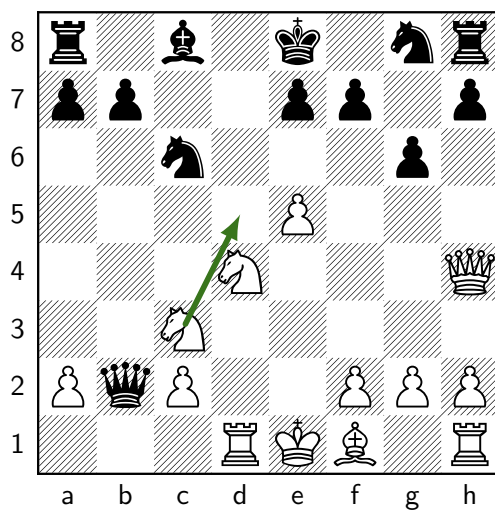


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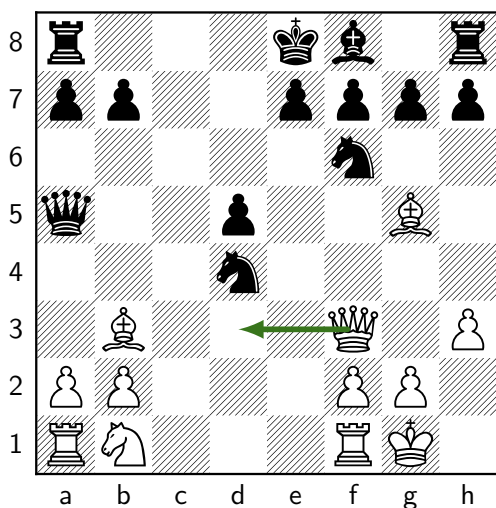
Figure 9: **Move 15, White: cxd5.** (a) Our position; Melanie plays cxd5 (c4d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0019#3638` (ply 22) [structural distance $d=102$, QAP obj=64]: the master played e5; under μ our $Pc4$ plays the role of that game’s $Pe4$; cross-type roles: our $Nb1$ as their $Ra1$, our $Pb2$ as their $Nb1$; transferred move c4d5 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0003#78` (ply 24) [structural distance $d=100$, QAP obj=47]: the master played Be3; under μ our $Pc4$ plays the role of that game’s $Bc1$ (a **cross-type** correspondence); cross-type roles: our $pf7$ as their $Qd1$, our $Pe3$ as their $Ne1$; transferred move c4d5 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0028#2267` (ply 24) [structural distance $d=103$, QAP obj=21]: the master played f4; under μ our $Pc4$ plays the role of that game’s $Pf3$; cross-type roles: our $Bd3$ as their $Rd1$, our $Qc1$ as their $Bf1$; transferred move c4d5 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to cxd5 (summed QAP objective 179, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



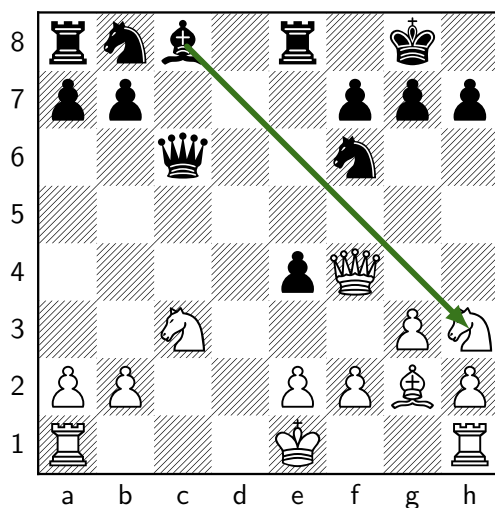
(a)



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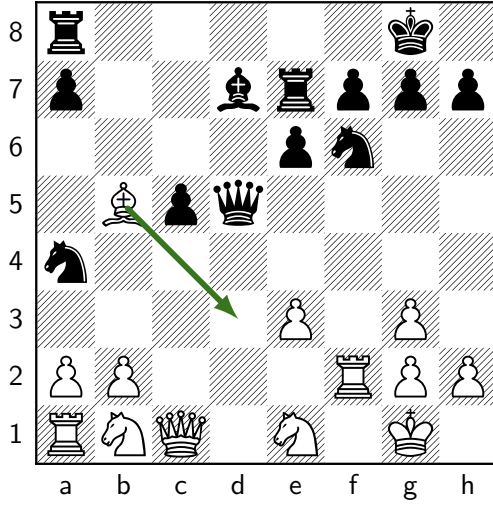


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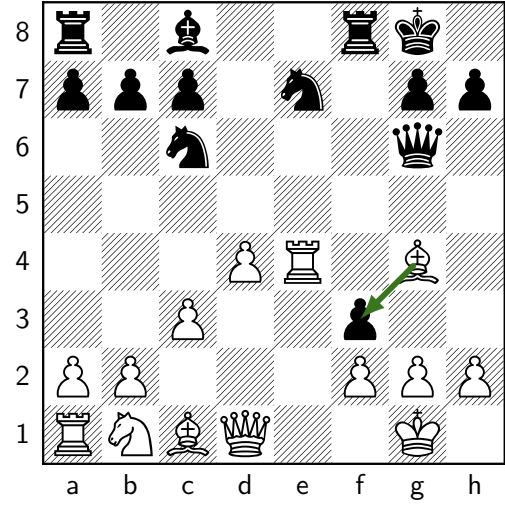


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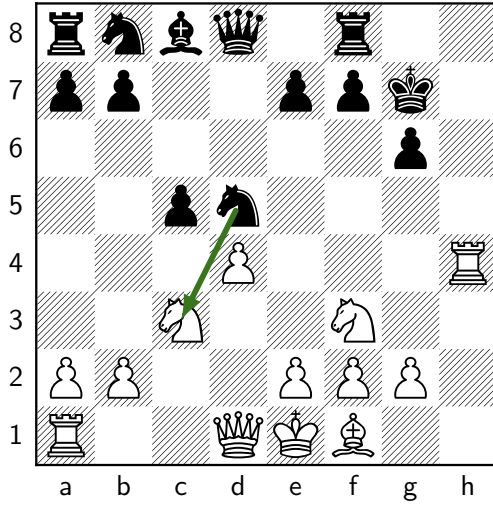
Figure 10: **Move 16, White: Bb5.** (a) Our position; Melanie plays Bb5 (d3b5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0010#911` (ply 22) [structural distance $d=99$, QAP obj=38]: the master played Nd5; under μ our Bd3 plays the role of that game’s Nc3 (a **cross-type** correspondence); cross-type roles: our Ne1 as their Rd1, our Pg2 as their Bf1; transferred move d3b5 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0024#1116` (ply 22) [structural distance $d=105$, QAP obj=29]: the master played Qd3; under μ our Bd3 plays the role of that game’s Qf3 (a **cross-type** correspondence); cross-type roles: our Nb1 as their Ra1, our Pb2 as their Nb1; transferred move d3b5 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0022#798` (ply 25) [structural distance $d=102$, QAP obj=57]: the master played Bxh3; under μ our Rf2 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our Bd3 as their Nb8, our Pa2 as their Re8; transferred move f2f3. Synthesis: 2 retrieved analogs transferred to Bb5 (summed QAP objective 67, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



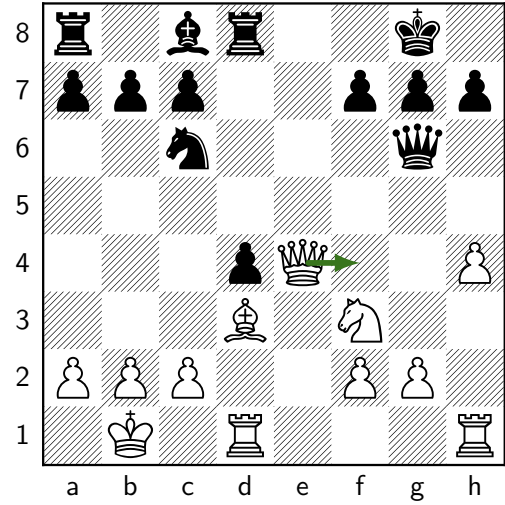
(a)



(b)

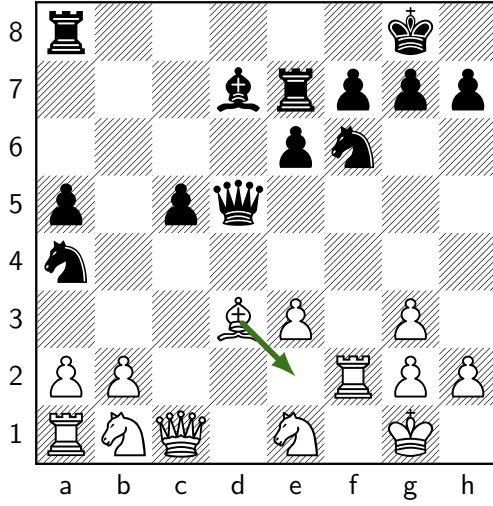


(c)

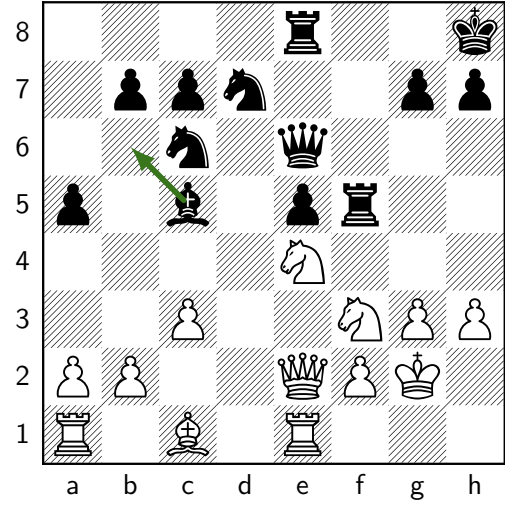


(d)

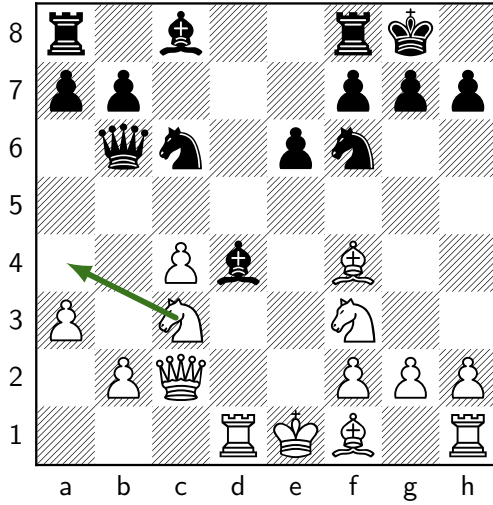
Figure 11: **Move 17, White: Bd3.** (a) Our position; Melanie plays Bd3 (b5d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0013#2836` (ply 24) [structural distance $d=117$, QAP obj=45]: the master played Bxf3; under μ our Bb5 plays the role of that game’s Bg4; cross-type roles: our pc5 as their Nb1, our Qc1 as their Bc1; transferred move b5d3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0020#2899` (ply 21) [structural distance $d=116$, QAP obj=27]: the master played Nxc3; under μ our Bb5 plays the role of that game’s Nd4 (a **cross-type** correspondence); cross-type roles: our Qc1 as their Nb8, our Rf2 as their Bc8; transferred move b5d3 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0013#1283` (ply 32) [structural distance $d=115$, QAP obj=22]: the master played Qf4; under μ our Bb5 plays the role of that game’s Qe4 (a **cross-type** correspondence); cross-type roles: our Qc1 as their Rd1, our Ne1 as their Pb2; transferred move b5d3 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Bd3 (summed QAP objective 116, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



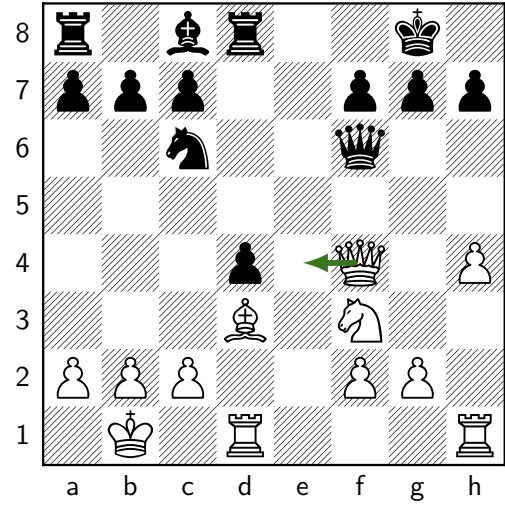
(a)



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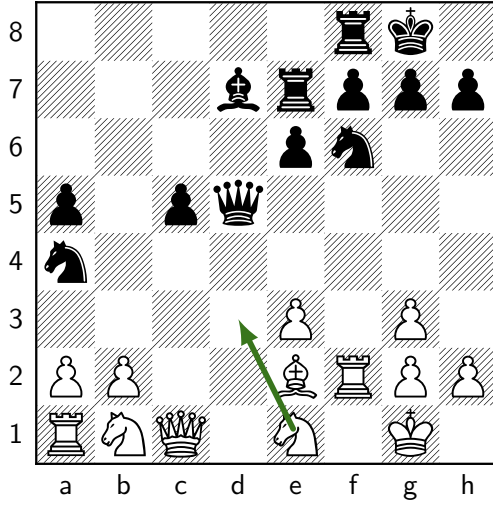


(c)

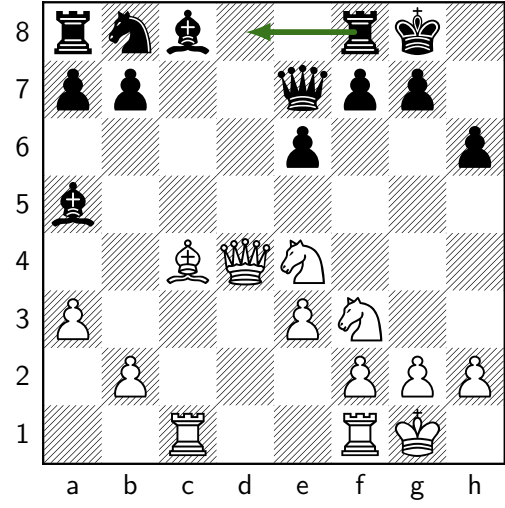


(d)

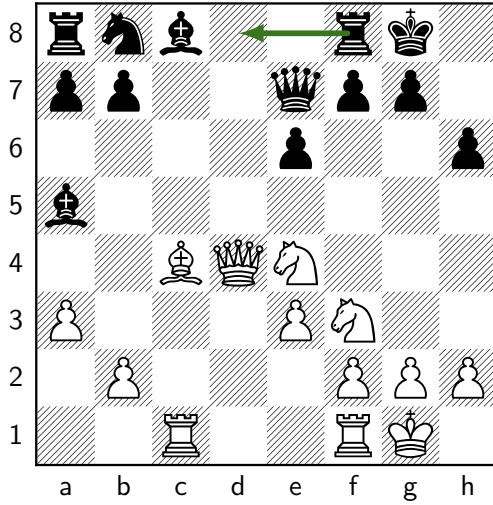
Figure 12: **Move 18, White: Be2.** (a) Our position; Melanie plays Be2 (d3e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0005#2029` (ply 33) [structural distance $d=87$, QAP obj=40]: the master played Bb6; under μ our Bd3 plays the role of that game’s Bc4; cross-type roles: our nf6 as their Re8, our Rf2 as their Nd7; transferred move d3e2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0025#17` (ply 22) [structural distance $d=103$, QAP obj=16]: the master played Na4; under μ our Bd3 plays the role of that game’s Nc3 (a **cross-type** correspondence); cross-type roles: our Ph2 as their Rd1, our pf7 as their Bf1; transferred move d3e2 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0013#1283` (ply 34) [structural distance $d=99$, QAP obj=15]: the master played Qe4; under μ our Bd3 plays the role of that game’s Qf4 (a **cross-type** correspondence); cross-type roles: our Pg2 as their Rd1, our na4 as their Pg2; transferred move d3e2 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Be2 (summed QAP objective 110, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



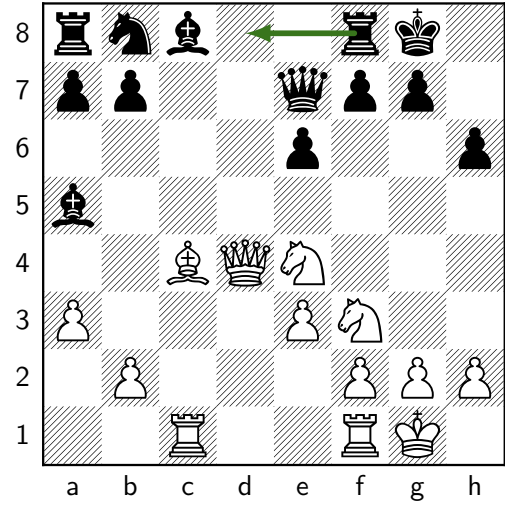
(a)



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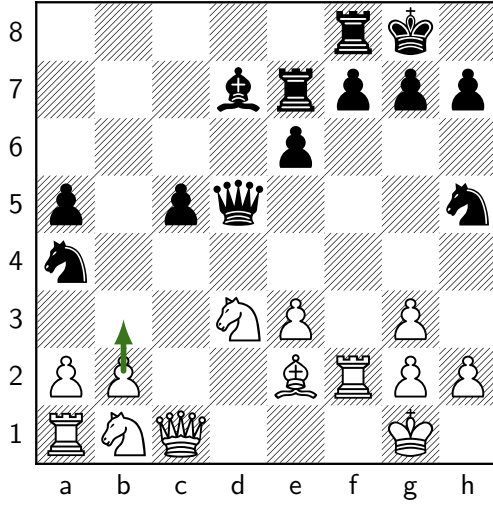


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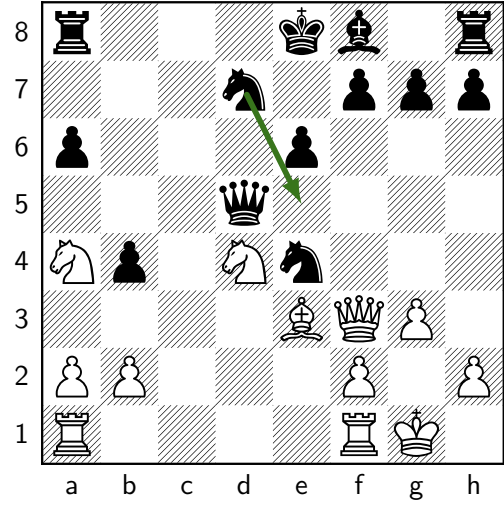


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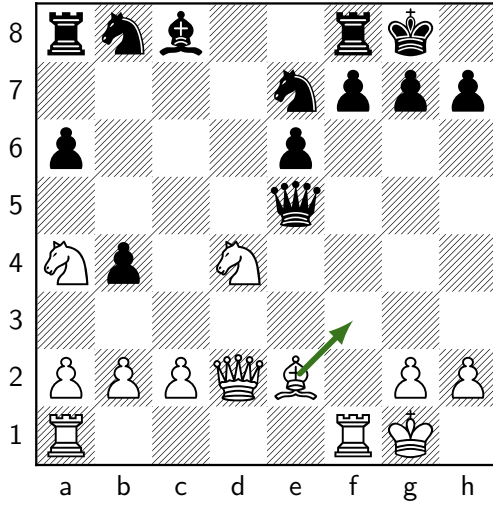
Figure 13: **Move 19, White: Nd3.** (a) Our position; Melanie plays Nd3 (e1d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0003#380` (ply 25) [structural distance $d=103$, QAP obj=51]: the master played Rd8; under μ our *Ne1* plays the role of that game’s *Rf1* (a **cross-type** correspondence); cross-type roles: our *Rf2* as their *Bc8*, our *re7* as their *Ph6*; transferred move e1d3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0005#3656` (ply 25) [structural distance $d=103$, QAP obj=51]: the master played Rd8; under μ our *Ne1* plays the role of that game’s *Rf1* (a **cross-type** correspondence); cross-type roles: our *Rf2* as their *Bc8*, our *re7* as their *Ph6*; transferred move e1d3 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0017#2166` (ply 25) [structural distance $d=103$, QAP obj=51]: the master played Rd8; under μ our *Ne1* plays the role of that game’s *Rf1* (a **cross-type** correspondence); cross-type roles: our *Rf2* as their *Bc8*, our *re7* as their *Ph6*; transferred move e1d3 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Nd3 (summed QAP objective 280, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



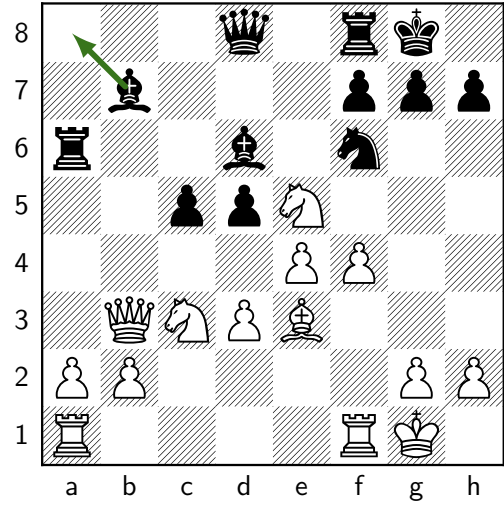
(a)



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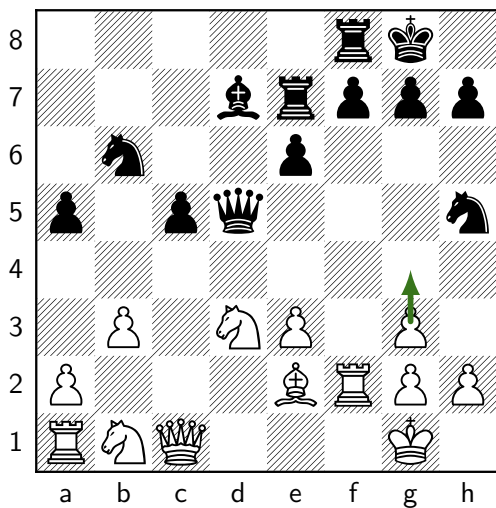


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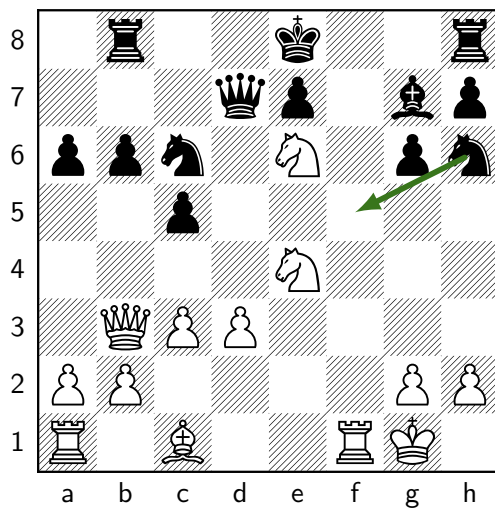


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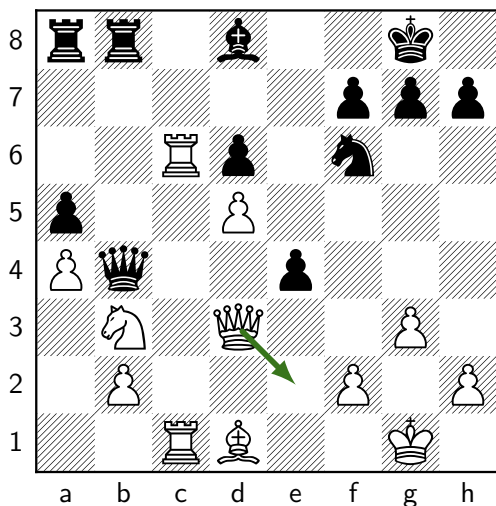
Figure 14: **Move 20, White: b3.** (a) Our position; Melanie plays b3 (b2b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0019#1415` (ply 27) [structural distance $d=105$, QAP obj=63]: the master played Ne5; under μ our *Pb2* plays the role of that game’s *Nd2* (a **cross-type** correspondence); cross-type roles: our *Qc1* as their *Ra8*, our *Rf2* as their *Bf8*; transferred move b2b3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0022#3156` (ply 30) [structural distance $d=100$, QAP obj=70]: the master played Bf3; under μ our *Nd3* plays the role of that game’s *Be2* (a **cross-type** correspondence); cross-type roles: our *Pg2* as their *Ra1*, our *Nb1* as their *Pc2*; transferred move d3f4. (d) Retrieved from `lichess_elite_2024-09_p0025#3756` (ply 29) [structural distance $d=105$, QAP obj=61]: the master played Ba8; under μ our *Nd3* plays the role of that game’s *Bb2* (a **cross-type** correspondence); cross-type roles: our *Nb1* as their *Qd8*, our *Qc1* as their *Rf8*; transferred move d3e1. Synthesis: 1 retrieved analog transferred to b3 (summed QAP objective 63, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



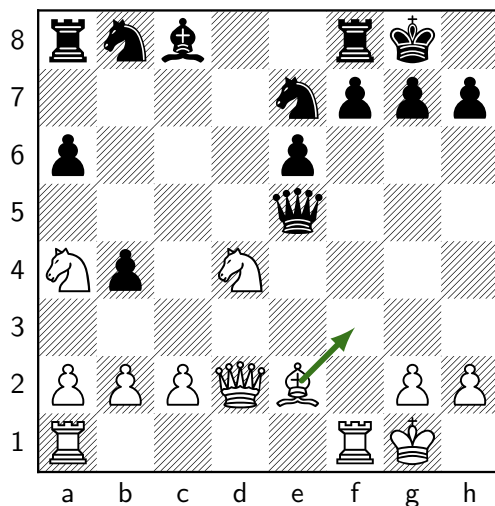
(a)



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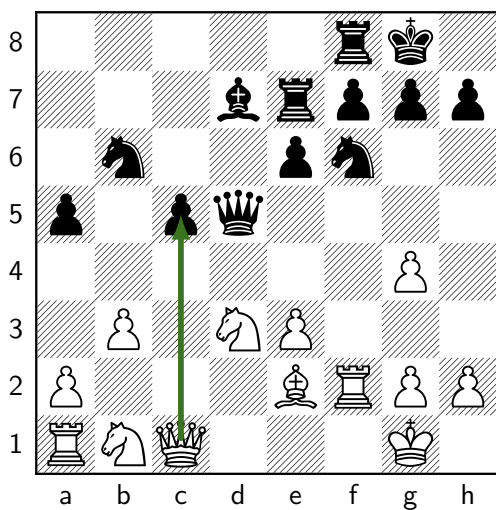


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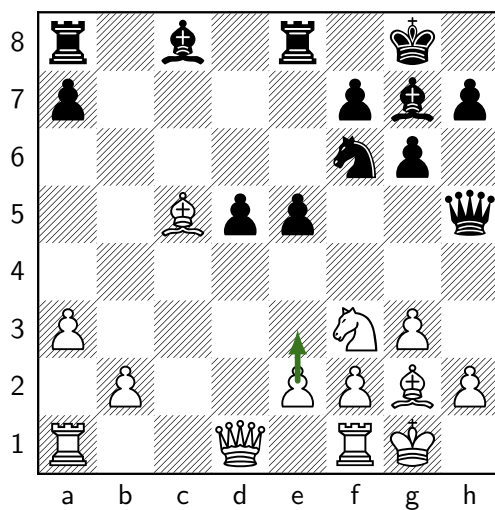


(d)

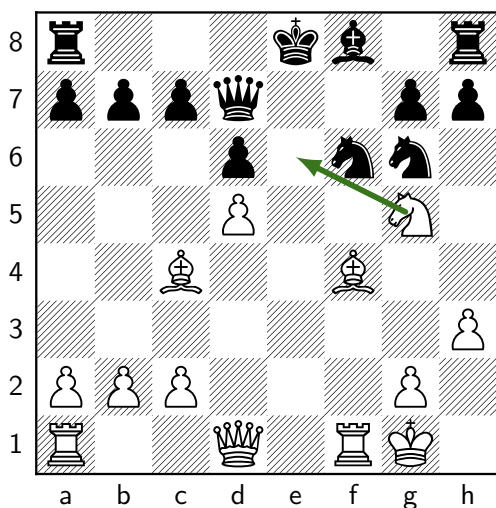
Figure 15: **Move 21, White: g4.** (a) Our position; Melanie plays g4 (g3g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0022#2394` (ply 33) [structural distance $d=77$, QAP obj=53]: the master played Nf5; under μ our $Pg3$ plays the role of that game’s $Nh3$ (a **cross-type** correspondence); cross-type roles: our $Qc1$ as their $Rh8$, our $pc5$ as their $Bg7$; transferred move g3g4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0016#327` (ply 48) [structural distance $d=72$, QAP obj=9]: the master played Qe2; under μ our $Pg3$ plays the role of that game’s $Qd3$ (a **cross-type** correspondence); cross-type roles: our $Qc1$ as their $Rc1$, our $Pa2$ as their $Bd1$; transferred move g3g4 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0022#3156` (ply 30) [structural distance $d=92$, QAP obj=71]: the master played Bf3; under μ our $Nd3$ plays the role of that game’s $Be2$ (a **cross-type** correspondence); cross-type roles: our $nb6$ as their $Ra1$, our $Ra1$ as their $Pc2$; transferred move d3f4. Synthesis: 2 retrieved analogs transferred to g4 (summed QAP objective 61, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



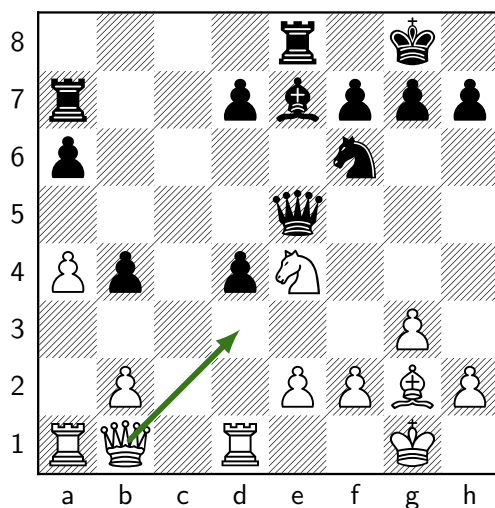
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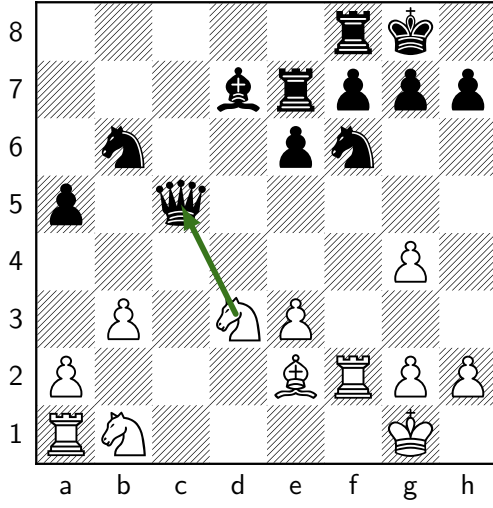


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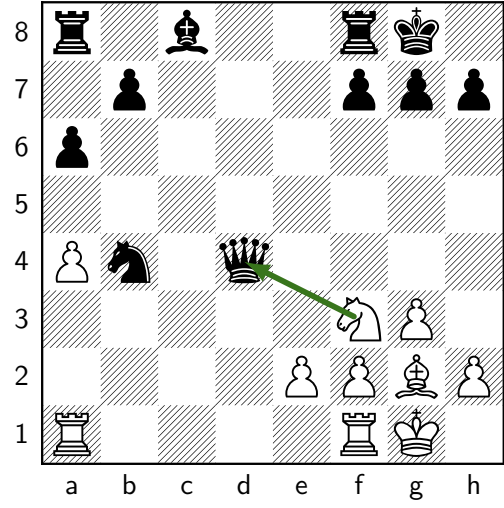


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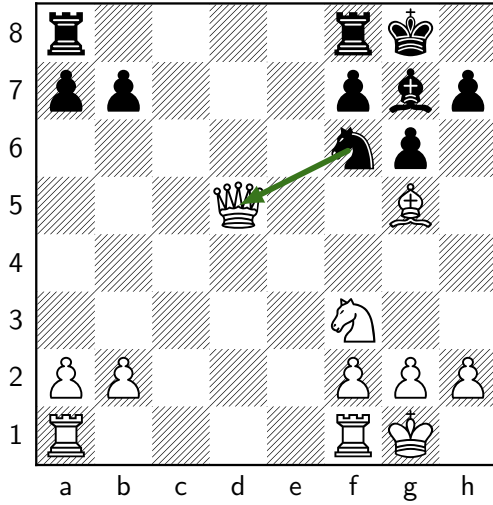
Figure 16: **Move 22, White: Qxc5.** (a) Our position; Melanie plays Qxc5 (c1c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0024#884` (ply 28) [structural distance $d=97$, QAP obj=55]: the master played e3; under μ our *Be2* plays the role of that game’s *Pe2* (a **cross-type** correspondence); cross-type roles: our *Qc1* as their *Ra1*, our *Ph2* as their *Qd1*; transferred move e2f3. (c) Retrieved from `lichess_elite_2024-09_p0028#2662` (ply 28) [structural distance $d=102$, QAP obj=44]: the master played Ne6; under μ our *Rf2* plays the role of that game’s *Ng5* (a **cross-type** correspondence); cross-type roles: our *Nb1* as their *Ra1*, our *Pe3* as their *Qd1*; transferred move f2f3. (d) Retrieved from `lichess_elite_2024-09_p0013#2677` (ply 44) [structural distance $d=86$, QAP obj=26]: the master played Qd3; under μ our *Nb1* plays the role of that game’s *Qb1* (a **cross-type** correspondence); cross-type roles: our *Qc1* as their *Rd1*, our *nf6* as their *pb4*; transferred move b1c3. Synthesis: no retrieved analog transferred to Qxc5 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



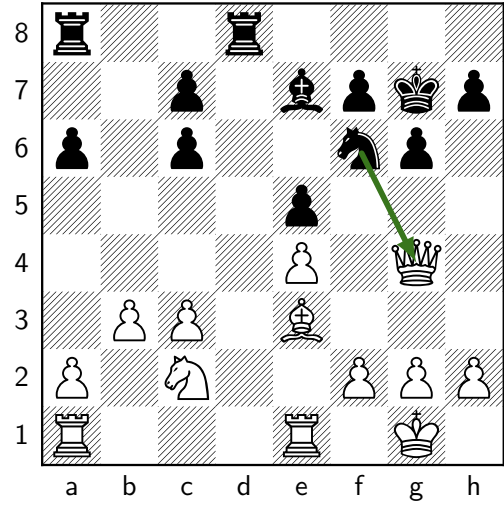
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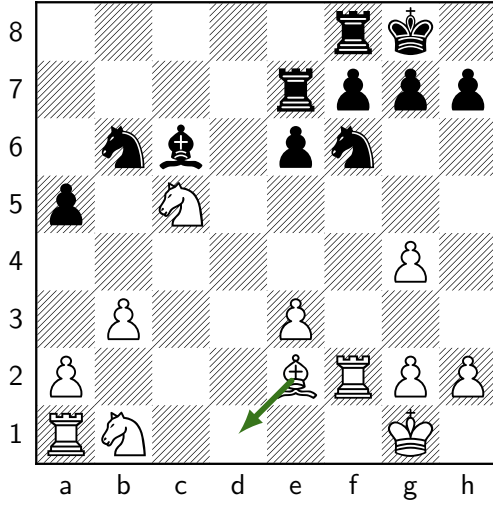


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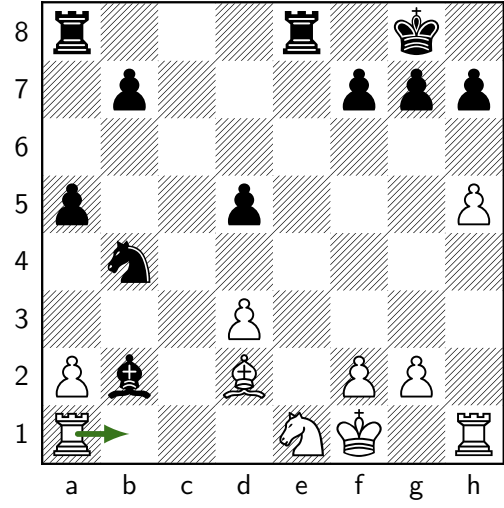


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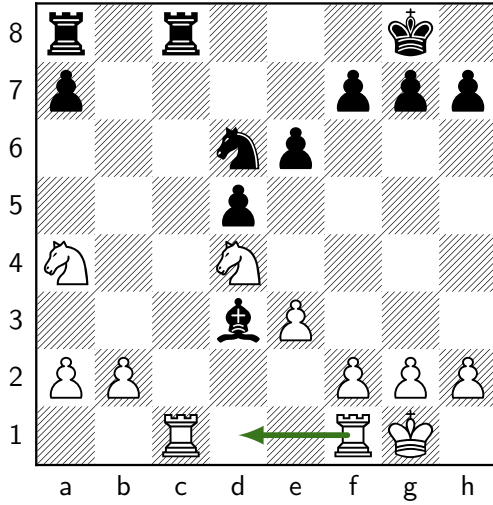
Figure 17: **Move 23, White: Nxc5**. (a) Our position; Melanie plays Nxc5 (d3c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0024#1530` (ply 32) [structural distance $d=97$, QAP obj=92]: the master played Nxd4; under μ our Nd3 plays the role of that game’s Nf3; cross-type roles: our Pe3 as their Ra1, our Pb3 as their Rf1; transferred move d3c5 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0017#3836` (ply 31) [structural distance $d=93$, QAP obj=92]: the master played Nxd5; under μ our Nd3 plays the role of that game’s Nf3; cross-type roles: our Nb1 as their Ra8, our Pb3 as their Rf8; transferred move d3c5 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0026#1220` (ply 35) [structural distance $d=82$, QAP obj=78]: the master played Nxc5; under μ our Nd3 plays the role of that game’s Nf3; cross-type roles: our pa5 as their Ra8, our Nb1 as their Pc7; transferred move d3c5 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Nxc5 (summed QAP objective 553, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



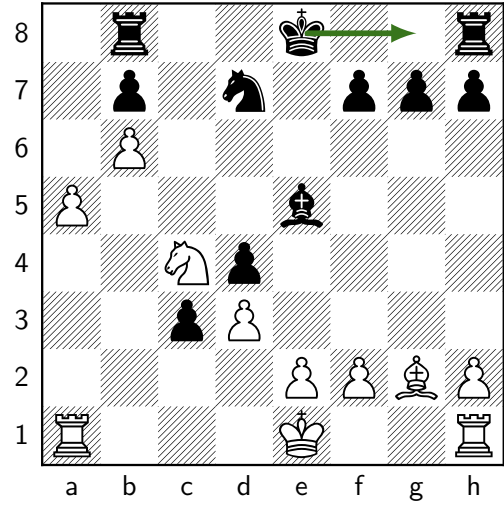
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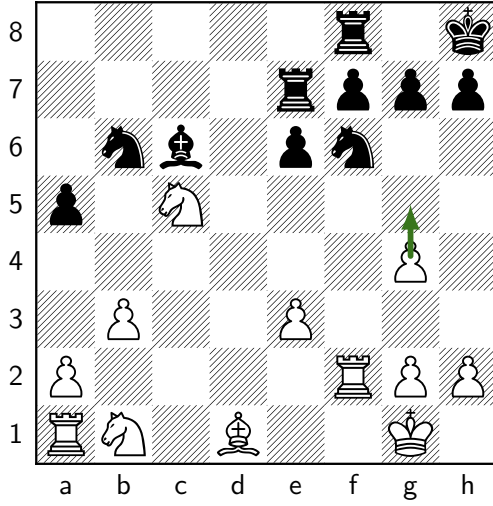


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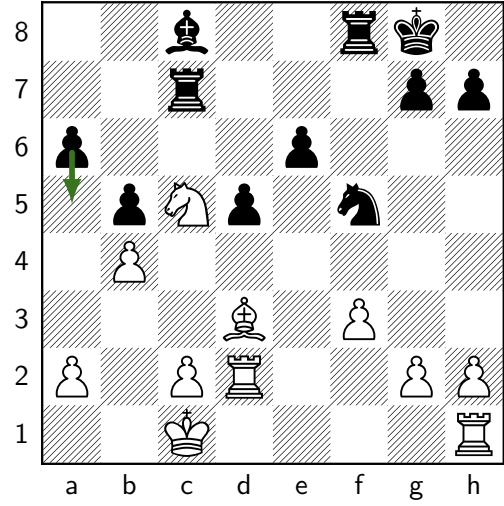


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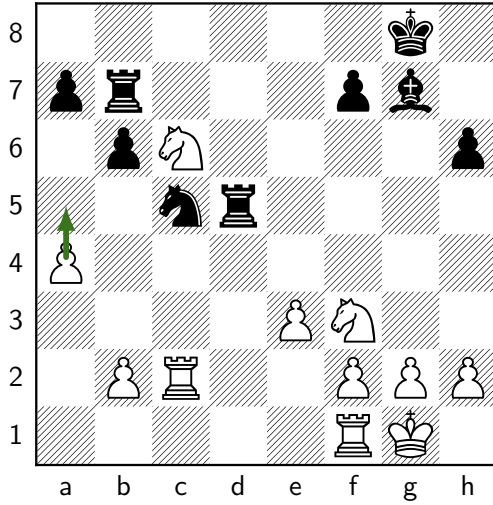
Figure 18: **Move 24, White: Bd1.** (a) Our position; Melanie plays Bd1 (e2d1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0023#1615` (ply 40) [structural distance $d=64$, QAP obj=17]: the master played Rb1; under μ our *Be2* plays the role of that game’s *Ra1* (a **cross-type** correspondence); cross-type roles: our *Pg2* as their *Ne1*, our *Pa2* as their *Rh1*; transferred move e2d1 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0010#1425` (ply 34) [structural distance $d=66$, QAP obj=15]: the master played Rfd1; under μ our *Be2* plays the role of that game’s *Rf1* (a **cross-type** correspondence); cross-type roles: our *Ra1* as their *Pa2*, our *Pg4* as their *Na4*; transferred move e2d1 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0020#998` (ply 39) [structural distance $d=63$, QAP obj=31]: the master played 0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *bc6* as their *Rb8*, our *nf6* as their *Be5*; transferred move g1f1. Synthesis: 2 retrieved analogs transferred to Bd1 (summed QAP objective 32, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



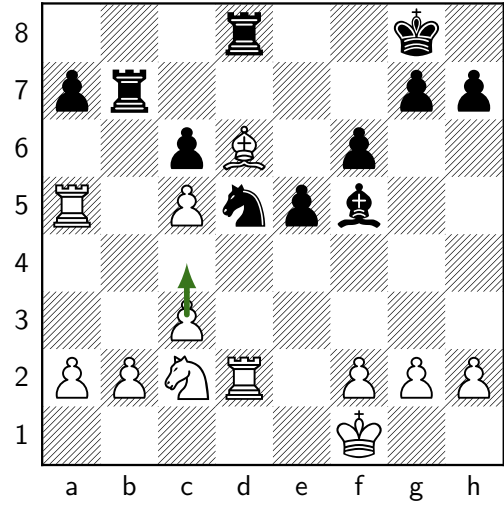
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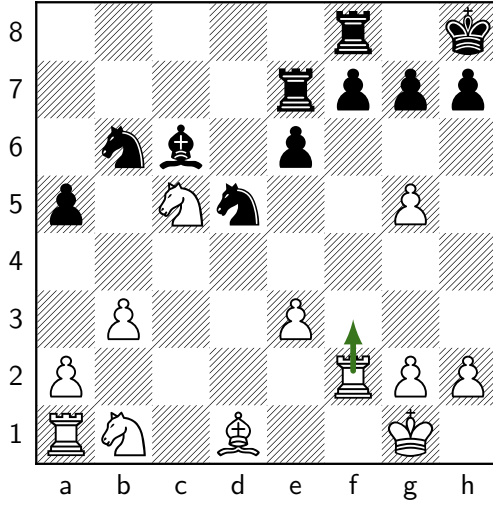


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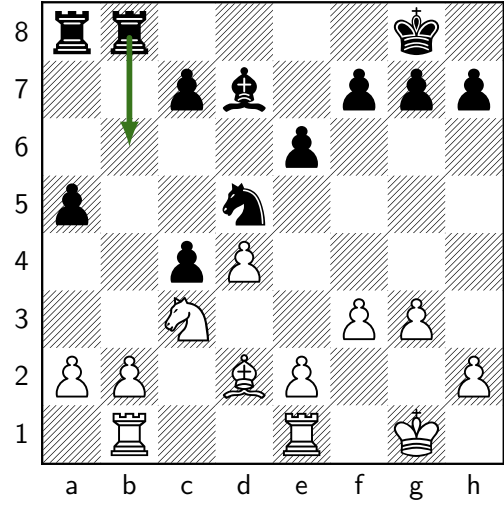


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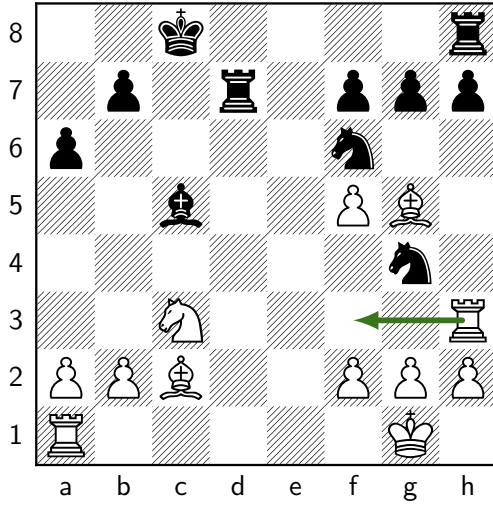
Figure 19: **Move 25, White: g5.** (a) Our position; Melanie plays g5 (g4g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0022#897` (ply 43) [structural distance $d=68$, QAP obj=63]: the master played a5; under μ our $Pg4$ plays the role of that game’s $Pa3$; cross-type roles: our $Pg2$ as their $Bc8$, our $pe6$ as their $Nf5$; transferred move g4g5 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0017#18` (ply 48) [structural distance $d=64$, QAP obj=56]: the master played a5; under μ our $Pg4$ plays the role of that game’s $Pa4$; cross-type roles: our $Pe3$ as their $Rf1$, our $nb6$ as their $Pg2$; transferred move g4g5 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0010#3930` (ply 52) [structural distance $d=71$, QAP obj=43]: the master played c4; under μ our $Pg4$ plays the role of that game’s $Pc3$; cross-type roles: our $Nb1$ as their $Pa2$, our $Bd1$ as their $Pb2$; transferred move g4g5 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to g5 (summed QAP objective 180, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



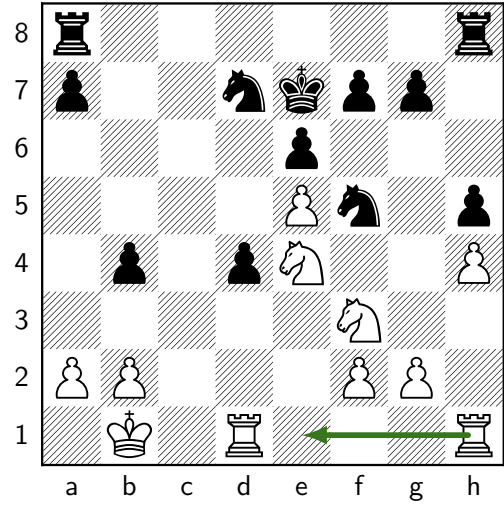
(a)



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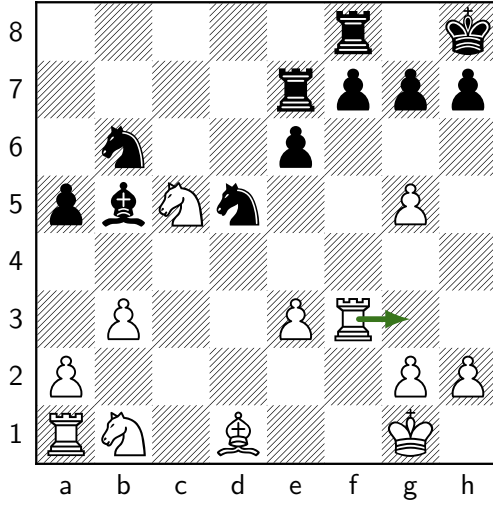


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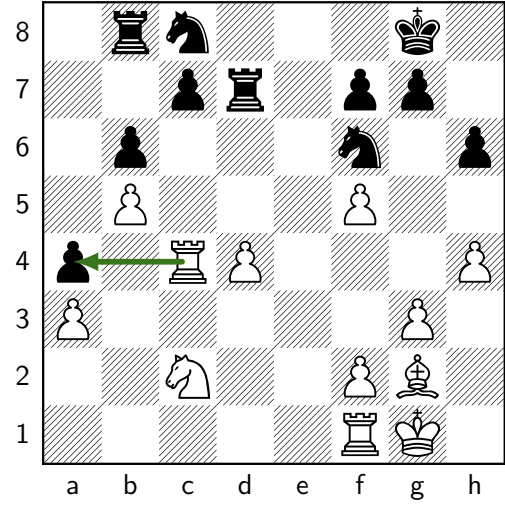


(d)

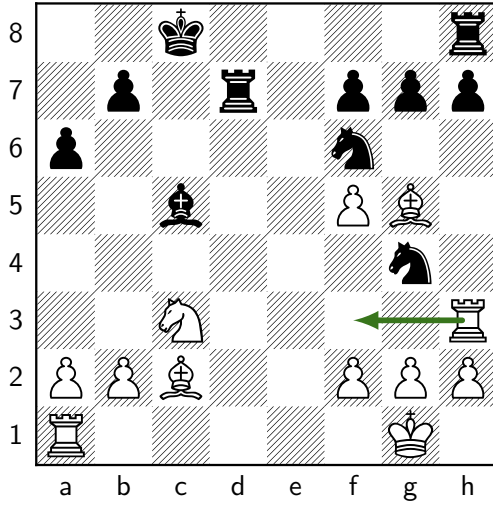
Figure 20: **Move 26, White: Rf3.** (a) Our position; Melanie plays Rf3 (f2f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2025-11_p0000#758` (ply 35) [structural distance $d=60$, QAP obj=29]: the master played Rb6; under μ our Rf2 plays the role of that game’s Rb1; cross-type roles: our Pg5 as their Ra8, our Ra1 as their Pc7; transferred move f2f3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0010#1305` (ply 36) [structural distance $d=58$, QAP obj=25]: the master played Rf3; under μ our Rf2 plays the role of that game’s Rh3; cross-type roles: our nd5 as their Pb2, our Pg5 as their Bc2; transferred move f2f3 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0018#3473` (ply 40) [structural distance $d=74$, QAP obj=63]: the master played Rhe1; under μ our Pg2 plays the role of that game’s Rh1 (a **cross-type** correspondence); cross-type roles: our Rf2 as their Nf3, our re7 as their Ne4; transferred move g2g3. Synthesis: 2 retrieved analogs transferred to Rf3 (summed QAP objective 55, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



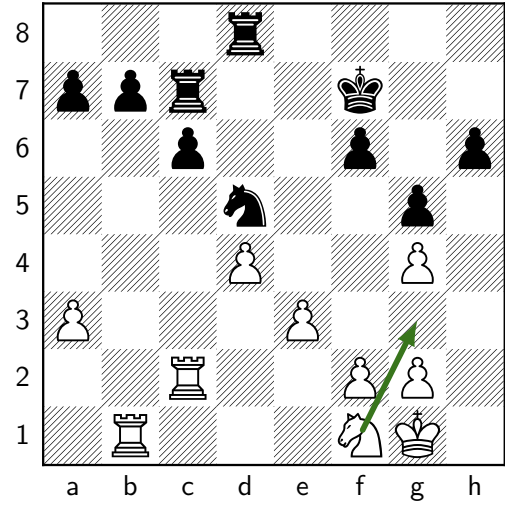
(a)



(b)

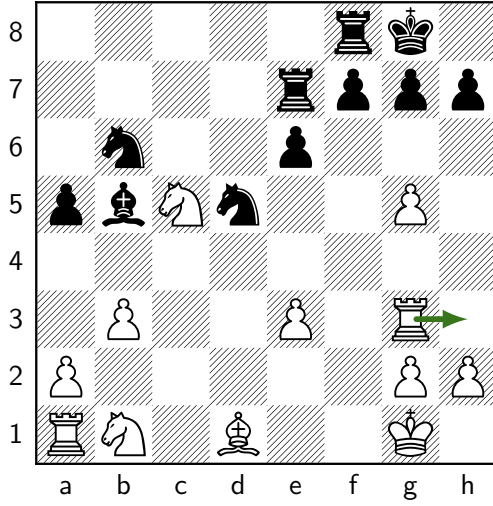


(c)

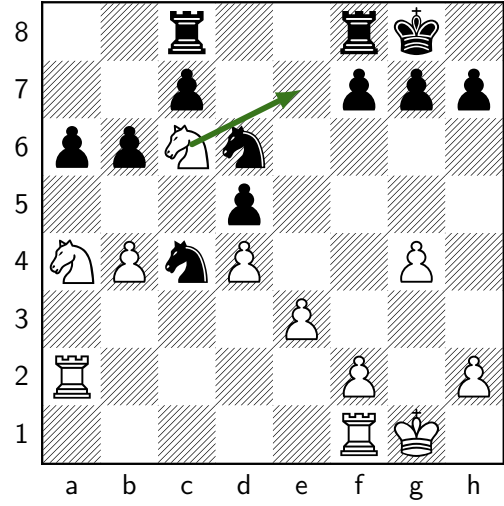


(d)

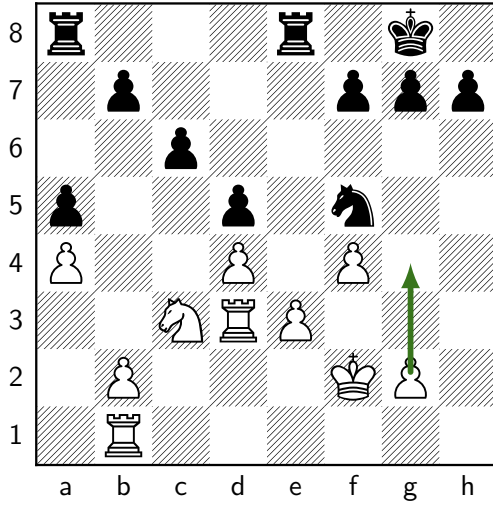
Figure 21: **Move 27, White: Rg3**. (a) Our position; Melanie plays Rg3 (f3g3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0019#1177` (ply 66) [structural distance $d=49$, QAP obj=52]: the master played Rxa4; under μ our Rf3 plays the role of that game’s Rc4; cross-type roles: our Pg2 as their Rf1, our pe6 as their Nc2; transferred move f3g3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0010#1305` (ply 36) [structural distance $d=51$, QAP obj=32]: the master played Rf3; under μ our Rf3 plays the role of that game’s Rh3; cross-type roles: our Bd1 as their Ra1, our nd5 as their Pa2; transferred move f3g3 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0019#2694` (ply 52) [structural distance $d=64$, QAP obj=66]: the master played Ng3; under μ our Ph2 plays the role of that game’s Nf1 (a **cross-type** correspondence); cross-type roles: our Nc5 as their Rb1, our Ra1 as their Pf2; transferred move h2h3. Synthesis: 2 retrieved analogs transferred to Rg3 (summed QAP objective 84, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



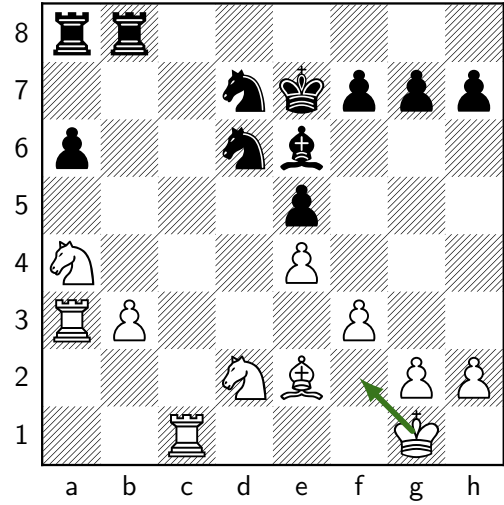
(a)



(b)



(c)



(d)

Figure 22: **Move 28, White: Rh3.** (a) Our position; Melanie plays Rh3 (g3h3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0013#3761` (ply 46) [structural distance $d=57$, QAP obj=43]: the master played Ne7+; under μ our Rg3 plays the role of that game’s Nc6 (a **cross-type** correspondence); cross-type roles: our Pg2 as their Rf1, our Nb1 as their Ra2; transferred move g3h3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0019#3307` (ply 44) [structural distance $d=78$, QAP obj=61]: the master played g4; under μ our Pg5 plays the role of that game’s Pg2; cross-type roles: our Nb1 as their Pf4, our bb5 as their pa5; transferred move g5g6. (d) Retrieved from `lichess_elite_2024-09_p0025#1014` (ply 44) [structural distance $d=77$, QAP obj=42]: the master played Kf2; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our bb5 as their Rc1, our Rg3 as their Nd2; transferred move g1f2. Synthesis: 1 retrieved analog transferred to Rh3 (summed QAP objective 43, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

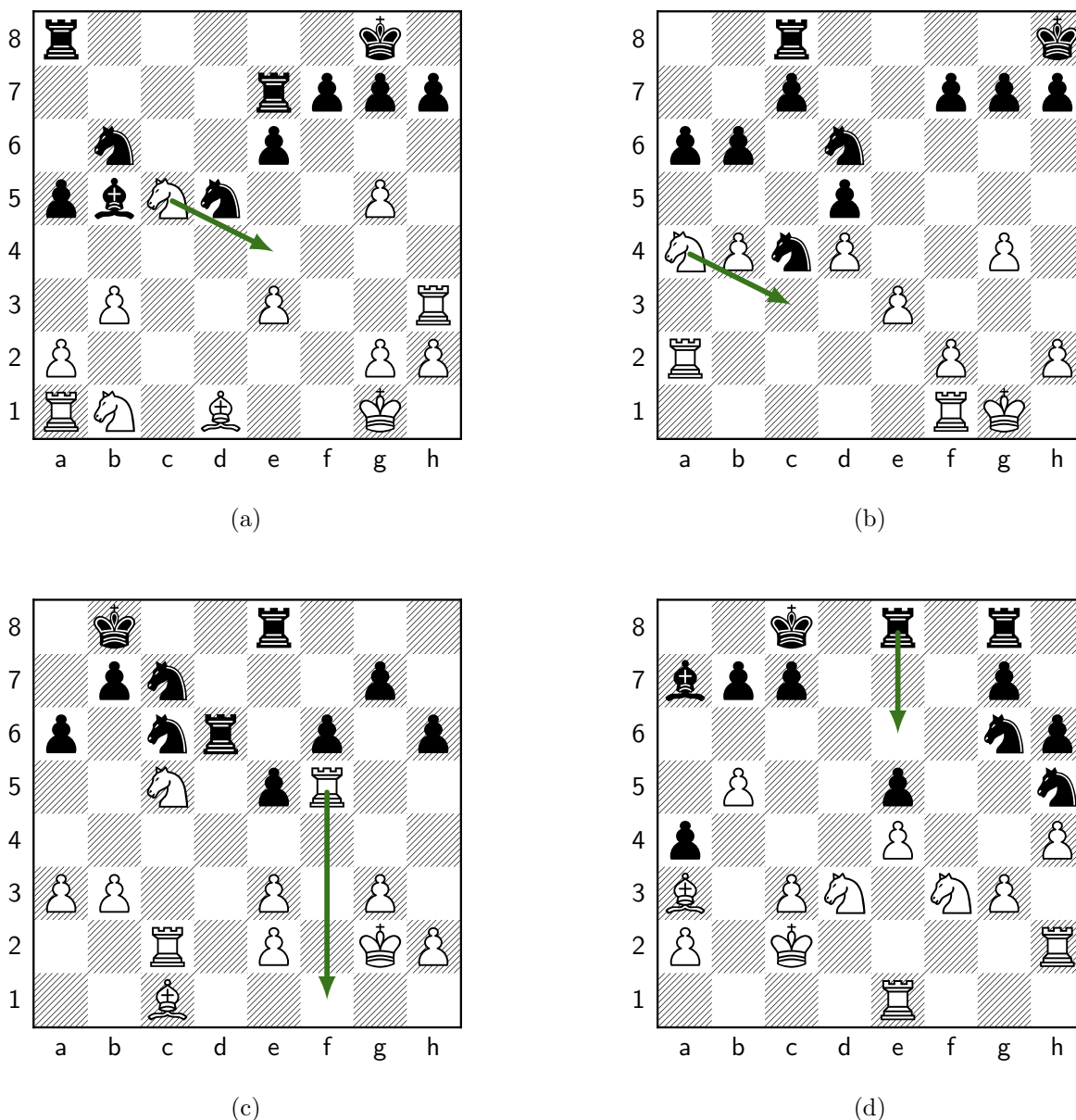
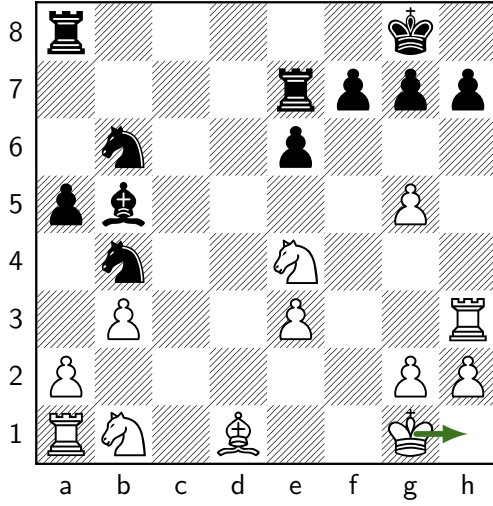
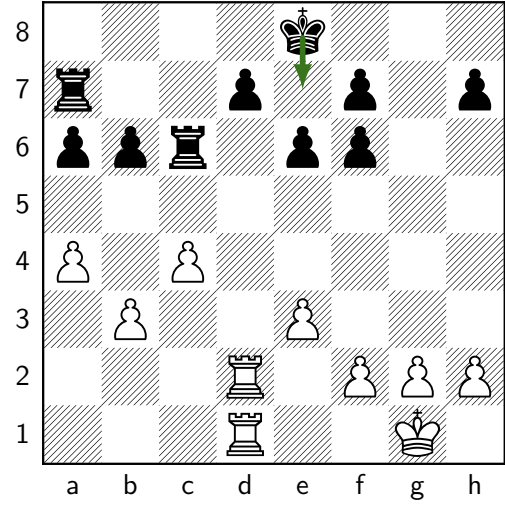


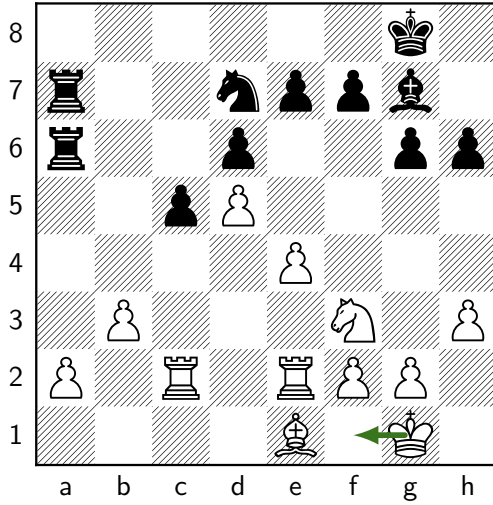
Figure 23: **Move 29, White: Ne4.** (a) Our position; Melanie plays Ne4 (c5e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0013#3761` (ply 50) [structural distance $d=66$, QAP obj=43]: the master played Nc3; under μ our Nc5 plays the role of that game’s Na4; cross-type roles: our bb5 as their Rf1, our Nb1 as their Ra2; transferred move c5e4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0012#1537` (ply 46) [structural distance $d=61$, QAP obj=33]: the master played Rf1; under μ our Nc5 plays the role of that game’s Rf5 (a **cross-type** correspondence); cross-type roles: our Nb1 as their Bc1, our nb6 as their Rc2; transferred move c5e4 (**endorses this move**). (d) Retrieved from `lichess_elite_2025-11_p0000#2724` (ply 65) [structural distance $d=60$, QAP obj=67]: the master played Re6; under μ our Rh3 plays the role of that game’s Re1; cross-type roles: our Nb1 as their Rg8, our Pg5 as their Ba7; transferred move h3h4. Synthesis: 2 retrieved analogs transferred to Ne4 (summed QAP objective 76, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



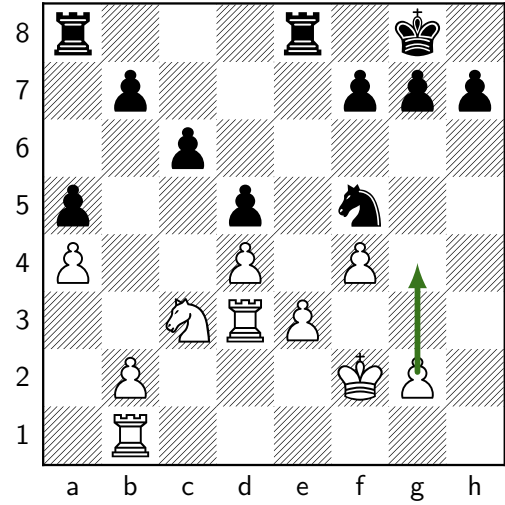
(a)



(b)

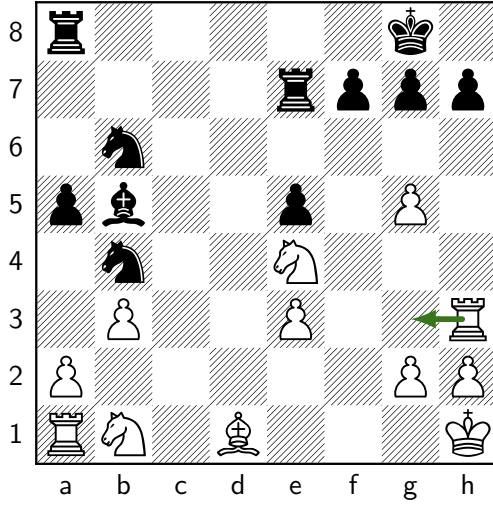


(c)

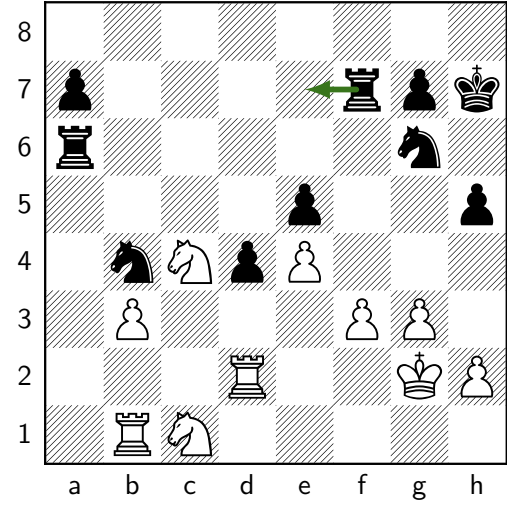


(d)

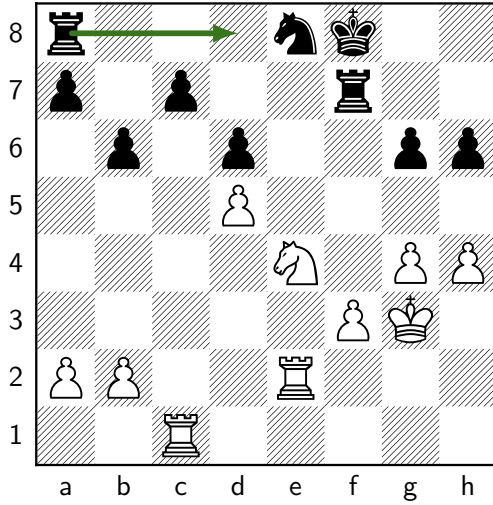
Figure 24: **Move 30, White: Kh1.** (a) Our position; Melanie plays Kh1 (g1h1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0031#2467` (ply 49) [structural distance $d=63$, QAP obj=29]: the master played Ke7; under μ our Kg1 plays the role of that game’s Ke1; cross-type roles: our Ra1 as their Ph7, our Bd1 as their Pb6; transferred move g1h1 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0021#2213` (ply 46) [structural distance $d=61$, QAP obj=25]: the master played Kf1; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our nb4 as their Rc2, our Pg5 as their Re2; transferred move g1h1 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0019#3307` (ply 44) [structural distance $d=58$, QAP obj=49]: the master played g4; under μ our Ne4 plays the role of that game’s Pg2 (a **cross-type** correspondence); cross-type roles: our Bd1 as their Rb1, our re7 as their Pb2; transferred move e4c5. Synthesis: 2 retrieved analogs transferred to Kh1 (summed QAP objective 55, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



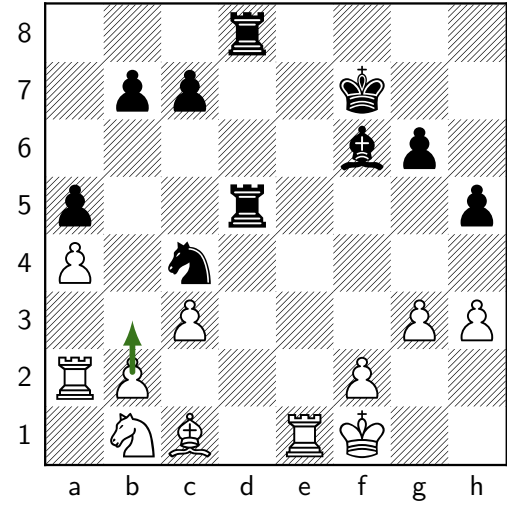
(a)



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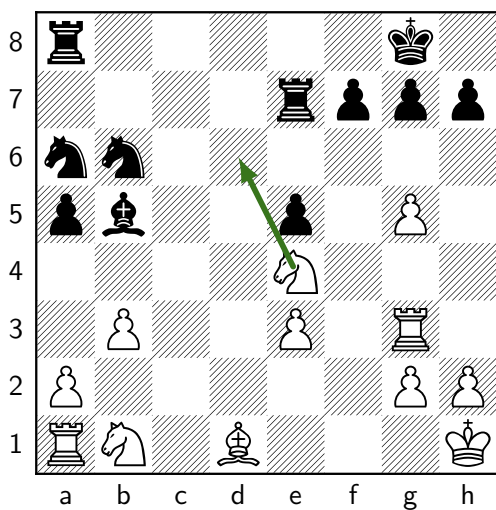


(c)

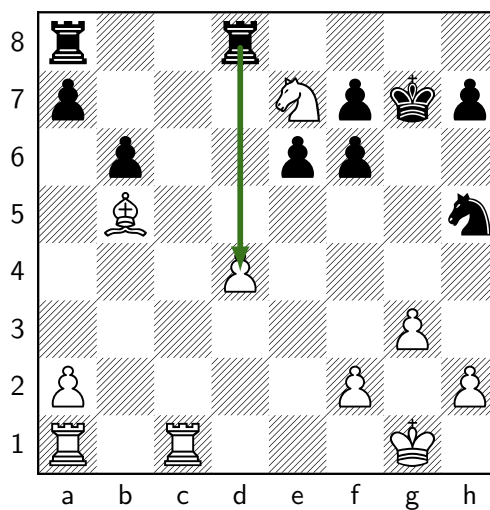


(d)

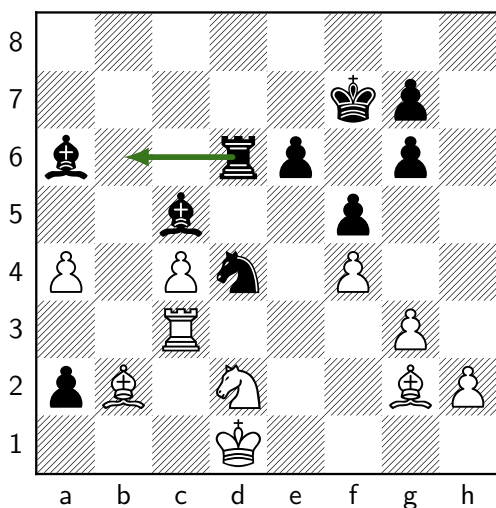
Figure 25: **Move 31, White: Rg3.** (a) Our position; Melanie plays Rg3 (h3g3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0026#3360` (ply 71) [structural distance $d=63$, QAP obj=42]: the master played R $e7$; under μ our R $h3$ plays the role of that game’s R $f2$; cross-type roles: our b $b5$ as their R $a6$, our R $a1$ as their N $b4$; transferred move h3g3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0021#4000` (ply 59) [structural distance $d=65$, QAP obj=33]: the master played R $d8$; under μ our B $d1$ plays the role of that game’s R $a1$ (a **cross-type** correspondence); cross-type roles: our P $b3$ as their N $e8$, our N $e4$ as their P $h6$; transferred move d1c2. (d) Retrieved from `lichess_elite_2024-09_p0025#3261` (ply 46) [structural distance $d=64$, QAP obj=30]: the master played b3; under μ our P $a2$ plays the role of that game’s P $b2$; cross-type roles: our b $b5$ as their N $b1$, our R $a1$ as their B $c1$; transferred move a2a3. Synthesis: 1 retrieved analog transferred to Rg3 (summed QAP objective 42, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



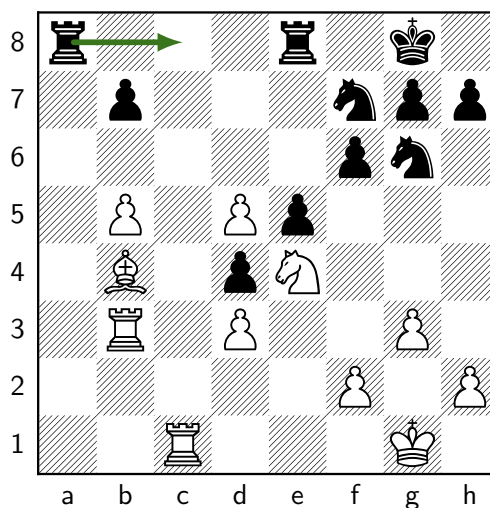
(a)



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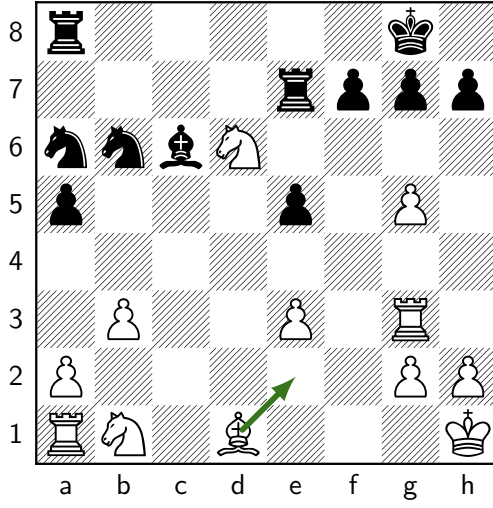


(c)

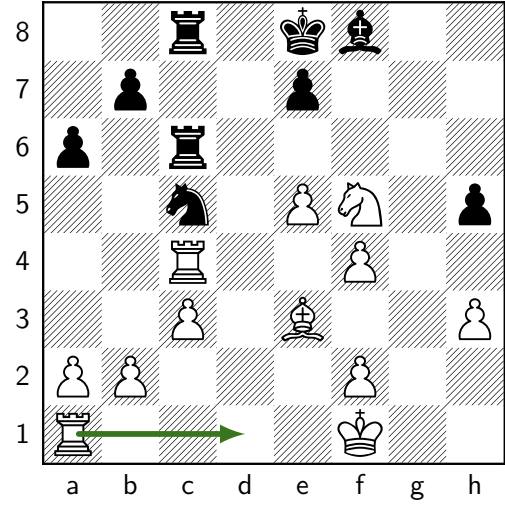


(d)

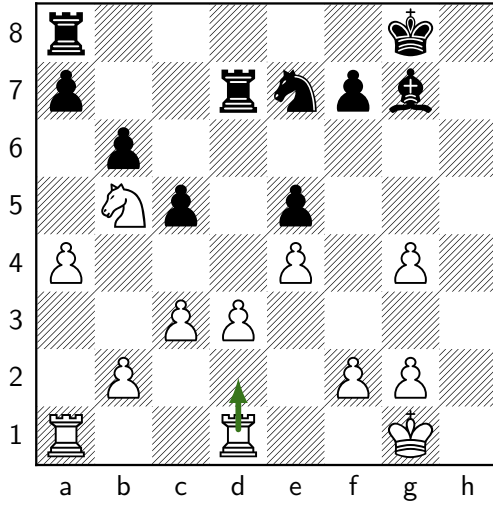
Figure 26: **Move 32, White: Nd6.** (a) Our position; Melanie plays Nd6 (e4d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0010#2610` (ply 47) [structural distance $d=77$, QAP obj=5]: the master played Rxd4; under μ our *Ne4* plays the role of that game’s *Rd1* (a **cross-type** correspondence); cross-type roles: our *bb5* as their *Ra8*, our *Pb3* as their *ne7*; transferred move e4d6 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0024#1167` (ply 55) [structural distance $d=82$, QAP obj=5]: the master played Rb6; under μ our *Rg3* plays the role of that game’s *Rd3*; cross-type roles: our *Ra1* as their *Pg7*, our *Pg5* as their *Ba6*; transferred move g3h3. (d) Retrieved from `lichess_elite_2024-09_p0026#495` (ply 55) [structural distance $d=71$, QAP obj=5]: the master played Rac8; under μ our *Ne4* plays the role of that game’s *Ra1* (a **cross-type** correspondence); cross-type roles: our *Pb3* as their *Re8*, our *Bd1* as their *Nf7*; transferred move e4d2. Synthesis: 1 retrieved analog transferred to Nd6 (summed QAP objective 5, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



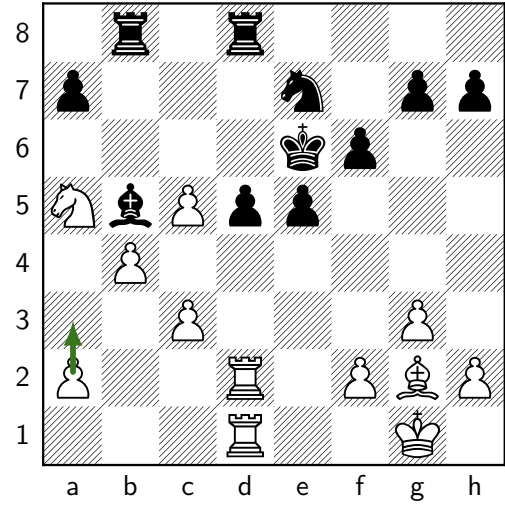
(a)



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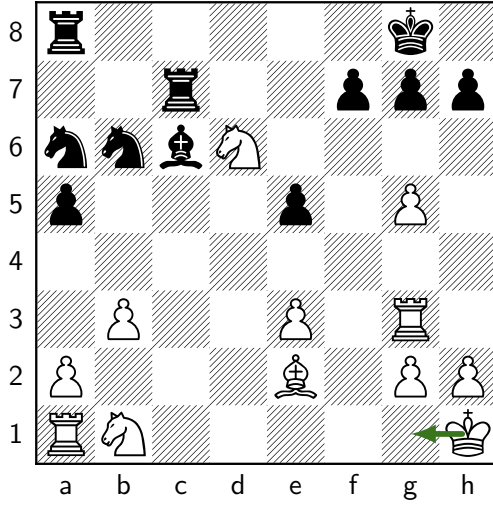


(c)

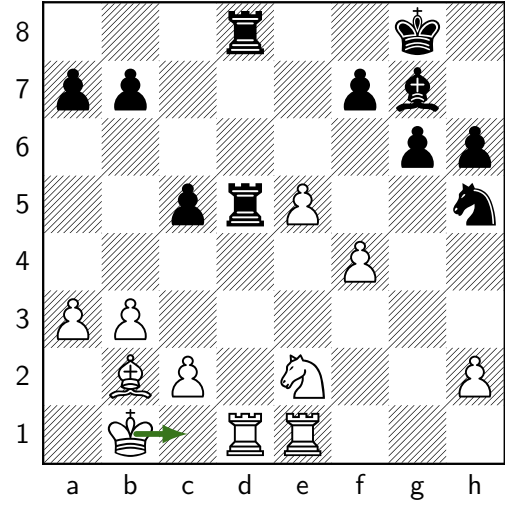


(d)

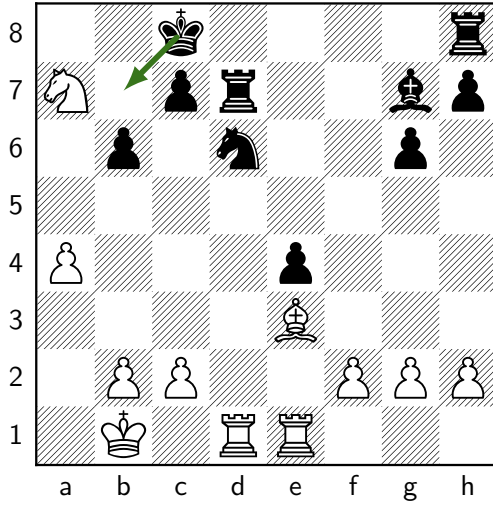
Figure 27: **Move 33, White: Be2.** (a) Our position; Melanie plays Be2 (d1e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0028#1513` (ply 52) [structural distance $d=69$, QAP obj=5]: the master played Rd1; under μ our Bd1 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Ra1 as their pa6, our pg7 as their rc6; transferred move d1e2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0003#1545` (ply 48) [structural distance $d=51$, QAP obj=57]: the master played Rd2; under μ our Rg3 plays the role of that game’s Rd1; cross-type roles: our Nb1 as their Ra1, our Bd1 as their Pb2; transferred move g3h3. (d) Retrieved from `lichess_elite_2024-09_p0018#238` (ply 60) [structural distance $d=65$, QAP obj=35]: the master played a3; under μ our Pe3 plays the role of that game’s Pa2; cross-type roles: our bc6 as their Rd2, our Ra1 as their Pf2; transferred move e3e4. Synthesis: 1 retrieved analog transferred to Be2 (summed QAP objective 5, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



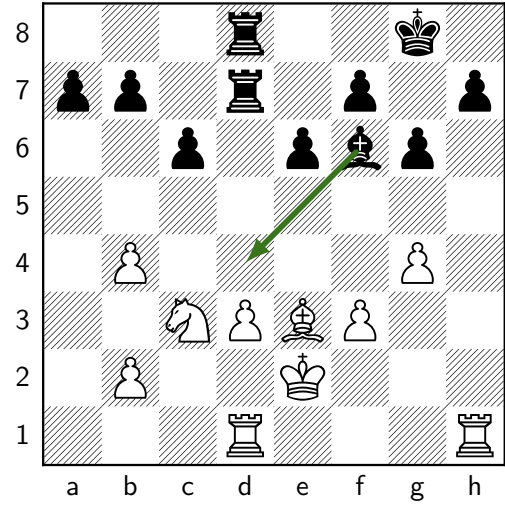
(a)



(b)

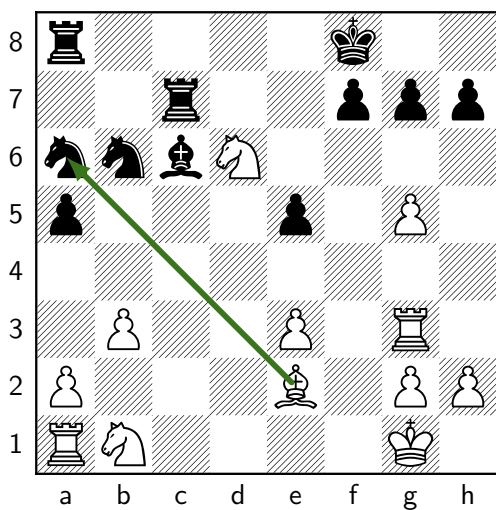


(c)

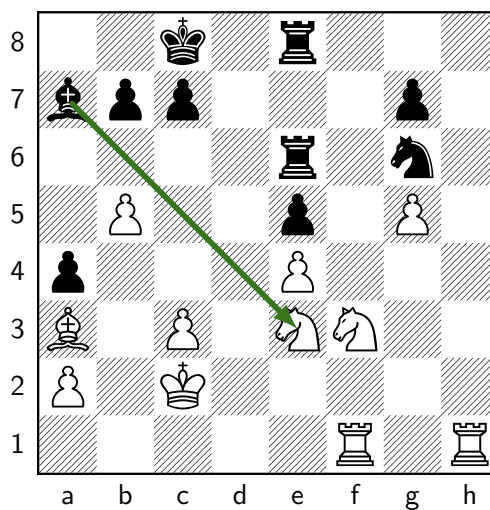


(d)

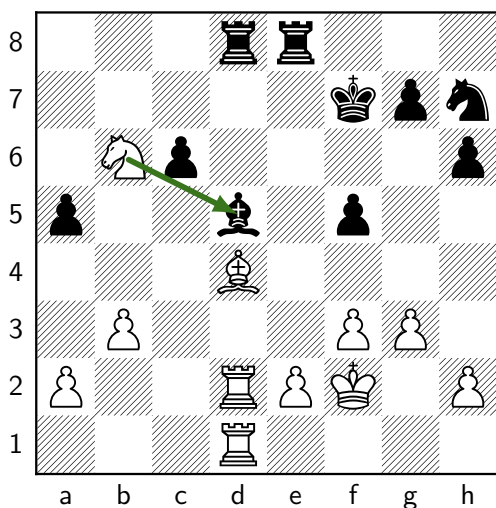
Figure 28: **Move 34, White: Kg1**. (a) Our position; Melanie plays Kg1 (**h1g1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0019#2066` (ply 52) [structural distance $d=65$, QAP obj=33]: the master played Kc1; under μ our *Kh1* plays the role of that game’s *Kb1*; cross-type roles: our *Be2* as their *Rd1*, our *Pb3* as their *Re1*; transferred move **h1g1** (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0013#924` (ply 55) [structural distance $d=73$, QAP obj=14]: the master played Kb7; under μ our *Kh1* plays the role of that game’s *Kc1*; cross-type roles: our *Pe3* as their *Rh8*, our *pa5* as their *na7*; transferred move **h1g1** (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0031#3603` (ply 57) [structural distance $d=71$, QAP obj=46]: the master played Bd4; under μ our *Be2* plays the role of that game’s *Bf3*; cross-type roles: our *ra8* as their *Pb7*, our *bc6* as their *Rd7*; transferred move **e2c4**. Synthesis: 2 retrieved analogs transferred to Kg1 (summed QAP objective 47, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



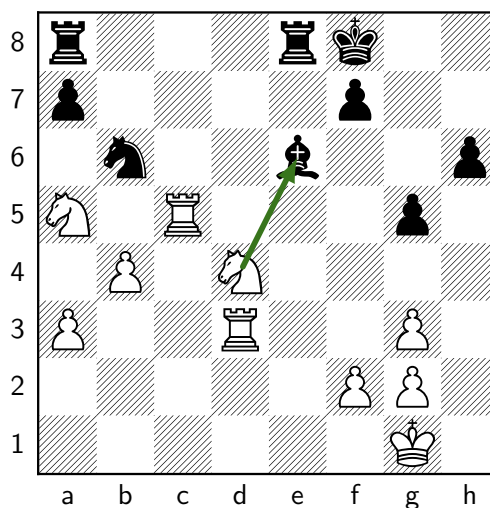
(a)



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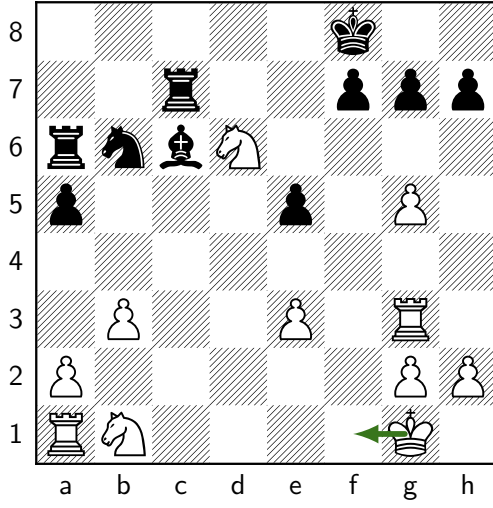


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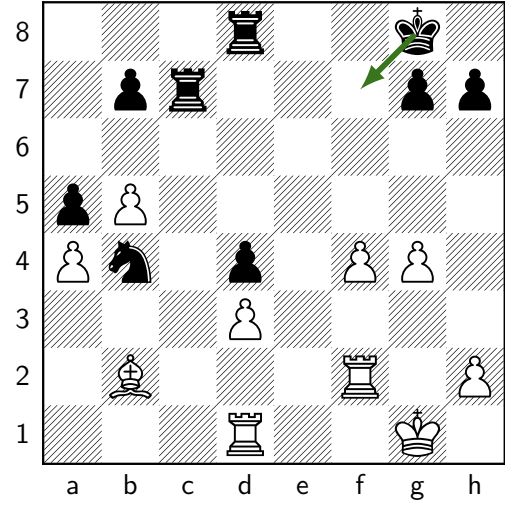


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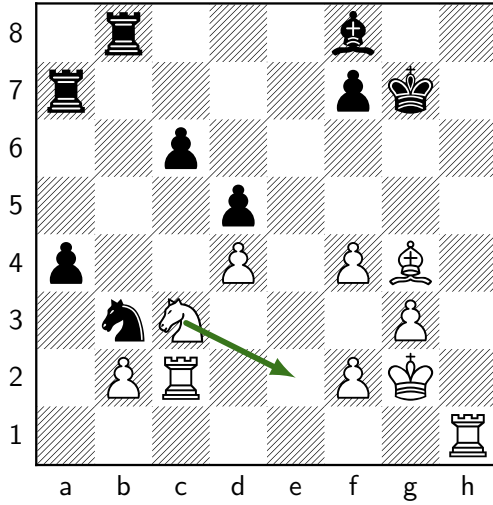
Figure 29: **Move 35, White: Bxa6**. (a) Our position; Melanie plays Bxa6 (e2a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2025-11_p0000#2724` (ply 85) [structural distance $d=62$, QAP obj=33]: the master played Bxe3; under μ our Be2 plays the role of that game’s Ba2; cross-type roles: our Nb1 as their Re8, our pe5 as their Re6; transferred move e2a6 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0025#3875` (ply 62) [structural distance $d=71$, QAP obj=17]: the master played Nxd5; under μ our Be2 plays the role of that game’s Nb6 (a **cross-type** correspondence); cross-type roles: our pg7 as their Rd1, our ph7 as their Rd2; transferred move e2a6 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0024#3686` (ply 62) [structural distance $d=75$, QAP obj=13]: the master played Nxe6+; under μ our Be2 plays the role of that game’s Nd4 (a **cross-type** correspondence); cross-type roles: our Nb1 as their Pf2, our Nd6 as their Pa3; transferred move e2a6 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Bxa6 (summed QAP objective 64, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



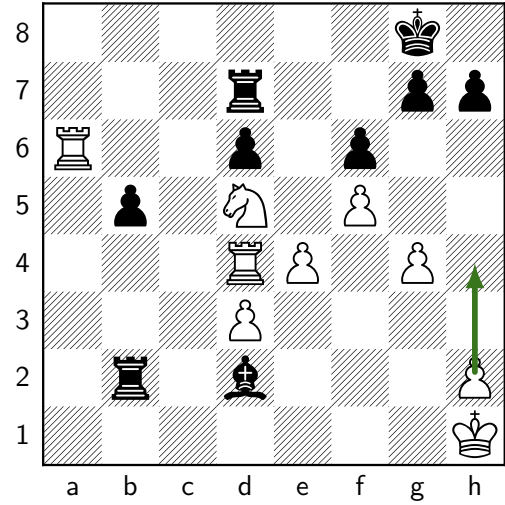
(a)



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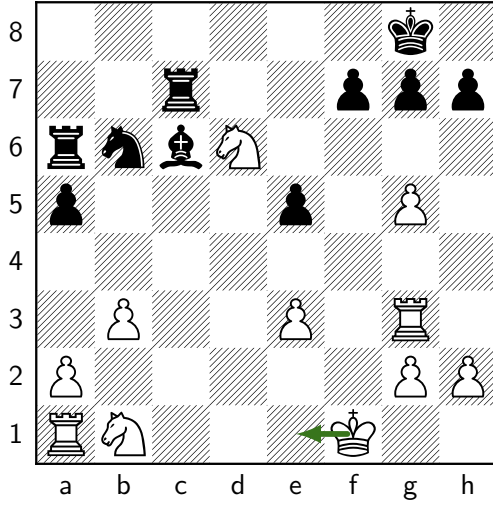


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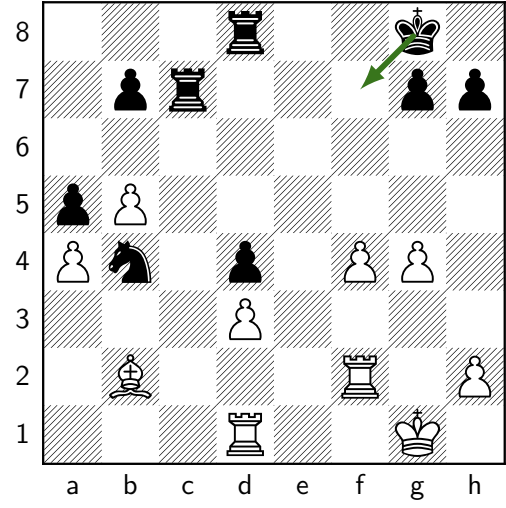


(d)

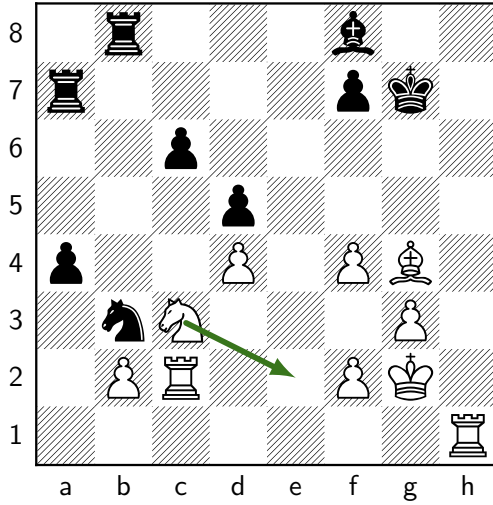
Figure 30: **Move 36, White: Kf1.** (a) Our position; Melanie plays Kf1 (g1f1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0021#2483` (ply 59) [structural distance $d=53$, QAP obj=32]: the master played Kf7; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our Pb3 as their Rd8, our Nb1 as their Ph7; transferred move g1f1 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0005#321` (ply 84) [structural distance $d=53$, QAP obj=44]: the master played Ne2; under μ our Nd6 plays the role of that game’s Nc3; cross-type roles: our pg7 as their Rh1, our bc6 as their nb3; transferred move d6e4. (d) Retrieved from `lichess_elite_2024-09_p0030#2152` (ply 72) [structural distance $d=62$, QAP obj=30]: the master played h4; under μ our Rg3 plays the role of that game’s Ph2 (a **cross-type** correspondence); cross-type roles: our Nb1 as their bd2, our ra6 as their Pd3; transferred move g3g4. Synthesis: 2 retrieved analogs transferred to Kf1 (summed QAP objective 63, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



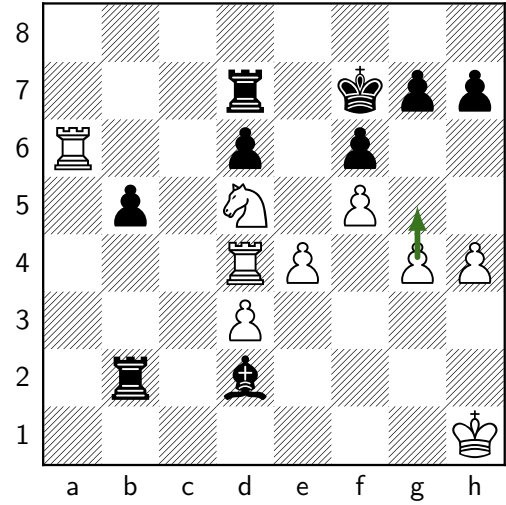
(a)



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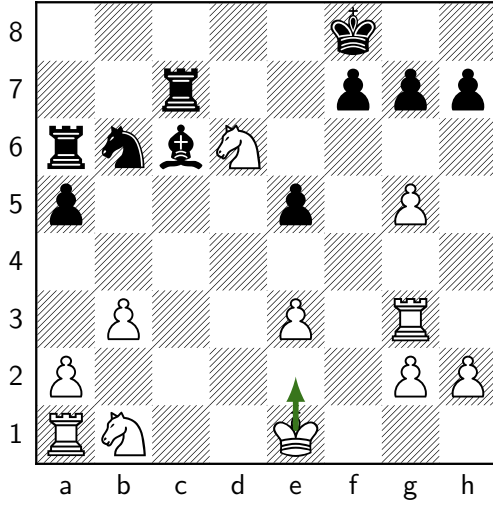


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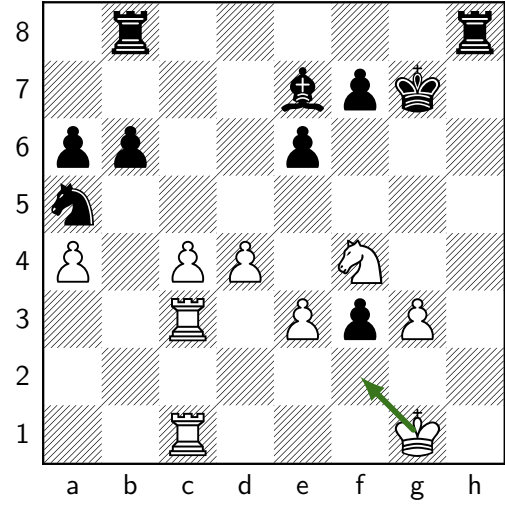


(d)

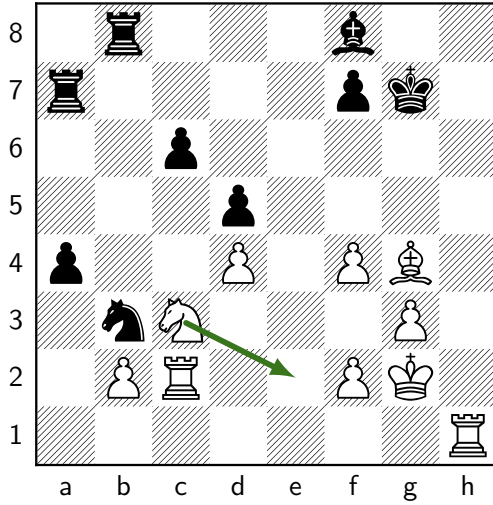
Figure 31: **Move 37, White: Ke1.** (a) Our position; Melanie plays Ke1 (f1e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0021#2483` (ply 59) [structural distance $d=52$, QAP obj=32]: the master played Kf7; under μ our Kf1 plays the role of that game’s Kg1; cross-type roles: our Pb3 as their Rd8, our Nb1 as their Pg7; transferred move f1e1 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0005#321` (ply 84) [structural distance $d=63$, QAP obj=44]: the master played Ne2; under μ our Nd6 plays the role of that game’s Nc3; cross-type roles: our bc6 as their Rh1, our Nb1 as their Rc2; transferred move d6e4. (d) Retrieved from `lichess_elite_2024-09_p0030#2152` (ply 74) [structural distance $d=61$, QAP obj=31]: the master played g5; under μ our Pa2 plays the role of that game’s Pg4; cross-type roles: our rc7 as their bd2, our Ph2 as their Rd4; transferred move a2a4. Synthesis: 1 retrieved analog transferred to Ke1 (summed QAP objective 32, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



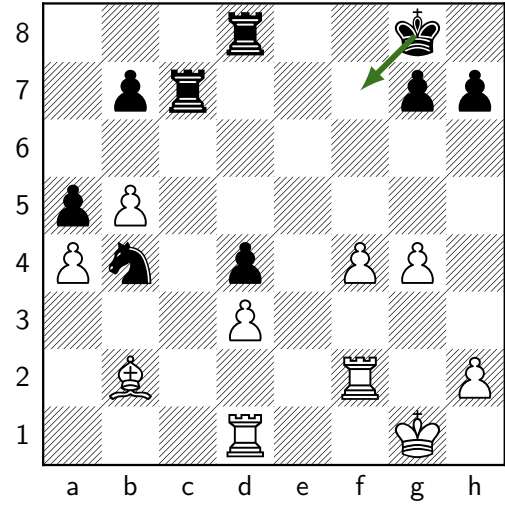
(a)



(b)

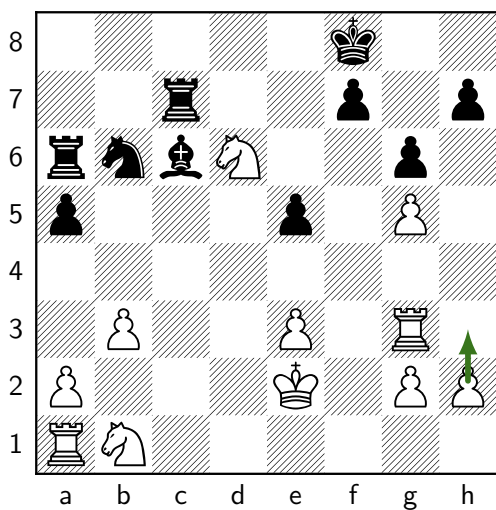


(c)

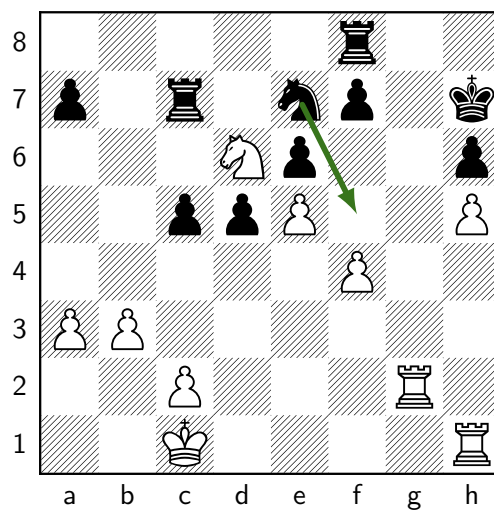


(d)

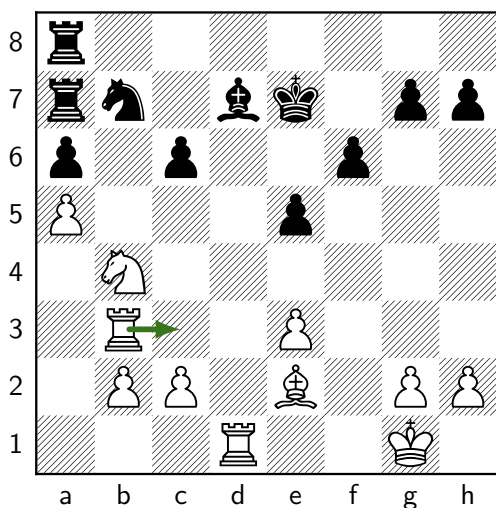
Figure 32: **Move 38, White: Ke2.** (a) Our position; Melanie plays Ke2 (e1e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0031#2568` (ply 76) [structural distance $d=69$, QAP obj=32]: the master played Kf2; under μ our Ke1 plays the role of that game’s Kg1; cross-type roles: our nb6 as their Rc1, our Ra1 as their Pe3; transferred move e1e2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0005#321` (ply 84) [structural distance $d=65$, QAP obj=44]: the master played Ne2; under μ our Nd6 plays the role of that game’s Nc3; cross-type roles: our Pe3 as their Rc2, our Ra1 as their Pf2; transferred move d6f5. (d) Retrieved from `lichess_elite_2024-09_p0021#2483` (ply 59) [structural distance $d=55$, QAP obj=32]: the master played Kf7; under μ our Ke1 plays the role of that game’s Kg1; cross-type roles: our Pg5 as their Rd8, our Nb1 as their Pg7; transferred move e1d2. Synthesis: 1 retrieved analog transferred to Ke2 (summed QAP objective 32, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



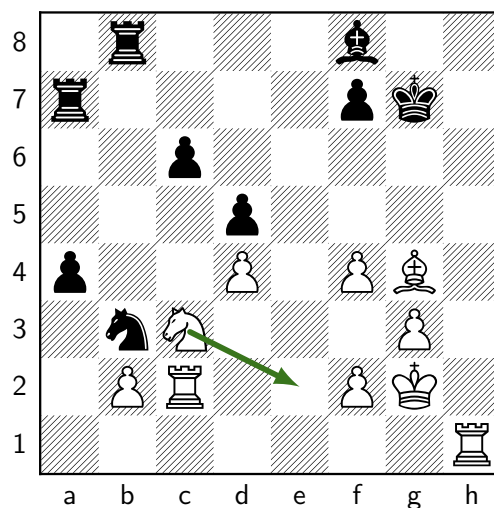
(a)



(b)

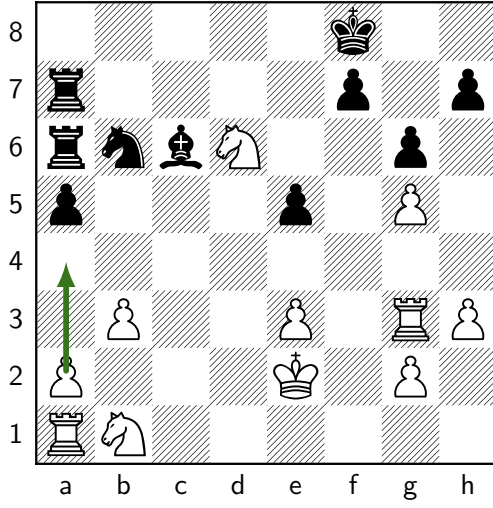


(c)

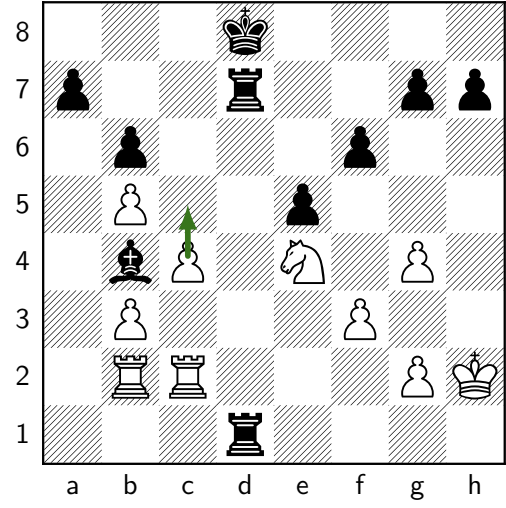


(d)

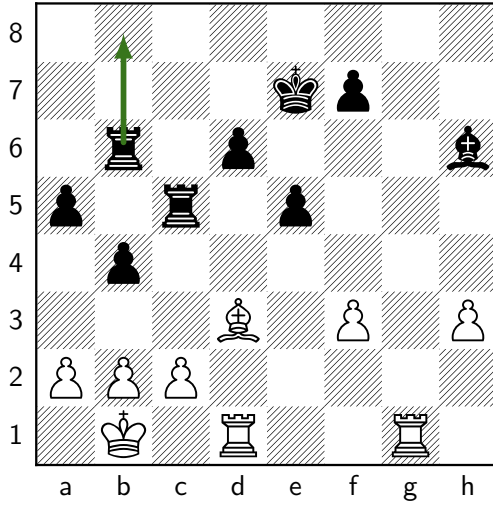
Figure 33: **Move 39, White: h3**. (a) Our position; Melanie plays **h3** (**h2h3**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0031#693` (ply 63) [structural distance $d=54$, QAP obj=23]: the master played **Nf5**; under μ our *Ph2* plays the role of that game’s *Ne2* (a **cross-type** correspondence); cross-type roles: our *bc6* as their *Pd5*, our *Rg3* as their *pc2*; transferred move **h2h3** (**endorses this move**). (c) Retrieved from `lichess_elite_2025-11_p0000#996` (ply 60) [structural distance $d=65$, QAP obj=58]: the master played **Rc3**; under μ our *Pb3* plays the role of that game’s *Rb3* (a **cross-type** correspondence); cross-type roles: our *Nd6* as their *Rd1*, our *nb6* as their *Pc2*; transferred move **b3b4**. (d) Retrieved from `lichess_elite_2024-09_p0005#321` (ply 84) [structural distance $d=56$, QAP obj=44]: the master played **Ne2**; under μ our *Nd6* plays the role of that game’s *Nc3*; cross-type roles: our *Pe3* as their *Rc2*, our *bc6* as their *nb3*; transferred move **d6e4**. Synthesis: 1 retrieved analog transferred to **h3** (summed QAP objective 23, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



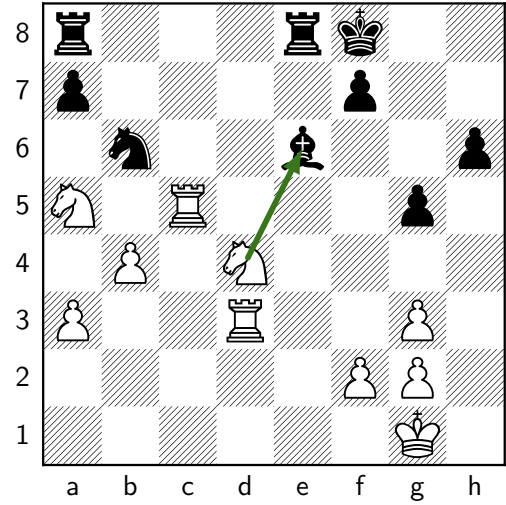
(a)



(b)

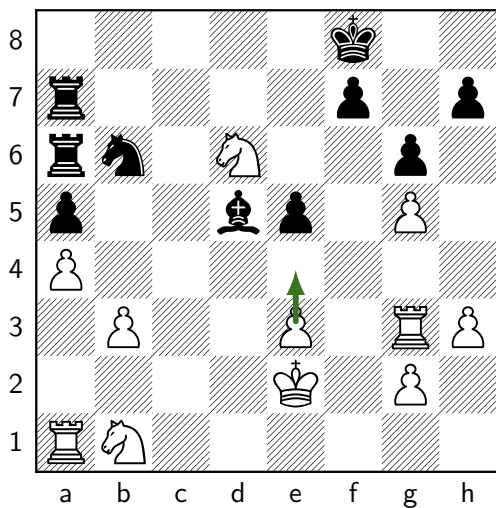


(c)

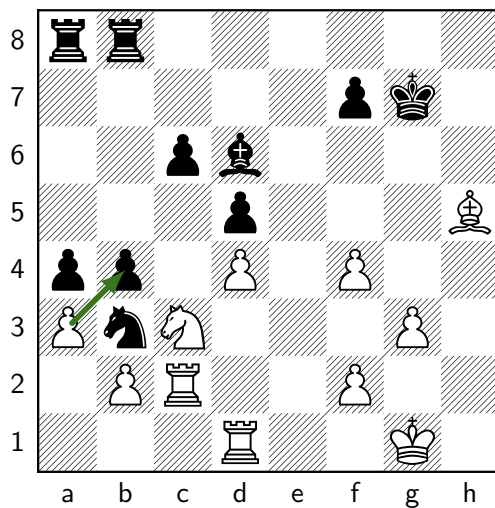


(d)

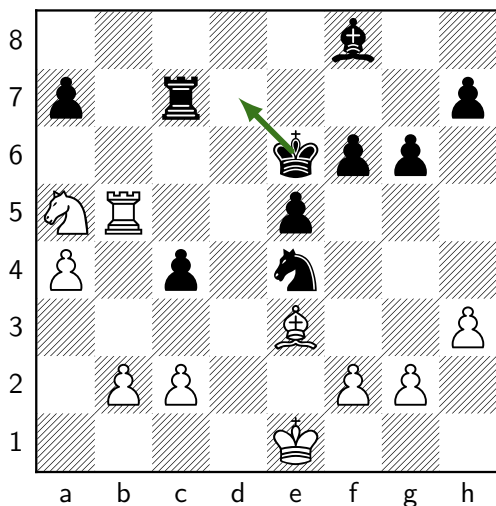
Figure 34: **Move 40, White: a4.** (a) Our position; Melanie plays a4 (a2a4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0022#960` (ply 66) [structural distance $d=39$, QAP obj=22]: the master played c5; under μ our *Pa2* plays the role of that game’s *Pc4*; cross-type roles: our *bc6* as their *rd1*, our *pa5* as their *Rc2*; transferred move a2a4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0018#1021` (ply 63) [structural distance $d=78$, QAP obj=18]: the master played Rb8; under μ our *Rg3* plays the role of that game’s *Rb3*; cross-type roles: our *Pb3* as their *Bh6*, our *ra6* as their *Pa5*; transferred move g3g4. (d) Retrieved from `lichess_elite_2024-09_p0024#3686` (ply 62) [structural distance $d=78$, QAP obj=11]: the master played Nxe6+; under μ our *Nd6* plays the role of that game’s *Nd4*; cross-type roles: our *Nb1* as their *Pf2*, our *Pa2* as their *Rd3*; transferred move d6f5. Synthesis: 1 retrieved analog transferred to a4 (summed QAP objective 22, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



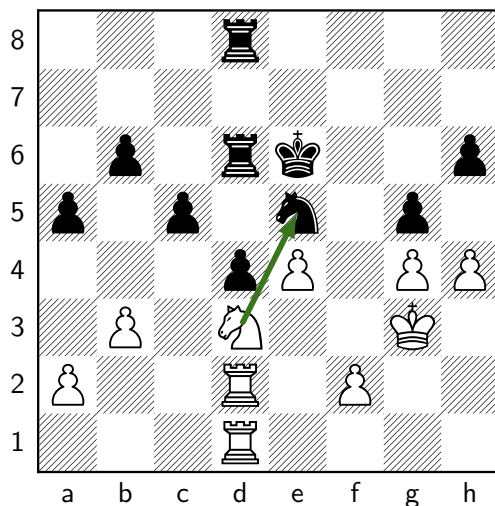
(a)



(b)

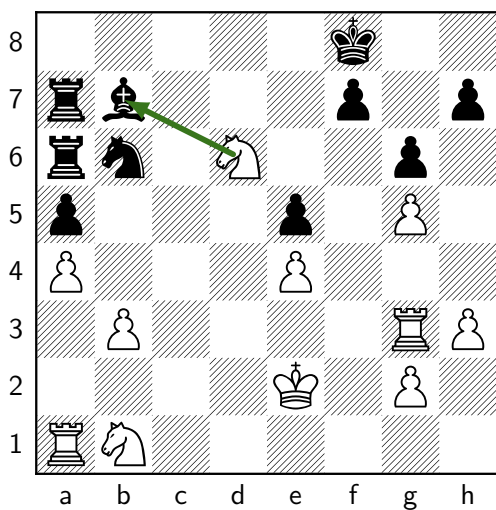


(c)

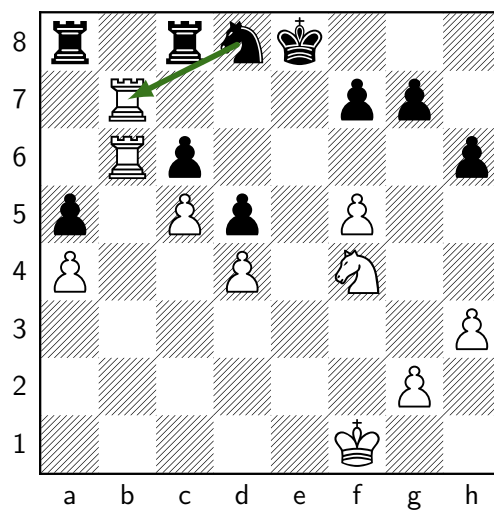


(d)

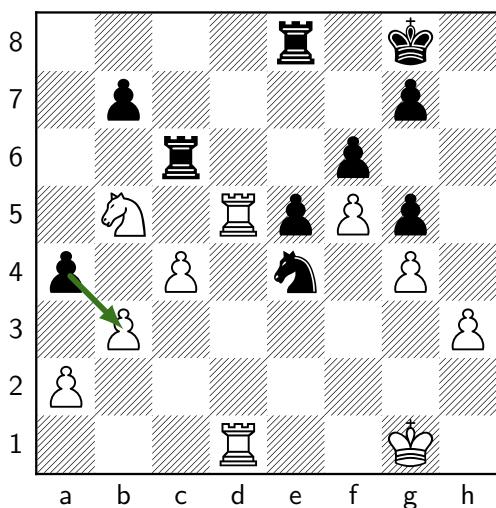
Figure 35: **Move 41, White: e4**. (a) Our position; Melanie plays **e4** (**e3e4**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0005#321` (ply 70) [structural distance $d=76$, QAP obj=6]: the master played **axb4**; under μ our *Pe3* plays the role of that game’s *Pa3*; cross-type roles: our *nb6* as their *Pb2*, our *pg6* as their *Rc2*; transferred move **e3e4** (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0030#3040` (ply 67) [structural distance $d=81$, QAP obj=63]: the master played **Kd7**; under μ our *Ke2* plays the role of that game’s *Ke3*; cross-type roles: our *Pe3* as their *Bf8*, our *bd5* as their *na5*; transferred move **e2d2**. (d) Retrieved from `lichess_elite_2024-09_p0024#3241` (ply 64) [structural distance $d=78$, QAP obj=23]: the master played **Nxe5**; under μ our *Nd6* plays the role of that game’s *Nd3*; cross-type roles: our *Pg2* as their *Rd1*, our *ra6* as their *Pb3*; transferred move **d6e8**. Synthesis: 1 retrieved analog transferred to **e4** (summed QAP objective 6, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



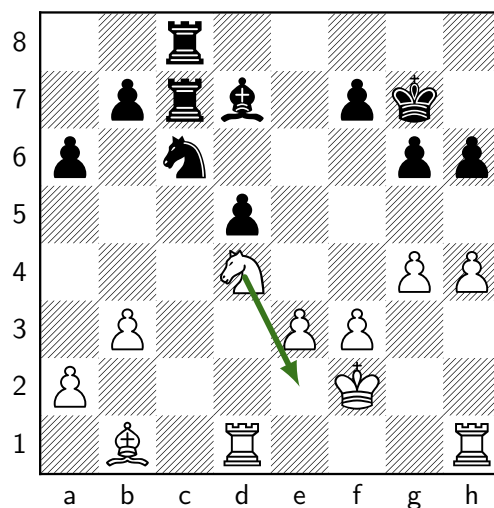
(a)



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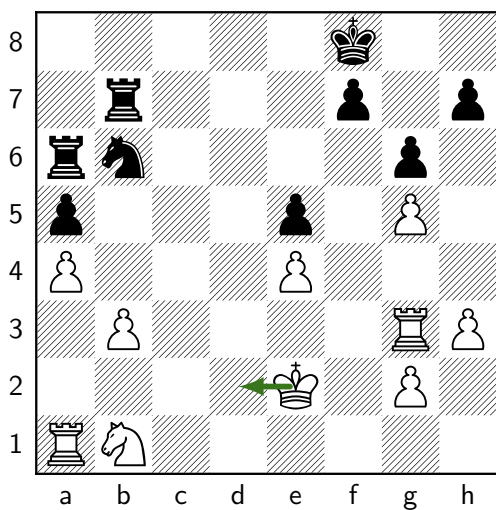


(c)

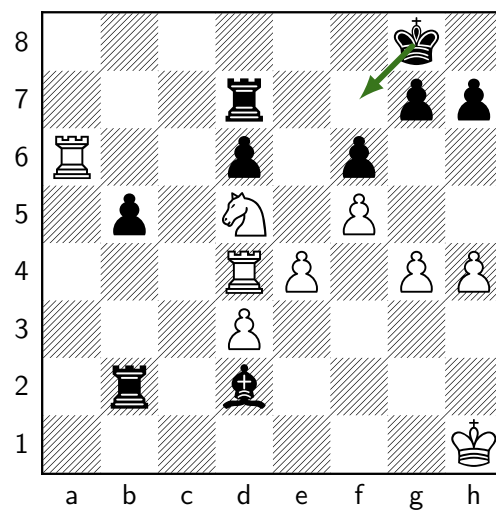


(d)

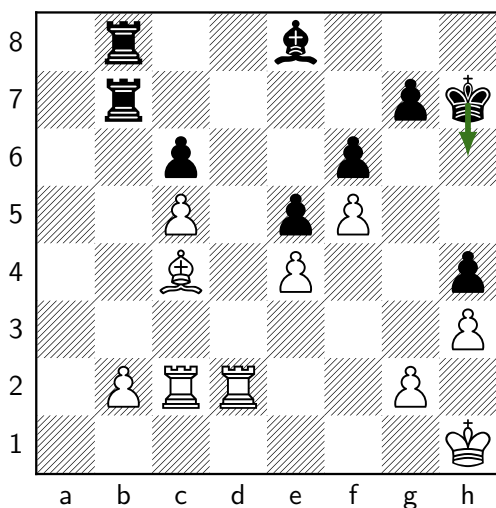
Figure 36: **Move 42, White: Nxb7.** (a) Our position; Melanie plays Nxb7 (d6b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0003#722` (ply 69) [structural distance $d=86$, QAP obj=38]: the master played Nxb7; under μ our Nd6 plays the role of that game’s Nd1; cross-type roles: our Nb1 as their Ra8, our bb7 as their rb7; transferred move d6b7 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0023#1908` (ply 69) [structural distance $d=88$, QAP obj=27]: the master played axb3; under μ our Nd6 plays the role of that game’s Pa5 (a **cross-type** correspondence); cross-type roles: our ra6 as their Pb7, our nb6 as their Rc6; transferred move d6b7 (**endorses this move**). (d) Retrieved from `twic1500#409` (ply 66) [structural distance $d=72$, QAP obj=31]: the master played Ne2; under μ our Nd6 plays the role of that game’s Nd4; cross-type roles: our ph7 as their Bb1, our Pb3 as their Rd1; transferred move d6b5. Synthesis: 2 retrieved analogs transferred to Nxb7 (summed QAP objective 65, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



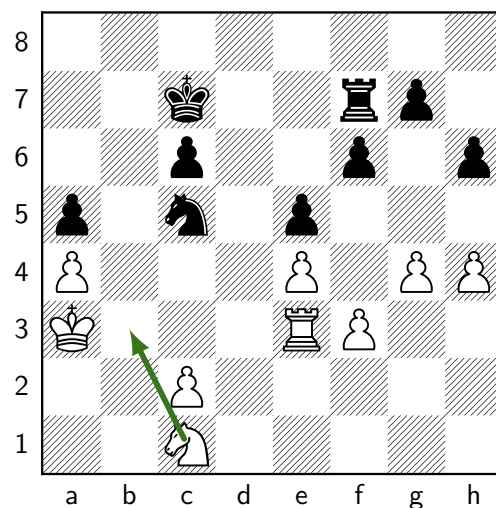
(a)



(b)

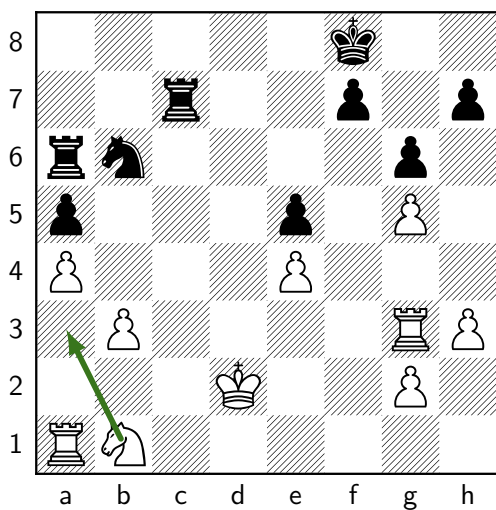


(c)

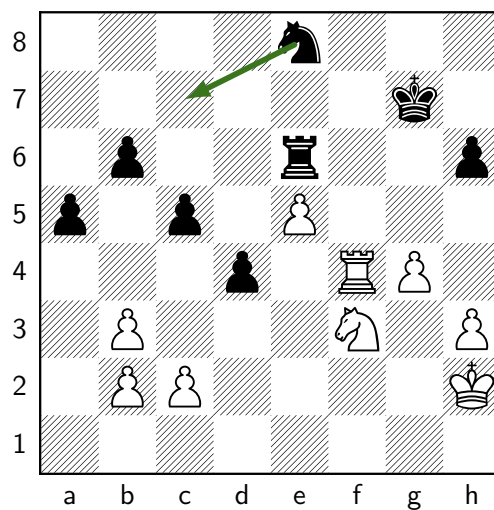


(d)

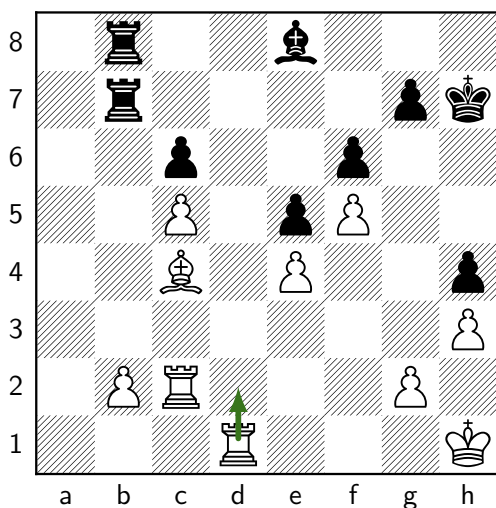
Figure 37: **Move 43, White: Kd2.** (a) Our position; Melanie plays Kd2 (e2d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0030#2152` (ply 73) [structural distance $d=54$, QAP obj=29]: the master played Kf7; under μ our *Ke2* plays the role of that game’s *Kg1*; cross-type roles: our *nb6* as their *ra6*, our *rb7* as their *pe4*; transferred move e2d2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0012#3553` (ply 73) [structural distance $d=69$, QAP obj=12]: the master played Kh6; under μ our *Ke2* plays the role of that game’s *Kh2*; cross-type roles: our *Pg2* as their *Be8*, our *nb6* as their *Rb7*; transferred move e2d2 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0021#2332` (ply 68) [structural distance $d=72$, QAP obj=37]: the master played Nb3; under μ our *Pb3* plays the role of that game’s *Nc1* (a **cross-type** correspondence); cross-type roles: our *Rg3* as their *Pc2*, our *ra6* as their *pc6*; transferred move b3b4. Synthesis: 2 retrieved analogs transferred to Kd2 (summed QAP objective 41, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



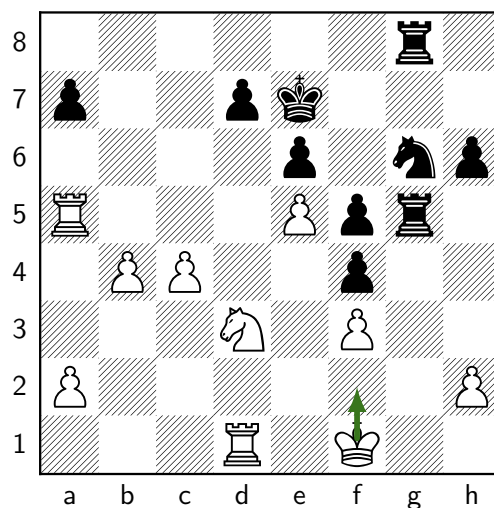
(a)



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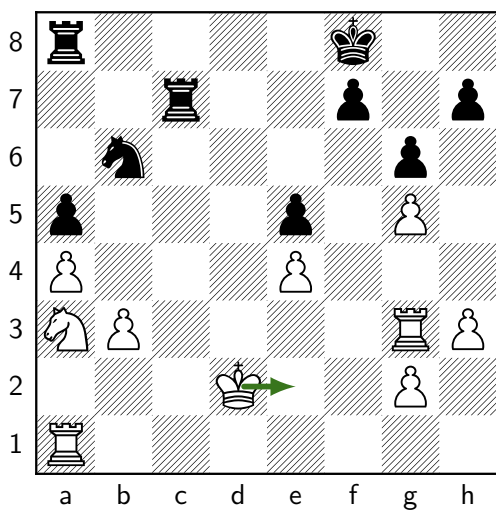


(c)

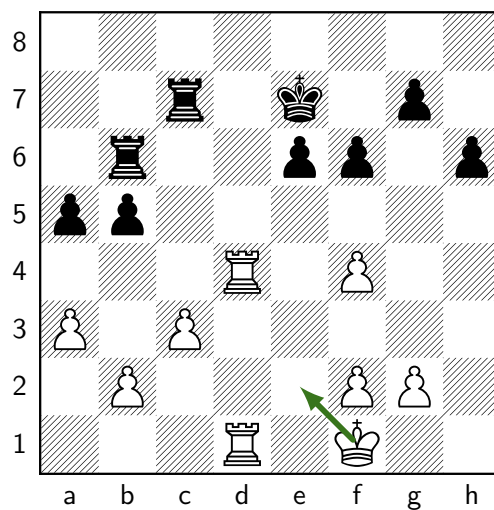


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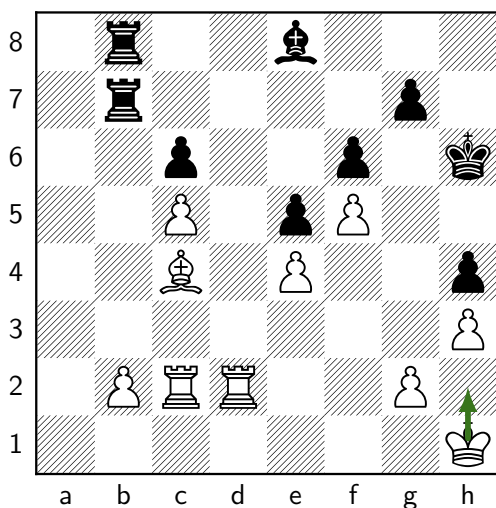
Figure 38: **Move 44, White: Na3**. (a) Our position; Melanie plays Na3 (b1a3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0017#874` (ply 87) [structural distance $d=64$, QAP obj=19]: the master played Nc7; under μ our Nb1 plays the role of that game’s Ne1; cross-type roles: our rc7 as their pe5, our nb6 as their rf4; transferred move b1a3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0012#3553` (ply 72) [structural distance $d=57$, QAP obj=13]: the master played Rdd2; under μ our Nb1 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our rc7 as their Pg2, our Pb3 as their Bc4; transferred move b1a3 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0022#2770` (ply 70) [structural distance $d=64$, QAP obj=31]: the master played Kf2; under μ our Kd2 plays the role of that game’s Kf1; cross-type roles: our Nb1 as their Rd1, our ra6 as their Pb4; transferred move d2e2. Synthesis: 2 retrieved analogs transferred to Na3 (summed QAP objective 31, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



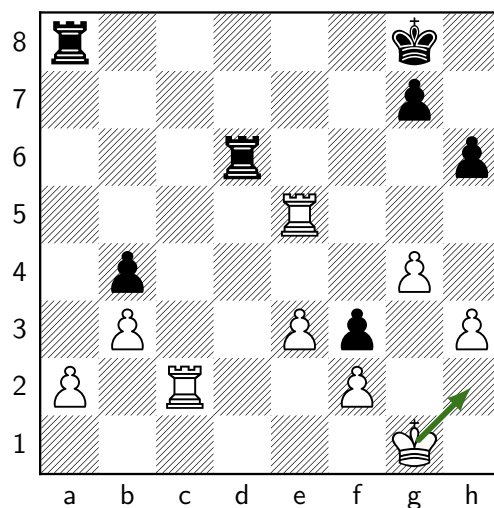
(a)



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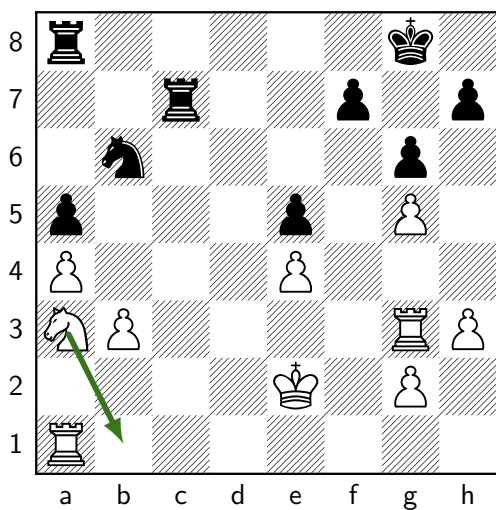


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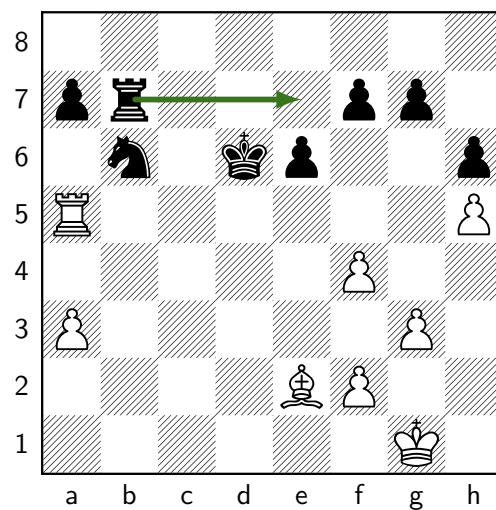


(d)

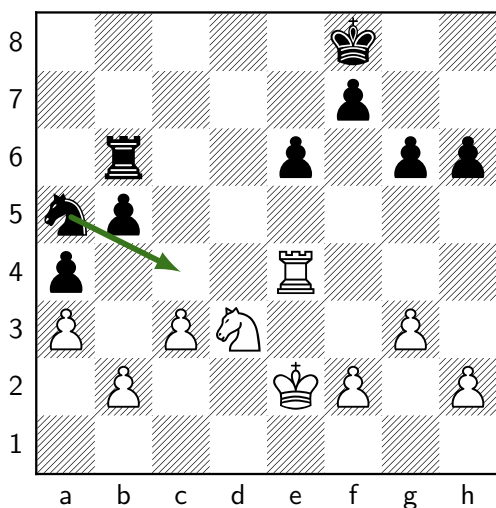
Figure 39: **Move 45, White: Ke2.** (a) Our position; Melanie plays Ke2 (d2e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0018#1830` (ply 72) [structural distance $d=58$, QAP obj=13]: the master played Ke2; under μ our *Kd2* plays the role of that game’s *Kf1*; cross-type roles: our *Pg2* as their *Rd1*, our *Ra1* as their *Pf2*; transferred move d2e2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0012#3553` (ply 74) [structural distance $d=43$, QAP obj=11]: the master played Kh2; under μ our *Kd2* plays the role of that game’s *Kh1*; cross-type roles: our *Na3* as their *Rc2*, our *Pb3* as their *Rd2*; transferred move d2e2 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0021#925` (ply 72) [structural distance $d=55$, QAP obj=9]: the master played Kh2; under μ our *Kd2* plays the role of that game’s *Kg1*; cross-type roles: our *Na3* as their *Rc2*, our *ra8* as their *Pf2*; transferred move d2e2 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Ke2 (summed QAP objective 33, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



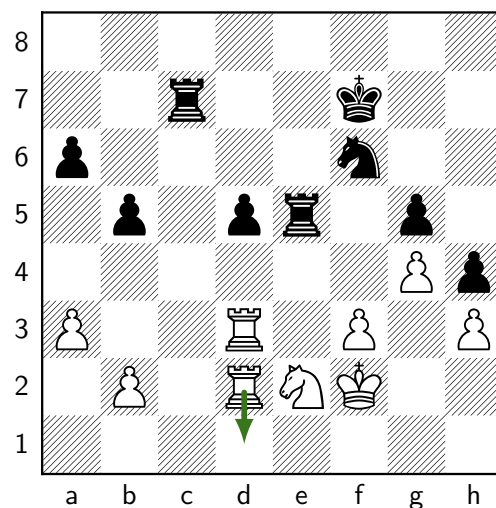
(a)



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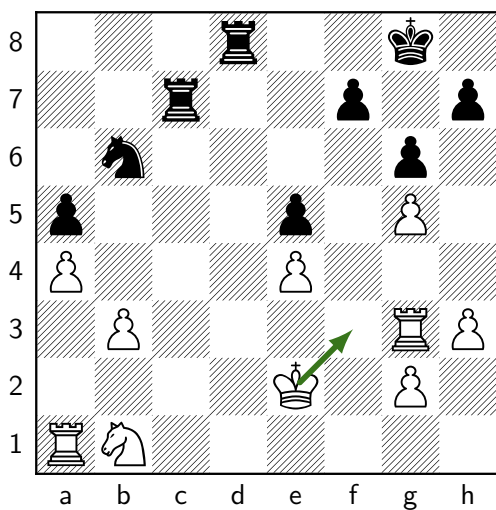


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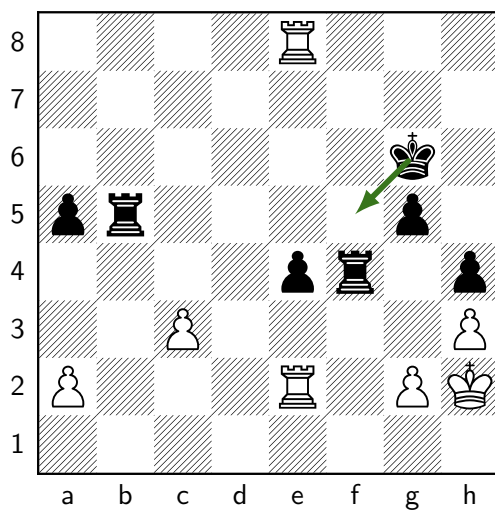


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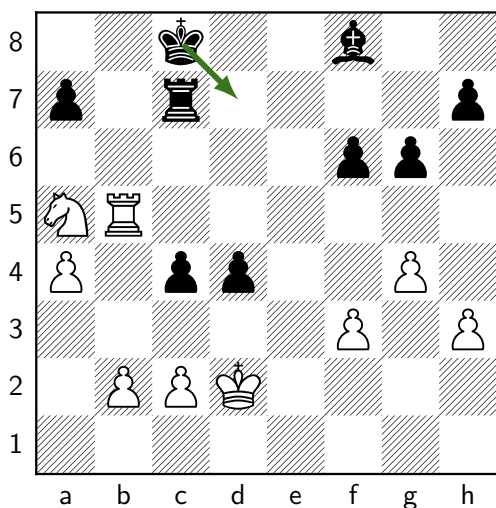
Figure 40: **Move 46, White: Nb1.** (a) Our position; Melanie plays Nb1 (a3b1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0017#2694` (ply 81) [structural distance $d=60$, QAP obj=21]: the master played Re7; under μ our Na3 plays the role of that game’s Rb2 (a **cross-type** correspondence); cross-type roles: our Pb3 as their Nb6, our nb6 as their ra5; transferred move a3b1 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0023#1518` (ply 85) [structural distance $d=62$, QAP obj=26]: the master played Nc4; under μ our Na3 plays the role of that game’s Na4; cross-type roles: our Pg2 as their Rb6, our Ra1 as their Ph6; transferred move a3c4. (d) Retrieved from `lichess_elite_2024-09_p0003#972` (ply 84) [structural distance $d=39$, QAP obj=15]: the master played Rd1; under μ our Na3 plays the role of that game’s Rd2 (a **cross-type** correspondence); cross-type roles: our Pa4 as their Ne2, our nb6 as their re5; transferred move a3c2. Synthesis: 1 retrieved analog transferred to Nb1 (summed QAP objective 21, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



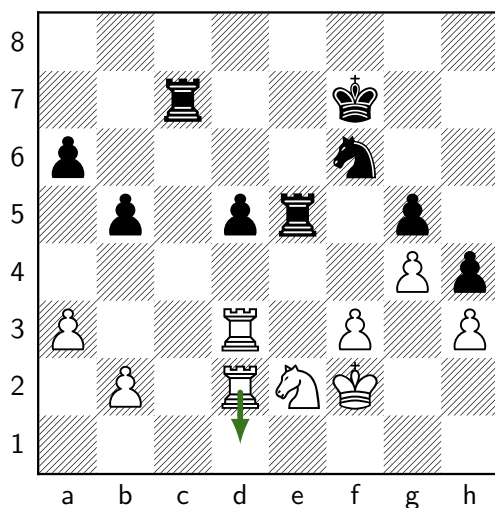
(a)



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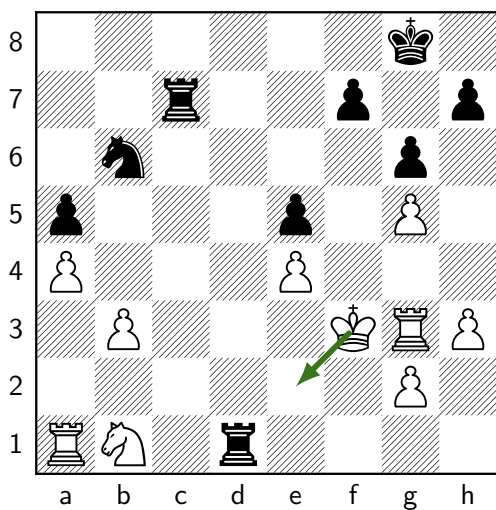


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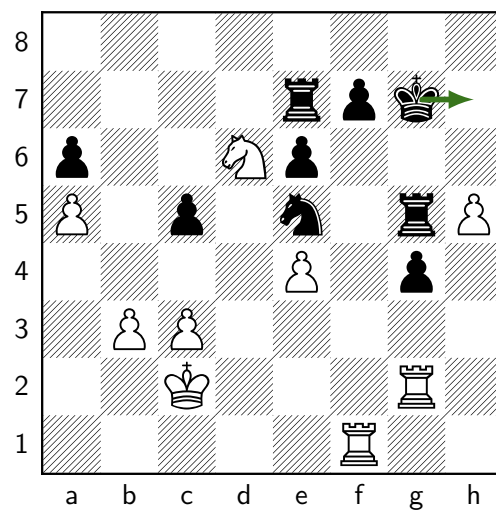


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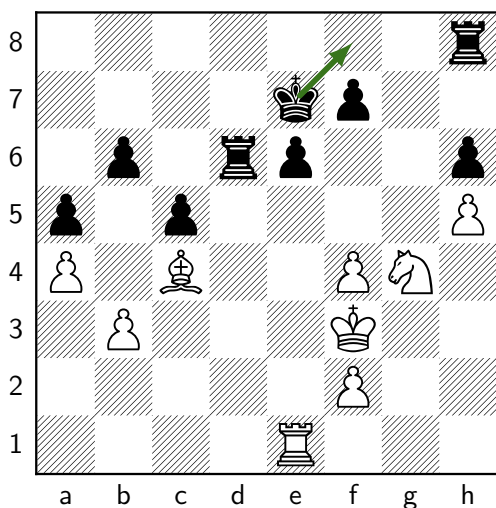
Figure 41: **Move 47, White: Kf3.** (a) Our position; Melanie plays Kf3 (e2f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0028#1137` (ply 91) [structural distance $d=71$, QAP obj=25]: the master played Kf5; under μ our *Ke2* plays the role of that game’s *Kg3*; cross-type roles: our *nb6* as their *re8*, our *pg6* as their *Rb5*; transferred move e2f3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0030#3040` (ply 79) [structural distance $d=67$, QAP obj=21]: the master played Kd7; under μ our *Ke2* plays the role of that game’s *Kc1*; cross-type roles: our *pf7* as their *Bf8*, our *Nb1* as their *Ph7*; transferred move e2e1. (d) Retrieved from `lichess_elite_2024-09_p0003#972` (ply 84) [structural distance $d=38$, QAP obj=16]: the master played Rd1; under μ our *Nb1* plays the role of that game’s *Rd2* (a **cross-type** correspondence); cross-type roles: our *Pg5* as their *Ne2*, our *rc7* as their *pd5*; transferred move b1a3. Synthesis: 1 retrieved analog transferred to Kf3 (summed QAP objective 25, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



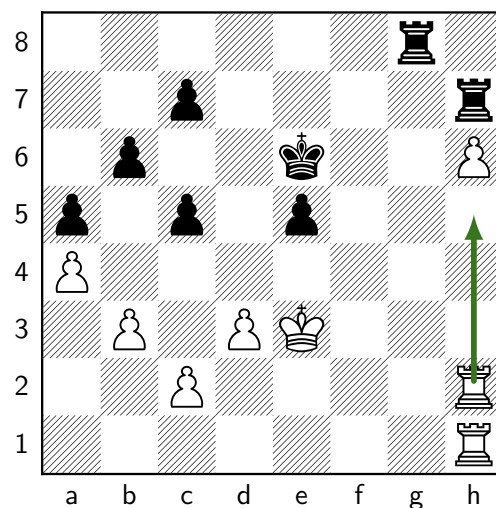
(a)



(b)



(c)



(d)

Figure 42: **Move 48, White: Ke2**. (a) Our position; Melanie plays Ke2 (f3e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0022#2070` (ply 99) [structural distance $d=75$, QAP obj=24]: the master played Kh7; under μ our Kf3 plays the role of that game’s Kg2; cross-type roles: our Pb3 as their Re7, our Rg3 as their pa5; transferred move f3e2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0003#2118` (ply 107) [structural distance $d=70$, QAP obj=18]: the master played Kf8; under μ our Kf3 plays the role of that game’s Ke2; cross-type roles: our Pb3 as their Rh8, our Nb1 as their Pe6; transferred move f3e2 (**endorses this move**). (d) Retrieved from `lichess_elite_2025-11_p0000#2370` (ply 86) [structural distance $d=65$, QAP obj=30]: the master played Rh5; under μ our Ra1 plays the role of that game’s Rh2; cross-type roles: our Pg2 as their Rh1, our nb6 as their Pb3; transferred move a1a2. Synthesis: 2 retrieved analogs transferred to Ke2 (summed QAP objective 43, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

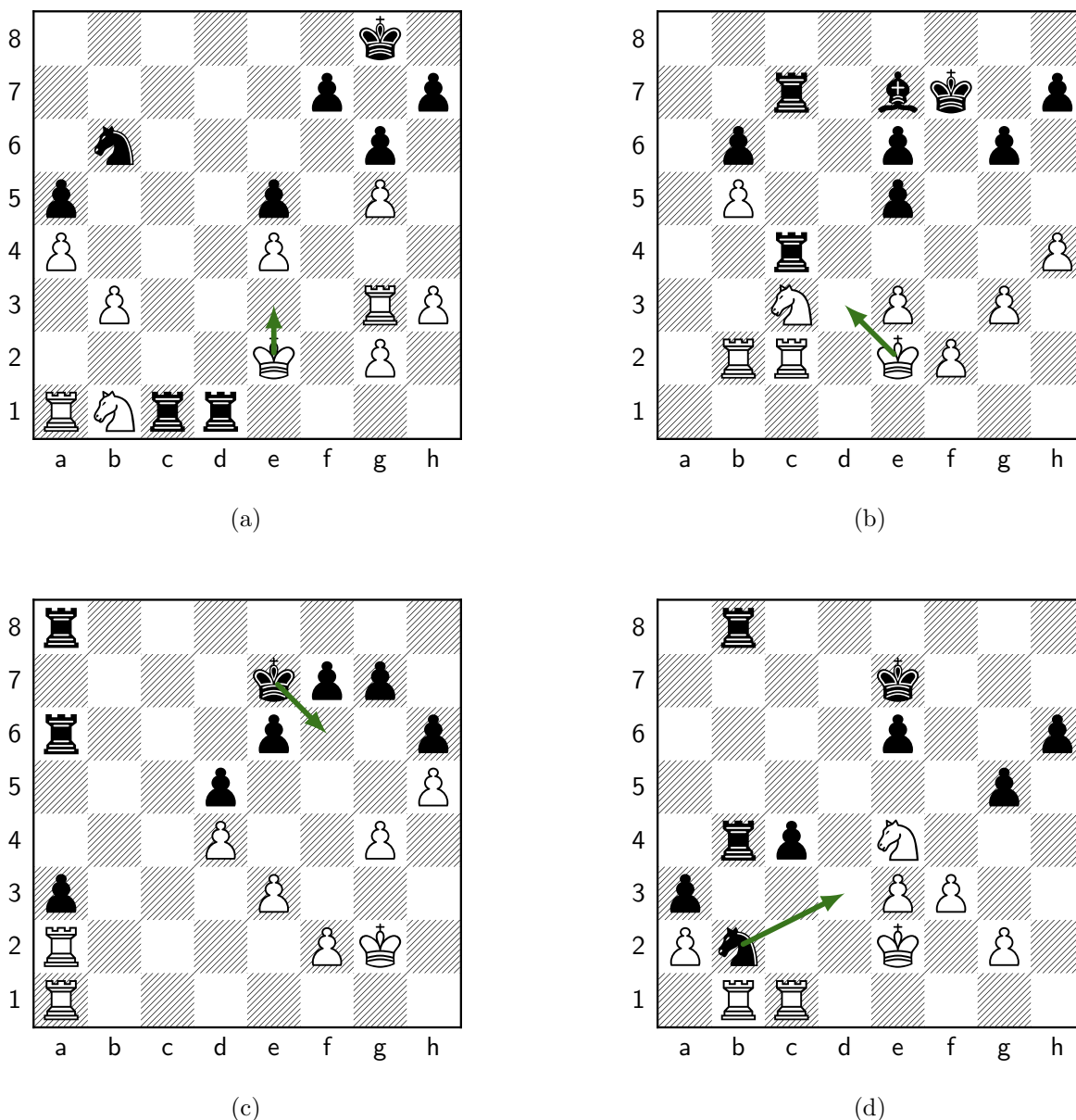


Figure 43: **Move 49, White: Ke3.** (a) Our position; Melanie plays Ke3 (e2e3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from lichess_elite_2024-09_p0031#3257 (ply 100) [structural distance $d=69$, QAP obj=39]: the master played Kd3; under μ our Ke2 plays the role of that game's Ke2; cross-type roles: our nb6 as their Pg3, our pf7 as their be7; transferred move e2e3 (**endorses this move**). (c) Retrieved from lichess_elite_2024-09_p0021#559 (ply 87) [structural distance $d=53$, QAP obj=18]: the master played Kf6; under μ our Ke2 plays the role of that game's Ke2; cross-type roles: our Pg2 as their Ra8, our Nb1 as their Pa3; transferred move e2e3 (**endorses this move**). (d) Retrieved from lichess_elite_2024-09_p0013#3532 (ply 81) [structural distance $d=56$, QAP obj=43]: the master played Nd3; under μ our Nb1 plays the role of that game's Nb7; cross-type roles: our Ph3 as their Rb8, our Rg3 as their pf3; transferred move b1a3. Synthesis: 3 retrieved analogs transferred to Ke3 (summed QAP objective 68, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

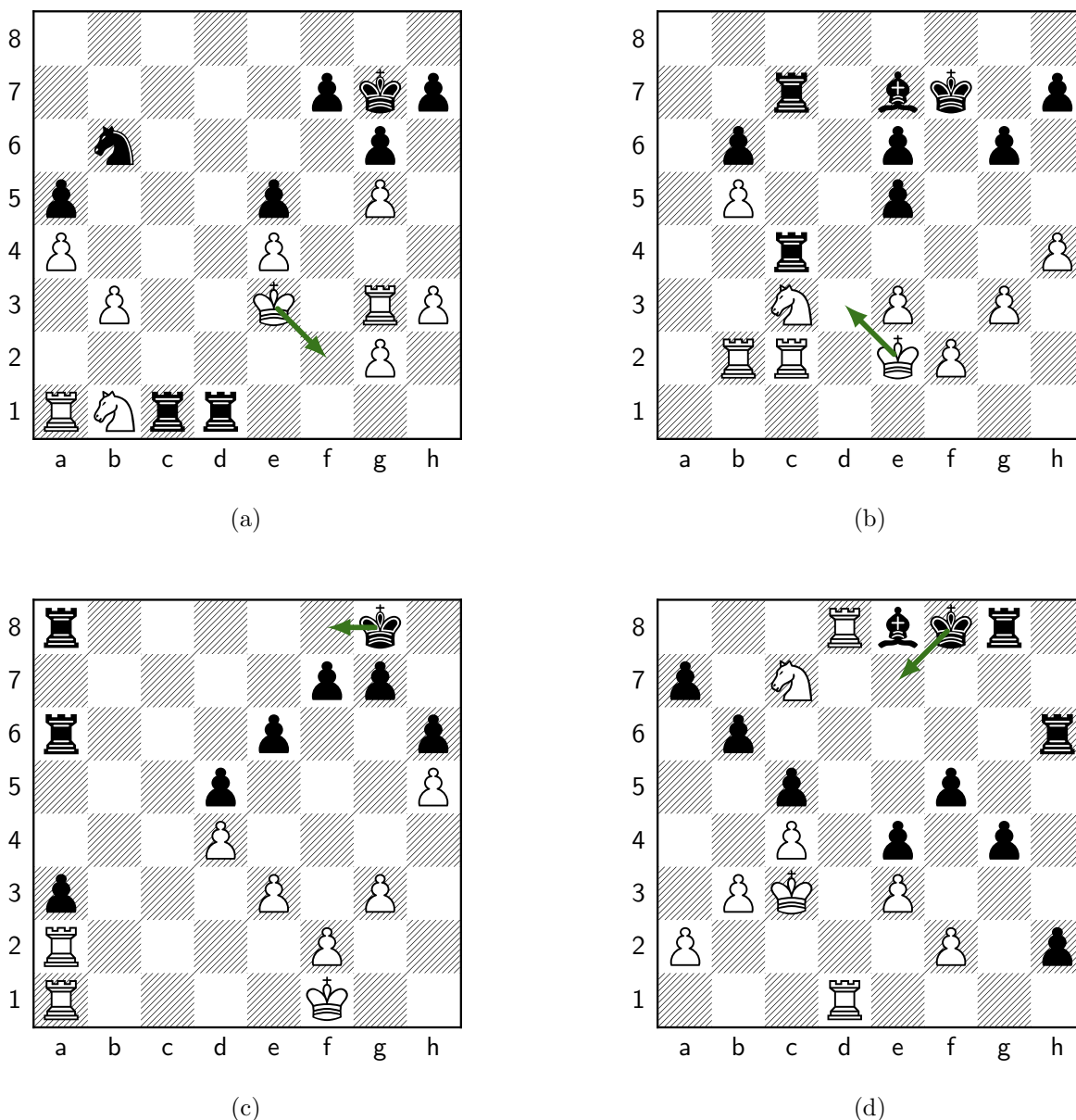
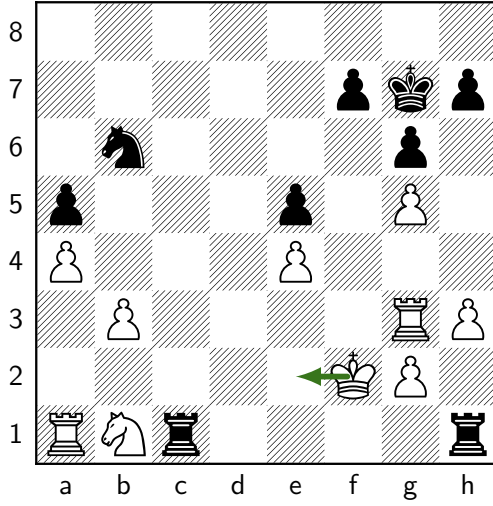
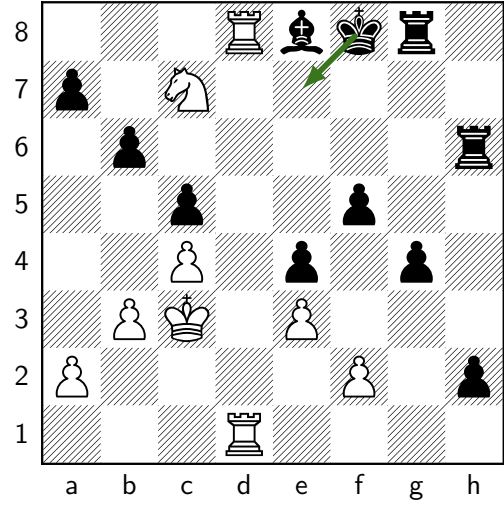


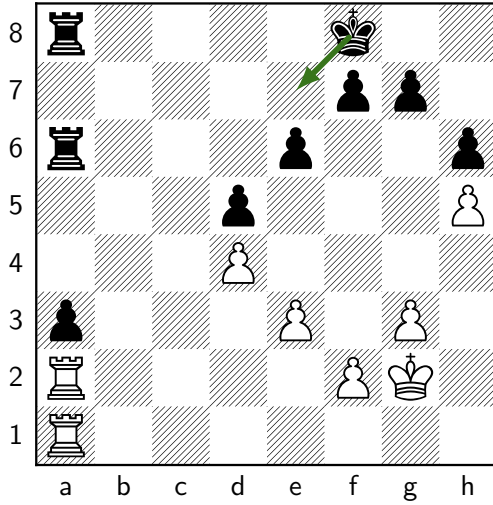
Figure 44: **Move 50, White: Kf2.** (a) Our position; Melanie plays Kf2 (e3f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from lichess_elite_2024-09_p0031#3257 (ply 100) [structural distance $d=61$, QAP obj=48]: the master played Kd3; under μ our Ke3 plays the role of that game's Ke2; cross-type roles: our Pg2 as their Rb2, our nb6 as their Pf2; transferred move e3f2 (**endorses this move**). (c) Retrieved from lichess_elite_2024-09_p0021#559 (ply 83) [structural distance $d=59$, QAP obj=22]: the master played Kf8; under μ our Ke3 plays the role of that game's Kg1; cross-type roles: our Nb1 as their Pa3; transferred move e3f2 (**endorses this move**). (d) Retrieved from lichess_elite_2024-09_p0021#3969 (ply 91) [structural distance $d=51$, QAP obj=47]: the master played Ke7; under μ our Ke3 plays the role of that game's Kf1; cross-type roles: our Nb1 as their Be8, our rc1 as their nc7; transferred move e3e2. Synthesis: 5 retrieved analogs transferred to Kf2 (summed QAP objective 132, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



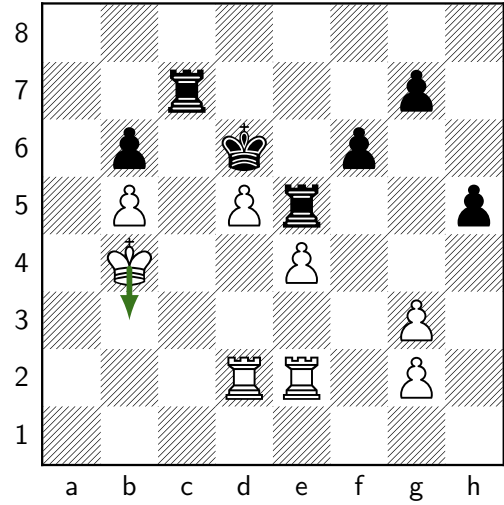
(a)



(b)

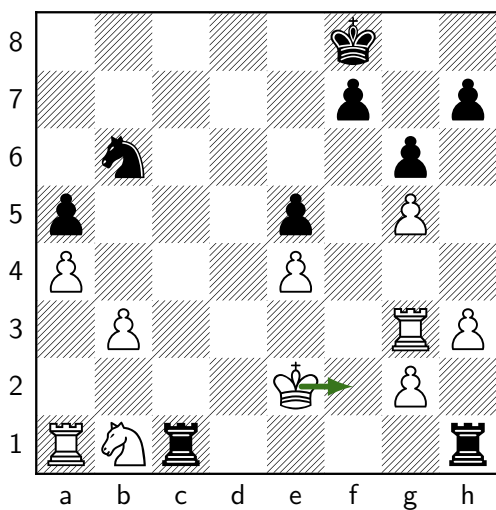


(c)

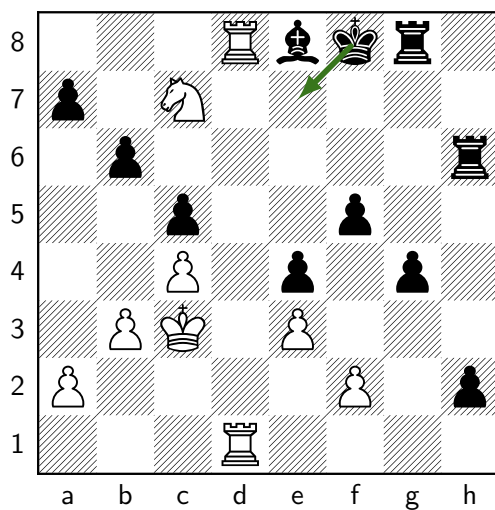


(d)

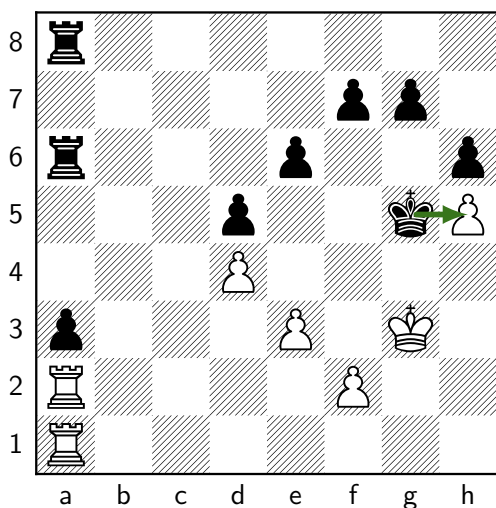
Figure 45: **Move 51, White: Ke2.** (a) Our position; Melanie plays Ke2 (f2e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0021#3969` (ply 91) [structural distance $d=63$, QAP obj=51]: the master played Ke7; under μ our Kf2 plays the role of that game’s Kf1; cross-type roles: our Nb1 as their Be8, our rc1 as their nc7; transferred move f2e2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0021#559` (ply 85) [structural distance $d=46$, QAP obj=22]: the master played Ke7; under μ our Kf2 plays the role of that game’s Kf1; cross-type roles: our nb6 as their Ph6, our Nb1 as their Pa3; transferred move f2e2 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0026#1677` (ply 94) [structural distance $d=71$, QAP obj=20]: the master played Kb3; under μ our Kf2 plays the role of that game’s Kb4; cross-type roles: our Nb1 as their Pe4, our Nb1 as their Pd5; transferred move f2e2 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Ke2 (summed QAP objective 116, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



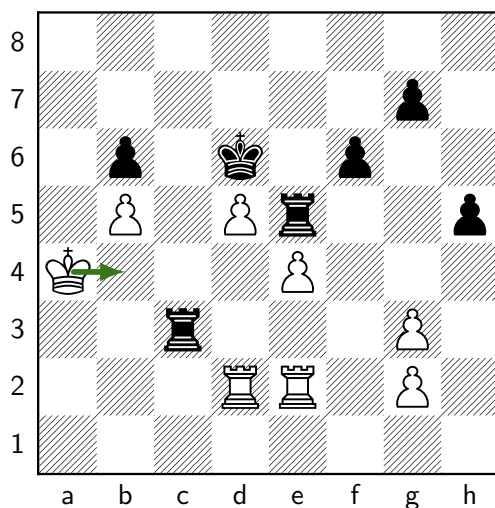
(a)



(b)

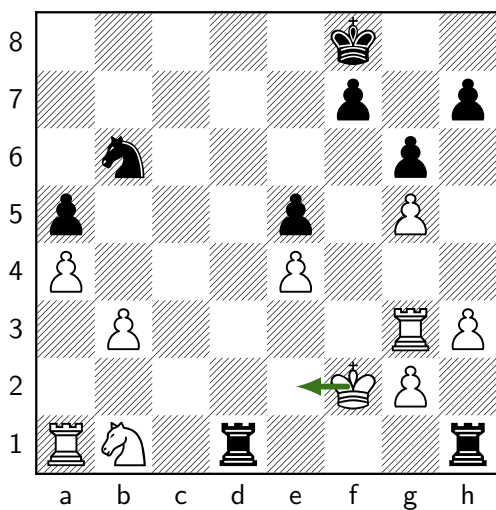


(c)

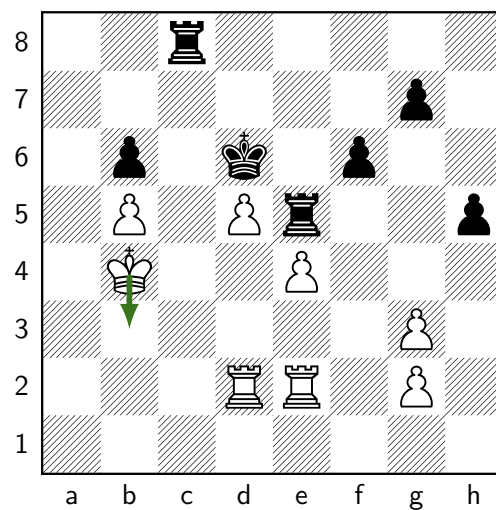


(d)

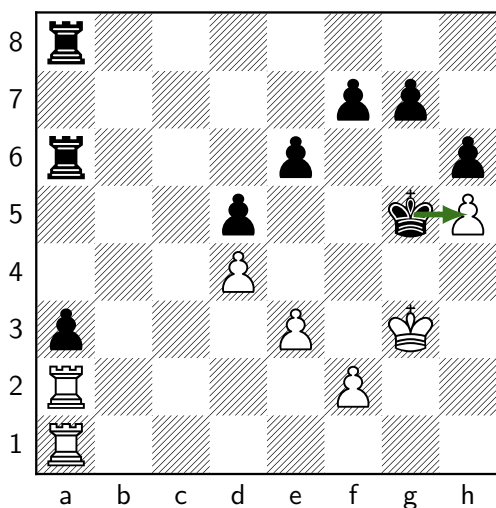
Figure 46: **Move 52, White: Kf2.** (a) Our position; Melanie plays Kf2 (e2f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0021#3969` (ply 91) [structural distance $d=72$, QAP obj=51]: the master played Ke7; under μ our Ke2 plays the role of that game’s Kf1; cross-type roles: our Nb1 as their Be8, our rc1 as their nc7; transferred move e2f2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0021#559` (ply 91) [structural distance $d=63$, QAP obj=22]: the master played Kxh5; under μ our Ke2 plays the role of that game’s Kg4; cross-type roles: our Nb1 as their Pa3; transferred move e2f2 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0026#1677` (ply 92) [structural distance $d=73$, QAP obj=20]: the master played Kb4; under μ our Ke2 plays the role of that game’s Ka4; cross-type roles: our Pg5 as their Rd2, our nb6 as their rc3; transferred move e2f2 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Kf2 (summed QAP objective 161, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



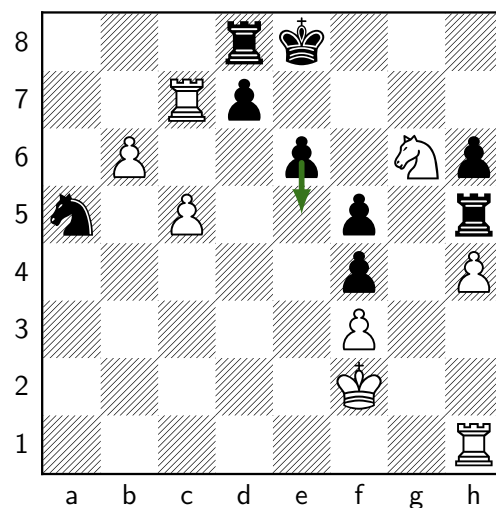
(a)



(b)



(c)



(d)

Figure 47: **Move 53, White: Ke2.** (a) Our position; Melanie plays Ke2 (f2e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0026#1677` (ply 102) [structural distance $d=70$, QAP obj=38]: the master played Kb3; under μ our Kf2 plays the role of that game’s Kb4; cross-type roles: our Nb1 as their Pe4, our Nb1 as their Pd5; transferred move f2e2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0021#559` (ply 91) [structural distance $d=67$, QAP obj=35]: the master played Kxh5; under μ our Kf2 plays the role of that game’s Kg4; cross-type roles: our Pg2 as their Ra8, our Nb1 as their Pa3; transferred move f2e2 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0022#2770` (ply 101) [structural distance $d=61$, QAP obj=39]: the master played e5; under μ our Pb3 plays the role of that game’s Pe3; cross-type roles: our pg6 as their Na5, our Nb1 as their ph4; transferred move b3b4. Synthesis: 5 retrieved analogs transferred to Ke2 (summed QAP objective 184, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

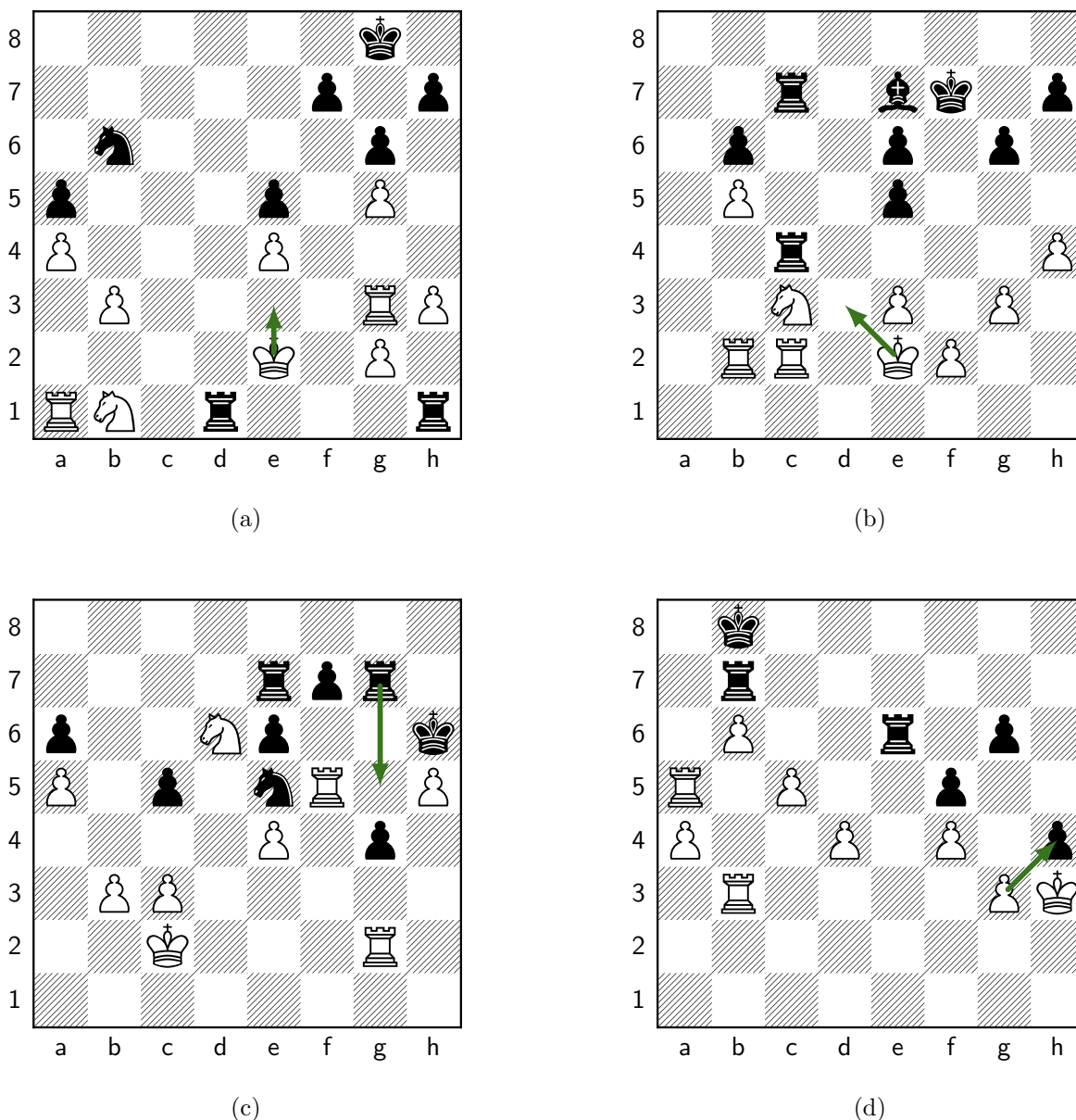
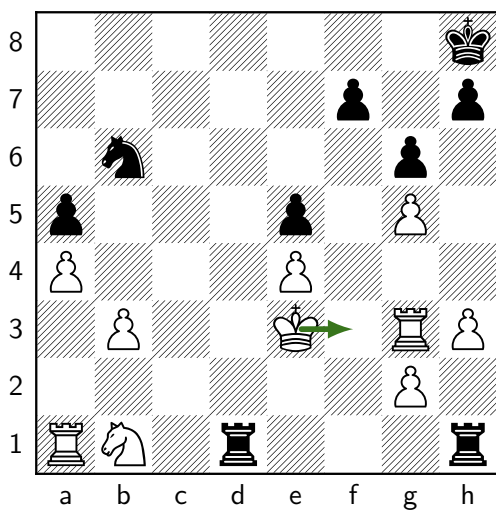
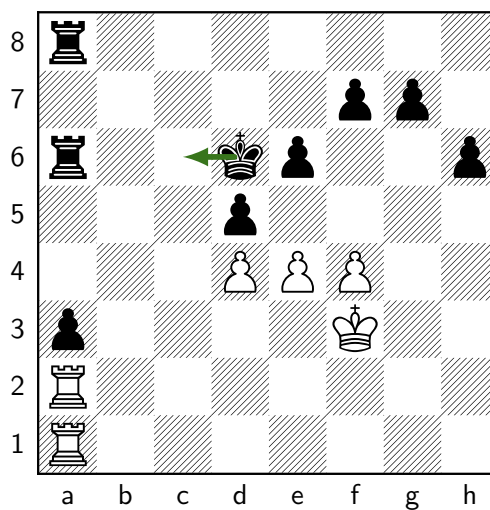


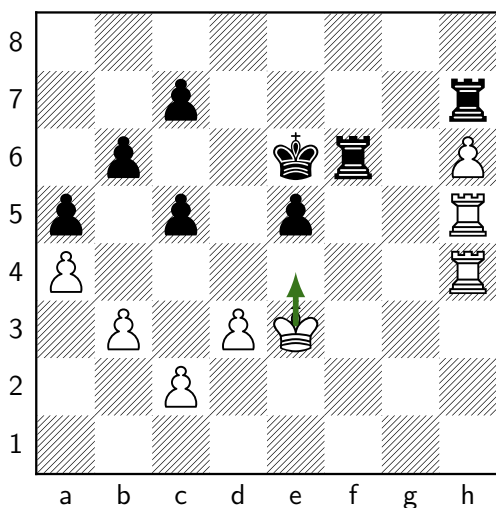
Figure 48: **Move 54, White: Ke3.** (a) Our position; Melanie plays Ke3 (e2e3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from lichess_elite_2024-09_p0031#3257 (ply 100) [structural distance $d=68$, QAP obj=25]: the master played Kd3; under μ our Ke2 plays the role of that game's Ke2; cross-type roles: our Ph3 as their Rb2, our nb6 as their Pf2; transferred move e2e3 (**endorses this move**). (c) Retrieved from lichess_elite_2024-09_p0022#2070 (ply 95) [structural distance $d=61$, QAP obj=29]: the master played Rg5; under μ our Rg3 plays the role of that game's Rg2; cross-type roles: our nb6 as their Re7, our Nb1 as their Pf7; transferred move g3d3. (d) Retrieved from lichess_elite_2024-09_p0031#2623 (ply 102) [structural distance $d=73$, QAP obj=29]: the master played gxh4; under μ our Nb1 plays the role of that game's Pg3 (a **cross-type** correspondence); cross-type roles: our rh1 as their ph4, our Pg5 as their Ra5; transferred move b1d2. Synthesis: 2 retrieved analogs transferred to Ke3 (summed QAP objective 46, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



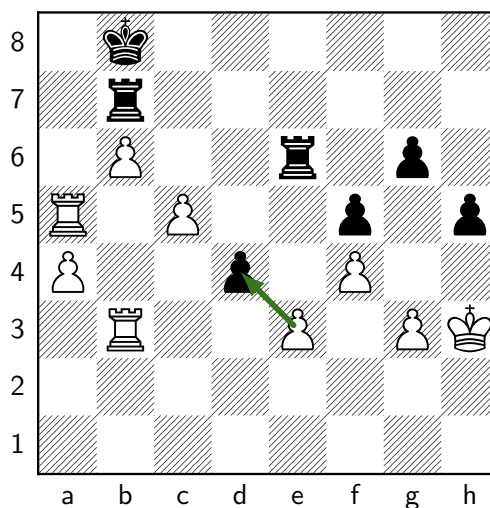
(a)



(b)

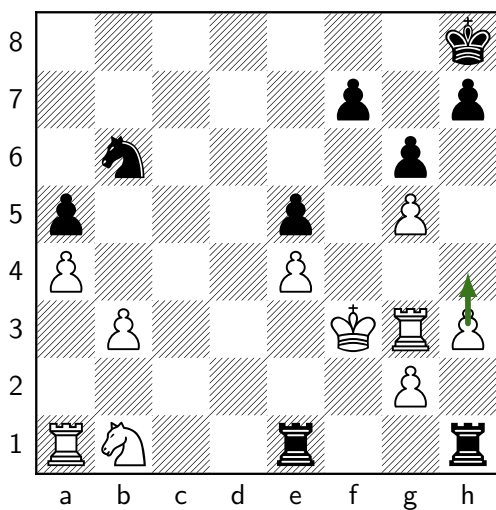


(c)

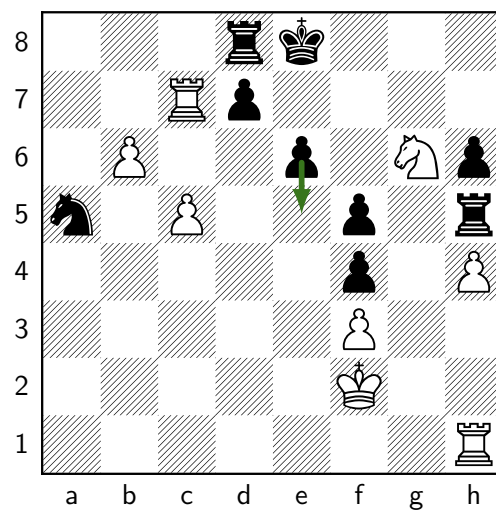


(d)

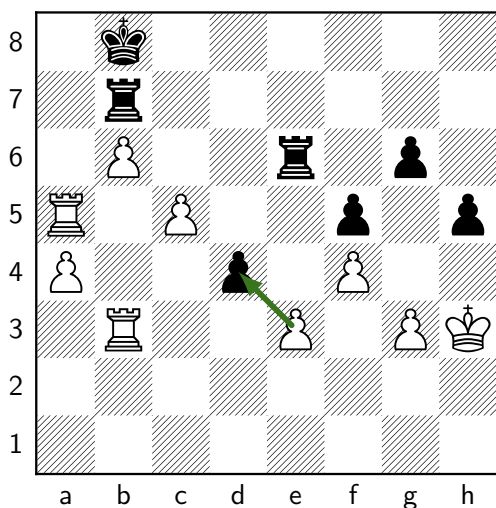
Figure 49: **Move 55, White: Kf3**. (a) Our position; Melanie plays Kf3 (e3f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0021#559` (ply 101) [structural distance $d=71$, QAP obj=41]: the master played Kc6; under μ our Ke3 plays the role of that game’s Kd3; cross-type roles: our Pg2 as their Ra8, our nb6 as their pe4; transferred move e3f3 (**endorses this move**). (c) Retrieved from `lichess_elite_2025-11_p0000#2370` (ply 112) [structural distance $d=69$, QAP obj=34]: the master played Ke4; under μ our Ke3 plays the role of that game’s Ke3; cross-type roles: our nb6 as their Rh5, our Nb1 as their Ph6; transferred move e3f3 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0031#2623` (ply 100) [structural distance $d=59$, QAP obj=39]: the master played exd4; under μ our Nb1 plays the role of that game’s Pe3 (a **cross-type** correspondence); cross-type roles: our rd1 as their pd4, our ph7 as their Ra5; transferred move b1c3. Synthesis: 2 retrieved analogs transferred to Kf3 (summed QAP objective 75, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



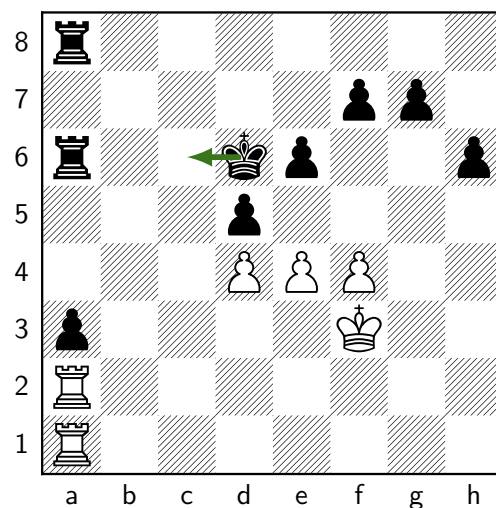
(a)



(b)

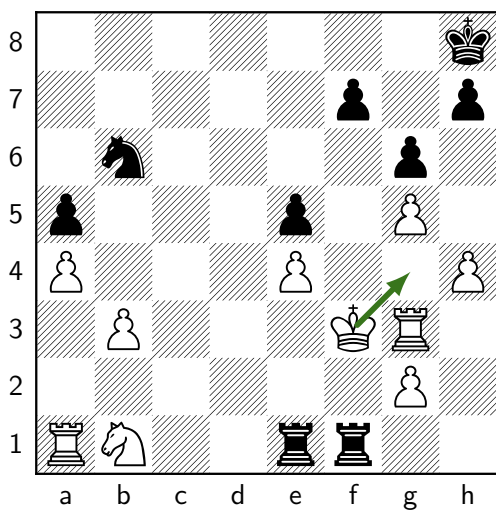


(c)

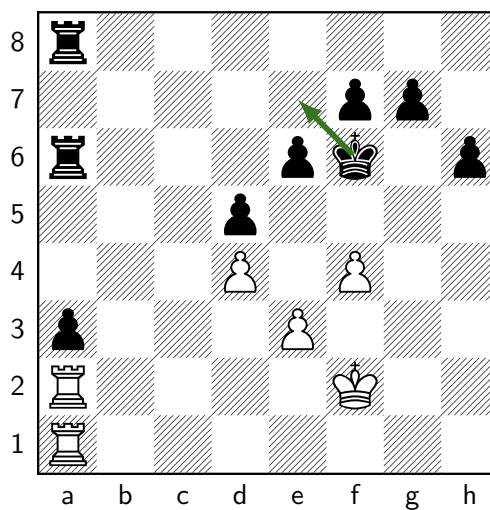


(d)

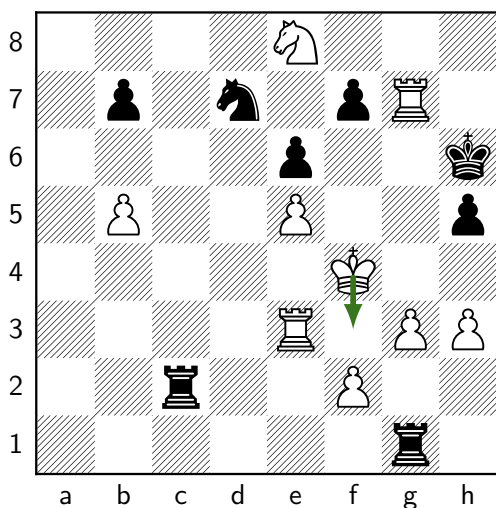
Figure 50: **Move 56, White: h4**. (a) Our position; Melanie plays h4 (h3h4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0022#2770` (ply 101) [structural distance $d=66$, QAP obj=61]: the master played e5; under μ our *Ph3* plays the role of that game’s *Pe3*; cross-type roles: our *Nb1* as their *Pd7*, our *rh1* as their *pb6*; transferred move h3h4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0031#2623` (ply 100) [structural distance $d=61$, QAP obj=59]: the master played exd4; under μ our *Nb1* plays the role of that game’s *Pe3* (a **cross-type** correspondence); cross-type roles: our *re1* as their *pd4*, our *Ph3* as their *Ra5*; transferred move b1c3. (d) Retrieved from `lichess_elite_2024-09_p0021#559` (ply 101) [structural distance $d=75$, QAP obj=57]: the master played Kc6; under μ our *Kf3* plays the role of that game’s *Kd3*; cross-type roles: our *Pg2* as their *Ra8*, our *Rg3* as their *Pf7*; transferred move f3g4. Synthesis: 1 retrieved analog transferred to h4 (summed QAP objective 61, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



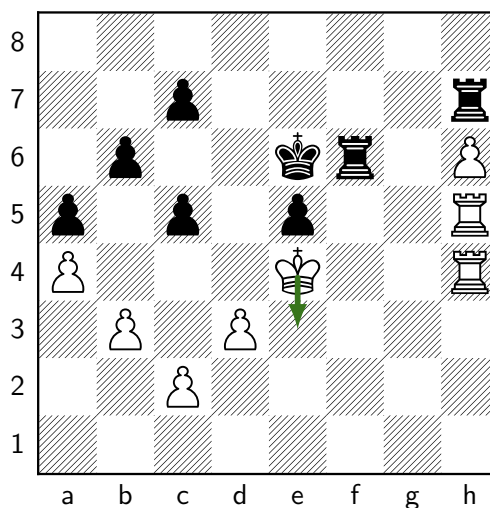
(a)



(b)

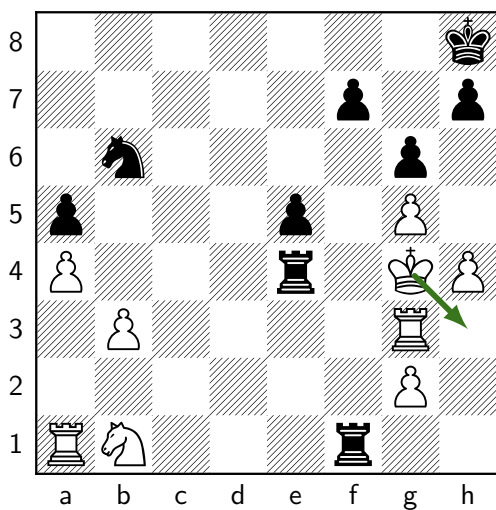


(c)

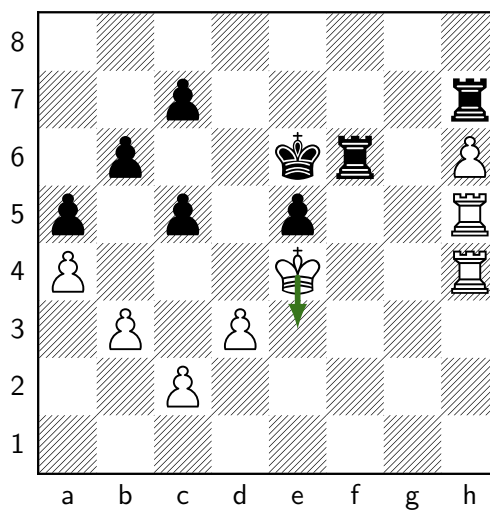


(d)

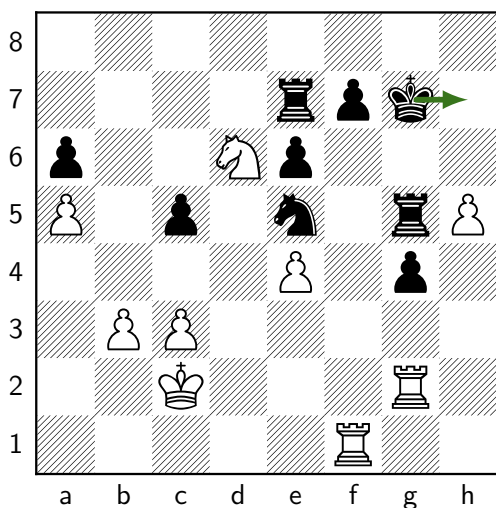
Figure 51: **Move 57, White: Kg4.** (a) Our position; Melanie plays Kg4 (f3g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0021#559` (ply 97) [structural distance $d=69$, QAP obj=28]: the master played Ke7; under μ our Kf3 plays the role of that game’s Kf3; cross-type roles: our Pb3 as their Ra8, our Rg3 as their Pf7; transferred move f3g4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0028#1547` (ply 98) [structural distance $d=69$, QAP obj=25]: the master played Kf3; under μ our Kf3 plays the role of that game’s Kf4; cross-type roles: our pg6 as their rc2, our Rg3 as their Pf2; transferred move f3g4 (**endorses this move**). (d) Retrieved from `lichess_elite_2025-11_p0000#2370` (ply 102) [structural distance $d=64$, QAP obj=25]: the master played Ke3; under μ our Kf3 plays the role of that game’s Ke4; cross-type roles: our Nb1 as their Ph6; transferred move f3g4 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Kg4 (summed QAP objective 154, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



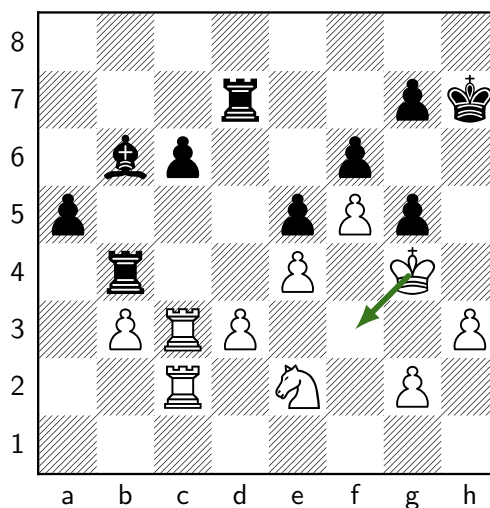
(a)



(b)



(c)



(d)

Figure 52: **Move 58, White: Kh3.** (a) Our position; Melanie plays Kh3 (**g4h3**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2025-11_p0000#2370` (ply 102) [structural distance $d=62$, QAP obj=21]: the master played Ke3; under μ our Kg4 plays the role of that game’s Ke4; cross-type roles: our Nb1 as their Ph6; transferred move g4h3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0022#2070` (ply 99) [structural distance $d=66$, QAP obj=17]: the master played Kh7; under μ our Kg4 plays the role of that game’s Kg2; cross-type roles: our nb6 as their Re7, our re4 as their Pa6; transferred move g4h3 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0012#59` (ply 102) [structural distance $d=65$, QAP obj=11]: the master played Kf3; under μ our Kg4 plays the role of that game’s Kg4; cross-type roles: our ph7 as their Ne2, our Pa4 as their Rc3; transferred move g4h3 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Kh3 (summed QAP objective 98, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.